Instructions

T-GSI

TURBINE - GENERATOR SUPERVISORY INSTRUMENTATION

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SAFETY SYMBOL LEGEND

********** * * * WARNING * * * **********	Commands attention to an operating procedure, practice, condition, or statement, which, if not strictly observed, could result in personal injury or death.
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NOTE	Commands attention to an essential operating or maintenance procedure, condition, or statement which must be highlighted.

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T-GSI INSTRUCTION BOOK

TABLE OF CONTENTS

		Page
1.0	INTRODUCTION	1
2.0	RECEIVING, HANDLING, AND STORAGE	3
2.1	GENERAL	3
2.2	RECEIVING AND HANDLING	3
2.3	STORAGE	3
3.0	EQUIPMENT DESCRIPTION	5
3.1	THE MASTER STATION	5
3.2	REMOTE STATIONS: TSI, TURBINE, And GENERATOR	7
3.2.1	THE TSI REMOTE STATION	8
3.2.2	THE TURBINE REMOTE STATION	8
3.2.3	THE GENERATOR REMOTE STATION	9
3.3	THE CONTROL CONSOLE	9
3.4	THE CRT MONITOR	10
3.5	THE TELEPRINTER (Optional)	10
4.0	SPECIFICATIONS	12
4.1	EQUIPMENT LOCATION	12
4.2	CABLES	12
4.2.1	MAXIMUM PERMISSIBLE LENGTH OF CONNECTING CABLES	12
4.2.2	ATTACHMENTS AND CABLE ACCESS	12
4.2.3	INTERCONNECTING CABLE BETWEEN CABINETS AND EQUIPMENT	12
4.3	ENVIRONMENT	12
4.3.1	TEMPERATURE	12
4.3.2	HUMIDITY	13
4.4	ACCURACY	13
4.4.1	INPUT POWER	13
4.4.2	SUSCEPTABILITY TO ELECTRICAL NOISE (Common Mode Rejection)	13
4.4.3	BAD SENSOR DETECTION AND INHIBITING	14
4.5	SCALES	14
4.6	RECORDING INTERVAL	15
4.6.1	UPDATING (Response Time)	15
4.6.2	LOG RECORDING INTERVAL	15
4.7	MEMORY CAPACITY	15
4.7.1	TAPE	15
4.7.2	DISKDRIVE	15
4.7.3	TREND DISPLAY	15
4.8	I/O DEVICES	16
4.8.1	CRT DISPLAY	16
4.8.1.1	CRT	16
4.8.1.2	CRT CONTROL CONSOLE	16
4.8.2	TAPE RECORDER	16
4.8.3	MAGNETIC DISK UNIT	16
4.8.4	TELEPRINTER PORT	16
4.8.5	DATA LINK TO CUSTOMER COMPUTER	16

TABLE OF CONTENTS (Continued)

		Page
5.0	INSTALLATION	17
5.1	GENERAL	17
5.1.1	DIAGNOSTICS	18
5.2	MASTER CABINET	21
5.2.1	EQUIPMENT INSTALLATION	21
5.2.2	CHECKOUT AND TEST	21
5.2.2.1	POWER CHECKS	21
5.2.2.1.1	GROUND CHECK	21
5.2.2.1.2	115 VAC POWER CHECK	21
5.2.2.1.3	230 VAC POWER CHECK	21
5.2.2.1.4	LOADED POWER SUPPLY CHECK	22
5.2.2.2	DISPLAY (CRT) CONTROLLER VERIFICATION	22
5.2.2.3	MAIN CONTROLLER VERIFICATION	28
5.2.2.4	HISTORICAL CONTROLLER VERIFICATION	32
5.2.2.5	DISPLAY SECTION	36
5.2.2.5.1	HCMA CHECK (PTCA, PCPA)	36
5.2.2.5.2	LED TEST ON CONTROL CONSOLE	36
5.2.2.6	COLOR CRT CHECK	36
5.2.2.6.1	POWER UP OF DISPLAY GENERATOR	36
5.2.2.6.2	POWER UP OF COLOR CRT	37
5.2.2.7	TELEPRINTER PORT CHECK	38
5.2.2.8	MAIN SECTION	38
5.2.2.8.1	HCMA CHECK (PCPA)	38
5.2.2.8.2	HMHA CHECK (PDLA)	38
5.2.2.8.3	HARACHECK	38
5.2.2.8.4	MAIN/DISPLAY CONTROLLER ON LINE CHECKOUT	38
5.2.2.9	HISTORICAL SECTION	39
5.2.2.9.1	POWER UP OF DISK DRIVE	39
5.2.2.9.2	POWER UP OF TAPE DECK	41
5.2.2.9.3	BRINGING HISTORICAL CONTROLLER ON LINE	43
5.3	CONTROL CONSOLE	44
5.3.1	EQUIPMENT INSTALLATION	44
5.3.2	CHECKOUT AND TEST	44
5.4	CRT MONITOR	44
5.4.1	EQUIPMENT INSTALLATION	44
5.4.2	CHECKOUT AND TEST	44
5.5	TELEPRINTER (OPTIONAL)	44
5.5.1	EQUIPMENT INSTALLATION	44
5.5.2	CHECKOUT AND TEST	44
5.6	TSI REMOTE CABINET	45
5.6.1	EQUIPMENT INSTALLATION	45
5.6.1.1	CONNECTIONS	45
5.6.1.1.1	GROUNDING	45
5.6.1.1.2	DATA LINK	45
5.6.1.1.3	ALARM-INITIATION CONTACTS CONNECTED IN PARALLEL	45
5.6.2	CHECKOUT AND TEST	45
5.6.2.1	POWER CHECKS	45
5.6.2.1.1	GROUND CHECK	45
5.6.2.1.2	115 VAC POWER CHECK	46
5.6.2.1.3	230 VAC POWER CHECK	46

TABLE OF CONTENTS (Continued)

		Page
5.6.2.2	ANALOG CHECK - TSI INPUTS	46
5.6.2.2.1	GENERAL PROCEDURE	46
5.6.2.2.2	POWER CHECKS	46
5.6.2.2.3	ANALOG INPUTS	47
5.6.2.2.4	ANALOG ADJUSTMENTS	47
5.6.2.3	LOADED POWER SUPPLY CHECK (Digital)	48
5.6.2.4	CONTROLLER VERIFICATION	48
5.6.2.5	INPUT/OUTPUT CHECK	52
5.6.2.5.1	CONTACTS	52
5.6.2.5.2	SPEED	52
5.6.2.5.3	THERMOCOUPLE INPUTS	52
5.6.2.5.4	4 - 20 MA INPUTS	53
5.6.2.5.5	VOLTAGE INPUTS	54
5.7	TURBINE REMOTE CABINET	54
5.7.1	EQUIPMENT INSTALLATION	54
5.7.1.1	GROUNDING	54
5.7.1.2	CONNECTIONS TO TRANSDUCERS	55
5.7.1.3	DATA LINK	55
5.7.2	CHECKOUT AND TEST	57
5.7.2.1	POWER CHECKS	57
5.7.2.1.1	GROUND CHECK	57
5.7.2.1.2	115 VAC POWER CHECK	57
5.7.2.1.3	230 VAC POWER CHECK	57
5.7.2.1.4	LOADED POWER SUPPLY CHECK	57
5.7.2.2	CONTROLLER VERIFICATION	58
5.7.2.3	INPUT/OUTPUT CHECK	58
5.7.2.3.1	CONTACTS	58
5.7.2.3.2	THERMOCOUPLE INPUTS	59
5.7.2.3.3	4 - 20 MA INPUTS	60
5.7.2.3.4	0 - 10 VOLT INPUTS	60
5.8	GENERATOR REMOTE CABINET	60
5.8.1	EQUIPMENT INSTALLATION	60
5.8.1.1	GROUNDING	60
5.8.1.2	CONNECTIONS TO TRANSDUCERS	61
5.8.1.3	DATA LINK	61
5.8.2	CHECKOUT AND TEST	61
5.8.2.1	POWER CHECKS	61
5.8.2.1.1	GROUND CHECK	61
5.8.2.1.2	115 VAC POWER CHECK	61
5.8.2.1.3	230 VAC POWER CHECK	61
5.8.2.1.4	LOADED POWER SUPPLY CHECK	62
5.8.2.2	CONTROLLER VERIFICATION	62
5.8.2.3	INPUT/OUTPUT CHECK	62
5.8.2.3.1	CONTACTS	63
5.8.2.3.2	THERMOCOUPLE INPUTS	63
5.8.2.3.3	4 - 20 MA INPUTS	64
5.8.2.3.4	0 - 10 VOLT INPUTS	64
5.8.2.3.5	RTDs	65
5.8.2.3.6	3 - PHASE POWER & FREQUENCY	66

TABLE OF **CONTENTS** (Continued)

		Page
5.9	SYSTEM CHECKS	66
5.9.1	EXERCISING REMOTES AND MASTER	66
5.9.2	DATA RETRIEVAL	66
6.0	OPERATION	68
6.1	INTRODUCTION	68
6.2	CONTROL CONSOLE KEY DESCRIPTION	68
6.3	DATA AND DISPLAY DESCRIPTIONS	70
6.3.1	CRT DISPLAY - GENERAL	70
6.3.2	COMPOSITE GRAPH DATA PAGE	70
6.3.3	GENERAL DATA PAGE	71
6.3.4	DATA GROUP PAGES	71
6.3.4.1	Bearing Data Page	71
6.3.4.2	Turbine Temperature Data Page	71
6.3.4.3	LP Turbine Data Page	71
6.3.4.4	Cooling Gas Temperatures Page	71
6.3.4.5	Stator RTD And TC Temperatures Page	72
6.3.4.6	Stator Water System Data Page	72
6.3.4.7	Excitation System Data Page	72
6.3.5	TREND DISPLAY	72
6.3.6	TREND CHANNEL DIRECTORY PAGE	73
6.3.7	SELECTIVE DATA PAGE	73
6.3.8	ALARM SUMMARY PAGE	73
6.3.9	DATA TABLE (LOG) DISPLAY	73
6.3.10	ALARM AND TRIP PROVISIONS	73
6.3.11	ALARM LOGIC: TURBINE-GENERATOR SHAFT VIBRATION	74
6.3.12	SYSTEM MESSAGES	74
6.3.12.1	Display Controller Messages	74
6.3.12.2	Main Controller Messages	75
6.3.12.3	Historical Controller Messages - Disk	75
6.3.12.4	Historical Controller Messages - Tape	75
6.3.12.5	Lights	75
6.3.13	JOURNAL BEARING TEMPERATURE SPIKE MONITOR	76
6.3.14	TAPERED LAND THRUST BEARING TEMPERATURE MONITOR	76
6.3.15	RATE OF MAIN STEAM AND REHEAT STEAM TEMPERATURE CHANGE ALARM	76
6.3.16	GENERATOR STATOR TEMPERATURE VARIABLE ALARM LIMIT	76
6.4	OPERATOR'S CONTROL CONSOLE INSTRUCTIONS	77
6.4.1	INTRODUCTION	77
6.4.2	CANCELLING A SEQUENCE	77
6.4.3	ENTERING NUMERIC DATA	77
6.4.4	OPERATOR ERRORS	78
6.4.5	PUSHBUTTON OPERATIONS	78
6.4.6	GENERAL RULES FOR USING THE FOLLOWING SEQUENCES	78
6.4.7	SEQUENCE TO DISPLAY A GENERAL DATA PAGE	78
6.4.8	SEQUENCE TO DISPLAY AN ALARM SUMMARY PAGE	79
6.4.9	SEQUENCE TO DISPLAY COMPOSITE DATA PAGE	79
6.4.10	SEQUENCE TO DISPLAY DATA GROUP	80
6.4.11	SEQUENCE TO ENTER A TIME SET	80
6.4.12	SEQUENCE TO ALARM ACKNOWLEDGE	83

TABLE OF CONTENTS (Continued)

		Page
6.4.13	SEQUENCE TO DISPLAY TREND DIRECTORY	a4
6.4.14	SEQUENCE TO DISPLAY TREND GRAPH	85
6.4.15	SEQUENCE TO DISPLAY SELECTIVE DATA	86
6.4.16	SEQUENCE TO DISPLAY CURRENT LOG	88
6.4.17	SEQUENCE TO DISPLAY AN HISTORICAL LOG	89
6.4.18	SEQUENCE TO READ TAPE	93
6.4.19	SEQUENCE TO DISPLAY NEXT HISTORICAL LOG	96
6.4.20	SEQUENCE TO PERFORM LED TEST ON PANEL	98
6.5	TELEPRINTER INSTRUCTIONS	99
6.5.1	INTRODUCTION	99
6.5.2	OPERATING INSTRUCTIONS	99
6.5.2.1	General Information	99
6.5.2.2	Edit Functions	99
6.5.2.3	Command Format	101
6.5.3	COMMAND DESCRIPTIONS	102
6.5.3.1	Current Log - CL [(subclass)]	102
6.5.3.2	Historical Log - HL (date/time) [(subclass)]	102
6.5.3.3	Next Log - NL [(subclass)]	102
6.5.3.4	Previous Log - PL [(subclass)]	102
6.5.3.5	Same Log - SL [(subclass)]	103
6.5.3.6	Tape Read - TR (date)	103
6.5.3.7	Set - SET BAUD/COLUMNS/LINES (number)	103
6.5.3.8	Alarm - ALARM	104
6.5.3.9	Add Channel - AC [(channel No.),(channel No.),...]	104
6.5.3.10	Delete Channel - DC [(channel No.),(channel No.) ....]	104
6.5.3.11	Help - HELP [(mnemonic)]	105
6.6	HISTORICAL DATA STORAGE INSTRUCTIONS	107
6.6.1	HISTORICAL CONTROLLER	107
6.6.1.1	POWERFAIL PROCEDURE	107
6.6.2	TAPE FUNCTIONS	107
6.6.2.1	DISMOUNTING A FULL TAPE	107
6.6.2.2	DISMOUNTING A PARTIALLY RECORDED TAPE	107
6.6.2.3	PERFORMING A TAPE READ ON AN HISTORICAL TAPE	108
6.6.2.4	TO DISMOUNT AN HISTORICAL TAPE.	109
6.6.2.5	REMOUNTING THE PARTIALLY RECORDED CURRENT TAPE	109
6.7	ALARM SET POINT INSTRUCTIONS	110
6.7.1	ALARM LIMIT OPERATION	110
6.7.2	SET UP CONDITIONS	110
6.7.3	ALARM LIMIT IDENTIFIERS	111
6.7.4	DISPLAY ALARM LIMIT SETTINGS	111
6.7.5	MODIFY ALARM LIMIT SETTINGS	112
6.7.6	RESETTING ALARM LIMIT SETTINGS	112
6.7.7	DISPLAY CHANNEL VALUE	113
6.7.8	MISCELLANEOUS ALARM LIMITS COMMANDS	113
7.0	MAINTENANCE	114
7.1	SYSTEM OVERVIEW	114
7.1.1	SIGNAL FLOW	114

TABLE OF CONTENTS (Continued)

Page

7.2 ROUTINE MAINTENANCE 123
7.2.1 MONTHLY 123
7.2.2 QUARTERLY 123
7.2.3 MAINTENANCE RECORDS 123
7.3 TROUBLESHOOTING 123
7.3.1 GENERAL 123
7.3.2 MASTER CABINET 124
7.3.2.1 MAIN CONTROLLER 124
7.3.2.2 DISPLAY CONTROLLER 129
7.3.2.3 HCMA CHECK (PTCA) 131
7.3.2.4 HMHA CHECK (PDLA) 131
7.3.2.5 HISTORICAL CONTROLLER 131
7.3.2.6 EXPLANATION OF DISK CRT MESSAGES 132
7.3.2.7 EXPLANATION OF TAPE CRT MESSAGES 134
7.3.2.7.1 TAPE DUMP CRT MESSAGES 135
7.3.2.8 DIAGNOSTIC COMMANDS 135
7.3.2.9 HISTORICAL CONTROLLER DIAGNOSTICS COMMANDS 135
7.3.2.10 DISK FUNCTIONS 136
7.3.2.11 TAPE FUNCTIONS 137
7.3.3 TSI REMOTE CABINET 137
7.3.3.1 CONTROLLER VERIFICATION-FAILURE TO PRINT OUT 137
7.3.3.2 ADJUSTMENT OF OVERVOLTAGE PROTECTORS 139
7.3.4 TURBINE REMOTE CABINET 139
7.3.4.1 CONTROLLER VERIFICATION-FAILURE TO PRINT OUT 139
7.3.5 GENERATOR REMOTE CABINET 141
7.3.5.1 CONTROLLER VERIFICATION-FAILURE TO PRINT OUT 141
7.3.6 CONTROL CONSOLE 142
7.3.7 CRT MONITOR 142
7.3.8 TELEPRINTER (OPTIONAL) 142
7.3.9 MODEM (DS512) 142
7.3.10 DS512 D.C. - SELF TEST 144
7.3.11 KOOLTRONIC ELECTRONIC CABINET AIR CONDITIONER (MODEL KAC8) 144
7.3.11.1 ROUTINE MAINTENANCE 144
7.3.11.1.1 UNUSUAL CONDITIONS 145
7.3.11.2 REFRIGERANT LOSS 146
7.3.11.3 CHARGING PROCEDURE 146
7.3.11.4 SPECIFICATIONS 147
7.3.11.5 BASIC RENEWAL PARTS 148
7.3.11.5.1 BASIC RENEWAL PARTS LISTING 148
7.4 CARD AND MODULE FUNCTION LISTS 148
7.4.1 GENERAL - DIRECTO-MATIC* PLUS 148
7.4.2 CARD FUNCTION LIST 151
7.4.3 MODULE FUNCTION LIST 158
7.4.4 OTHER MODULES 159
8.0 RENEWAL PARTS 160

INDEX 161
ADDENDUM. . . . . AF'TERPAGE 164

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LIST OF FIGURES

FIGURE	SUBJECT	PAGE
1	T-GSI COMPONENTS	6
2	CONTROL CONSOLE	11
3	MASTER STATION - RIGHT BAY - FRONT	24
4	MAINTENANCE MODULE	25
5	POWER UP PRINT OUT - DISPLAY	26
6	STATUS PRINT OUT - DISPLAY	27
7	ERROR PRINT OUT - DISPLAY	27
8	POWER UP PRINT OUT - MAIN	29
9	STATUS PRINT OUT - MAIN	30
10	ERROR PRINT OUT - MAIN	31
11	MASTER STATION - RIGHT BAY - REAR	33
12	POWER UP PRINT OUT - HISTORICAL	34
13	STATUS PRINT OUT - HISTORICAL	35
14	MASTER STATION - LEFT BAY - FRONT	42
15	POWER UP PRINT OUT - REMOTE CONTROLLER	49
16	STATUS PRINT OUT (with no errors) - REMOTE	50
17	STATUS PRINT OUT (with errors) -REMOTE	51
18	TRANSDUCER WIRING	56
19	BLOCK DIAGRAM - MASTER STATION	119
20	BLOCK DIAGRAM - TSI REMOTE	120
21	BLOCK DIAGRAM - TURBINE REMOTE	121
22	BLOCK DIAGRAM - GENERATOR REMOTE	122
23	DIRECTO-MATIC* PLUS - STANDARD MODULES	150

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1.0 INTRODUCTION

The Turbine-Generator Supervisory Instrumentation (T-GSI) System is a monitoring system designed to give operators improved capability for visually monitoring, and thus controlling, the operation of the turbine-generator.

The T-GSI system includes a combination disk-tape drive for immediate retrieval and long term storage of data, a color cathode ray tube (CRT) monitor, a push-button control console for the operator's use, a plant data link to the central plant computer and an optional teleprinter.

Because of the large number of sensors on the turbine and generator, and the resulting cabling requirements, the signals are conditioned and multiplexed at remote cabinets near the turbine-generator. They are then sent to the master cabinet in the relay room over a single, multi-wire communication data link cable from each remote station. This arrangement substantially reduces the amount of field run cable. The data can then be stored on magnetic disk and tape for recall, and immediately be displayed to the operator on the color CRT monitor.

As an early warning system, the T-GSI system provides increased operator awareness of the condition of the turbine-generator by using human engineered displays. From the control console, located on the control room, the operator can select from more than 15 individual pages for monitoring on the CRT and the data is displayed in a clear, compact, easily perceived format for improved operator effectiveness and control.

Refer to addendum for miscellaneous information or equipment variations.

The T-GSI system provides operating data on the following:

A. Turbine and Generator

1. vibration
2. speed
3. bearing metal temperature
4. journal bearing spike monitor

B. Turbine

1. shell, rotor, and differential expansion
2. control valve position
3. shaft eccentricity
4. temperatures: shell metal, water detection, main steam
5. rate of temperature change: main and reheat
6. exhaust pressure
7. thrust bearing temperature monitor
8. LP turbine water induction

C. Generator

1. stator slot and liquid header temperatures
2. stator water flow, pressure, and temperature
3. electrical operating parameters
4. cooling gas purity, pressure, and temperature
5. core monitor level
6. excitation system: status and operating parameters
7. generator load hold function

D. Boiler Feed Pump Turbine (if applicable)

1. shaft eccentricity
2. temperatures
3. vibration
4. speed

A list of the pages containing the above information is mounted on the operators' control console for easy reference.

2.0 RECEIVING, HANDLING, AND STORAGE

2.1 GENERAL

Upon receipt of the T-GSI shipment, examine it immediately for any damage that might have occurred in transit. If damage is found, a damage claim should be filed immediately with the transportation company; and the nearest Apparatus and Engineering Services Organization office of General Electric Company should be notified.

If the equipment is to be stored prior to installation, it should be left packed in the original packing material and containers. store in a dry area which is free of corrosive vapors and where the equipment will not be subjected to excessive shock, vibration, temperature, or humidity extremes.

2.2 RECEIVING AND HANDLING

The cabinets can be unpacked and inspected when storage is complete. If installation hazards are still present (dirt, moving of heavy materials, welding above, etc.) then unpacking should either be delayed, or the cabinets repacked and protected after inspection.

Check the cabinets, panels, and consoles for any dents, broken pushbuttons or lights, missing hinges, etc. Insure that the doors open easily and that no loose items are inside the cabinet. If any damage is found, contact the General Electric Apparatus and Engineering Services Organization department.

Upon completion of inspection, all possible precautions should be taken to keep the cabinets clean and undamaged. The doors should be kept closed at all times, and special care should be taken with the master cabinet, since the tape head requires a very clean environment.

2.3 STORAGE

A. SHORT TERM - PRIOR TO SYSTEM CHECKOUT.

During the installation process when excessive amounts of airborne particles are present, the cabinet doors should be kept closed. They should remain closed until the environment is clean and checkout of the system is to be started. This is most critical on the master cabinet because of the tape system head,

B. LONG TERM STORAGE

The cabinets are sealed by plastic when they are to be stored for an extended period of time. It is imperative that this seal not be broken (to check inside, etc.) until the storage period is over. As stated previously, store in a dry area which is free of corrosive vapors and where the equipment will not be subjected to excessive shock, vibration, temperature, or humidity extremes.

NOTES

3.0 EQUIPMENT DESCRIPTION

The T-GSI system consists of:

1. The Master Station
2. The TSI Remote cabinet
3. The Turbine Remote cabinet
4. The Generator Remote cabinet
5. The Control Console
6. The CRT Monitor
7. The Teleprinter (optional)

The T-GSI system components are depicted in Figure 1.

The packaging of the three remote stations and the master station is in NEMA type 12 cabinets. When specified, air conditioners are included to extract the electrical heat load out of the cabinets and each station is fitted with overtemperature relays to detect an air conditioner failure. These relays should be wired into the plant annunciator board to alarm in case such a failure occurs.

Heaters are provided in the remote cabinets for the purpose of raising the dew point in the event condensation occurs at the cabinet. The operator must physically turn these heaters on and off when needed: they do not work automatically. Each cabinet has a total of eight heaters, one on each door and wall, 500 Watts total; connected to the load side of the air conditioner breaker.

3.1 THE MASTER STATION

The Master Station is the central element of the T-GSI system. The station is made up from three subsystems: 3.1.1, Main controller, 3.1.2, Display Controller, and 3.1.3, Historical Controller. Other features include a plant computer data link and a teleprinter port.

3.1.1 Main Controller

The Main Controller is the controlling processor in a star configured arrangement of three remote stations. The principle task of the Main Controller is to gather all the inputs from the remote stations, alarm check the inputs, and then pass the input data to the Display Controller, the Historical Controller, the customer's annunciator and, if requested, the plant computer.

Information is passed to the plant computer using the T-GSI plant computer data link. The data link uses an asynchronous RS-232 full-duplex interface under link protocol cat. 3.1/F1 of ANSI standard X3.28. Modems are required if the data link cable length is greater than 50 feet or when there is a need for isolation between the plant computer and the T-GSI master station.

3.0 EQUIPMENT DESCRIPTION (Continued)

3.1.2 Display Controller

The main purpose of the Display Controller subsystem is to display T-GSI input data on the system CRT in a clear and concise format. A variety of display types, including present and historical input scans, categorized data group "pages", and a composite data display provides the operator with the flexibility of choosing the specific information to be examined. Other features include a trend display for up to three channels (operator selectable from the trend memory bank) and an alarm summary showing all present alarms - acknowledged and unacknowledged.

The Display Controller is also responsible for managing the communication with the control panel and the teleprinter. The teleprinter (optional) is used to provide a hard copy of a current or historic data log. The Display controller must interpret the teleprinter input commands and output the appropriate data log.

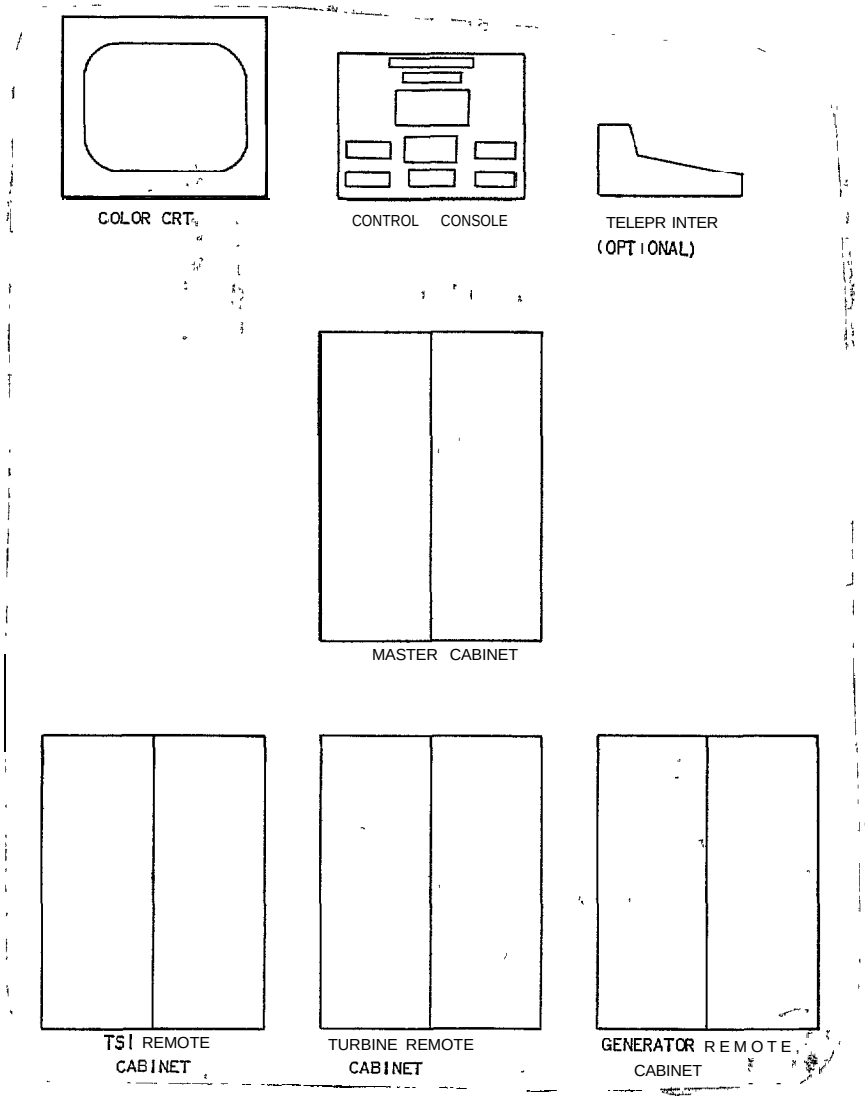


FIGURE 1: T-GSI COMPONENTS

3.0 EQUIPMENT DESCRIPTION (Continued)

3.1.2 Display Controller (Continued1

The connection port available for interfacing with a teleprinter (or video terminal) is a standard RS232C interface. Connections are made via a modem. Any buffered type teleprinter capable of operating at 300 baud and up to 9600 baud, with even parity, may be used. The default baud rate is either 300 or 1200 baud. The baud rate may be changed via the keyboard once the interface is made (see teleprinter operation section).

3.1.3 HISTORICAL CONTROLLER

The principle task of the Historical Controller is to receive a block of data from the Main Controller and to record the data to a disk and magnetic tape. The smallest unit of information stored on tape or disk is a log. A log consists of time and date, turbine number, and the magnitude of each point being monitored by the remote stations. The Historical Controller also handles requests made by the Display Controller. The requests require the Historical Controller to search the disk for a specified log (given a time and date), read the log, and send the log back to the Display Controller. The log specified must be found on the disk or the controller will state that the log was not found, even if the log is contained on tape (i.e. no search is made to tape). The tape read function 'permits the Historical Controller to transfer a block of data from an old tape to the disk. Once the data is transferred to the disk, the Historical Controller will permit searching.

3.2 REMOTE STATIONS: TSI, TURBINE & GENERATOR

To avoid repetition in the descriptions of the three remote stations, a general description is stated below. Separate equipment descriptions follow.

The purpose of the three remote stations is to include all the analog and digital hardware to input the data. Once the data has been input, it will be converted to a format that is usable by the master station, scaled and then stored.

Non-linear inputs: thermocouples, RTD's, hydrogen purity, and conductivity must be linearized before scaling. While doing the above task, the input data will also be tested to detect failed transducers. This data is sampled and updated once per second and is sent back to the master station on a high speed data link.

The high speed data link consists of a cable made up from three-three wire shielded triads: one for transmit data, one for receive data, and one for mike powered headsets.

3.0 EQUIPMENT DESCRIPTION (Continued)

3.2 REMOTE STATIONS: TSI, TURBINE, & GENERATOR (continued)

The data link is optically isolated to permit common mode isolation, up to 100 volts operational and spikes up to 1000 volts without damage, between the remote and master stations. The data link also has automatic error detection to insure the integrity of the data being sent to the master station.

3.2.1 THE TSI REMOTE STATION

This station receives the following signals from the main turbine:

vibration*
eccentricity*
shell expansion
rotor expansion
differential expansion
speed*
control valve position*
vibration phase angle

An asterisk (*) denotes signals available for monitoring on compatible boiler feed pump turbines; also available are selected temperatures. Boiler feed pump vibration can be monitored as long as a compatible signal is provided.

The main turbine (and BFPT) speed signal comes into the TSI cabinet as a digital input, for improved accuracy, and is converted to an analog signal. The other signals, except phase angle, are analog inputs that are converted to a digital signal,

This station also includes the TSI analog trip protection system that sends the high vibration (trip) signal to the EHC or TAC system.

3.2.2 THE TURBINE REMOTE STATION

This station receives or **calculates** the following signals from the main turbine:

all turbine and generator bearing temperatures
main steam shell metal temperatures
HP and LP water detection temperatures
exhaust presssure signals
rate of temperature change for main and reheat steam - calculated
maximum thrust bearing temperature and temperature differential - calculated

3.0 EQUIPMENT DESCRIPTION (Continued)

3.2.3 THE GENERATOR REMOTE STATION

This station receives and processes the following signals:

stator slot'temperature
liquid header temperatures
stator water flow
stator water pressure
stator water temperature
electrical operating parameters
cooling gas purity
cooling gas pressure
cooling gas temperature
core monitor level
excitation status
excitation operating parameters

The three line to line voltages, three line currents, megawatts, and megavars are generated by an analog signal conditioning card using the system PT and CT inputs. The system frequency reading is obtained by using one phase of the generator PT inputs by measuring the time for one cycle. The power factor reading is generated by software using the megawatt and megavar readings.

3.3 THE CONTROL CONSOLE

The control console serves as the interface between the operator and all the data gathered by the T-GSI system. Any display page or single data point can be monitored via the appropriate pushbutton(s). A drawing of a typical control console is given in Figure 2.

The console basically consists of five areas:

1. The keyboard entry display. It is located in the center of the panel. As communication is established with the T-GSI system, the monitor echoes the operator's information request and asks for specifics and/or a command. If the data is irretrievable, or if a request or format is incorrect, an appropriate status message will be displayed.

2. The system fault lights: main alert, display alert, and logging alert. These lights are located above the keyboard entry monitor and light when a malfunction occurs. They are controlled by a stall timer on each controller so that if a stall occurs due to a software or hardware problem, the alert light will come on and consequently, the yellow IMOK (I'm OK) light on the corresponding controller in the master station will go off.

3.0 EQUIPMENT DESCRIPTION (Continued)

3.3 THE CONTROL CONSOLE (continued)

3. CRT page selection. The following six pushbuttons are located in the lower left hand corner and are labelled: General Data, Alarm Summary, Composite Data, Data Groups, Tape Read, and Selective Data. They allow the operator to select from the various pages available. There are six other pushbuttons under the CRT page selection heading located in the lower right hand corner of the panel. They are: Current Log, Historical Log Search, Historical Log Next, scan Log Forward, Fast (scan speed - forward or reverse), and scan Log Reverse. For instructions on how to display data and the correct usage of these pushbuttons see the Operator Control console Instructions in the OPERATION section of this instruction book.

4. CRT page control. The pushbuttons under this heading are located below the keyboard entry monitor. They are: Cancel, Enter, Alarm Acknowledge, Delete Channel, the numbers 0 through 9, a dash (-), and a period (.). When requesting information, the Enter pushbutton will be used often. The numbers are used to identify certain data points, or pages, or on setting the time. The dash requests an LED test and Cancel allows the operator to start a new command. Alarm Acknowledge stops the red, flashing alarm points from flashing when that pushbutton is selected and the alarm summary page is displayed. Delete channel will remove displayed information from the CRT.

5. Trend Control. These pushbuttons are below the CRT page control, at bottom center. This area includes Trend Directory, Trend Display, and Time Set. The Trend Directory displays which points are trendable and their identification number. Trend Display is used to put certain points on or take certain points off a trending display. Time Set allows the operator to set the clock to either standard or daylight savings time. The T-GSI internal real time clock is kept at standard time. Although the time tag on the data stored on the tape and disk is from the real time clock, the information requested by the operator can be either standard or daylight savings time, and will be displayed as requested.

3.4 THE CRT MONITOR

The display unit is a 19 inch, full color CRT monitor that displays the information requested by the operator in a clear and concise form.

3.5 THE TELEPRINTER (Optional)

A teleprinter is an option. It is used to provide an instant hard copy of a current or historic data log.

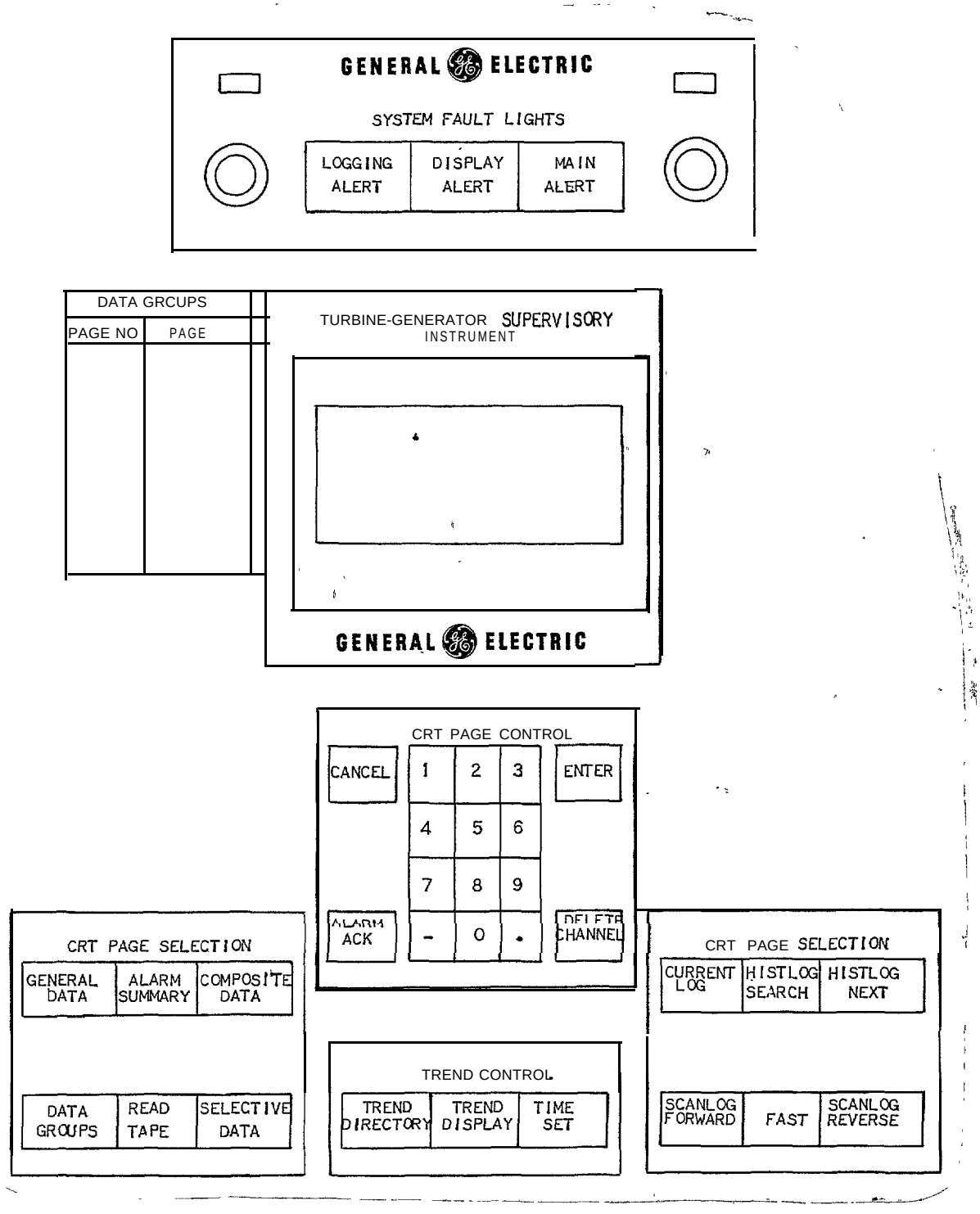


Figure 2: CONTROL CONSOLE.

4.0 SPECIFICATIONS

4.1 EQUIPMENT LOCATION

This equipment is designed to be installed in electric power generating plants. The CRT display and control console are ordinarily located in the control room. The master station should be located in a reasonably clean area where similar plant equipment is installed. The turbine remote should be located in close proximity to the turbine, the generator remote near the generator, and the TSI remote near the Electra-Hydraulic Control (EHC) system.

4.2 CABLES

4.2.1 MAXIMUM PERMISSIBLE LENGTH OF CONNECTING CABLES

The maximum length of cables from the master station to the CRT is 300 feet. It is recommended that the length of wiring from the detectors on the turbine to the TSI cabinet not exceed 650 feet, but in any case the loop resistance must not exceed 35 ohms. The maximum length of cables from the remote stations to the master cabinet shall not exceed 3000 feet.

4.2.2 ATTACHMENTS AND CABLE ACCESS

Customer terminal boards are provided within the enclosure. Interconnecting cabling can be brought into the cubicle at either the top or the bottom.

4.2.3 INTERCONNECTING CABLE BETWEEN CABINETS AND EQUIPMENT

Where special cables are needed, interconnection cables are provided by General Electric Company.

4.3 ENVIRONMENT

4.3.1 TEMPERATURE

The ambient temperature ranges for the cabinets are:

Master Station: 36 - 113°F, (2 - 45°C), when cabinet air conditioners are working properly. 50 - 90°F, (10 - 32.2°C), when cabinet air conditioners have failed, at which time THE CABINET DOORS SHOULD BE OPENED.

4.0 SPECIFICATIONS (Continued)

4.3.1 TEMPERATURE (continued)

Remote Stations: 35 - 113°F. In the case of air conditioner failure, THE CABINET DOORS SHOULD BE OPENED, and repair of the air conditioners should be undertaken immediately.

4.3.2 HUMIDITY

The remote cabinets electronics are capable of operating at a relative humidity of 15 to 95% without condensation. The master station, due to restrictions of the recording media, is capable of operating at a relative humidity of 20 to 80% without condensation.

4.4 ACCURACY

Data accuracy is maintained within +/- 1% on data accessed through the teleprinter, and data displayed on the CRT is accurate to within +/- 2% of full scale or better, Speed display and recorded accuracy is within +/- 1 RPM. Frequency display and recorded accuracy is within +/- 0.017 Hertz.

Note: The accuracy figure does not include errors in the detectors and associated TSI circuitry.

4.4.1 INPUT POWER

The equipment operates within specified accuracy from 115 VAC +/- 10% @60 +/- 0.5 Hertz or 50 +/- 0.5 Hertz as specified on the requisition. The equipment tolerates sustained frequency deviation up to +/- 5 Hertz from rated frequency without damage. In addition, power interruptions of at least 1 cycle (16 milliseconds) are also tolerated without misoperation of the system. The estimated power requirement is 15 KVA for the electronics and 15 KVA for air conditioners.

4.4.2 SUSCEPTABILITY TO ELECTRICAL NOISE (Common Mode Rejection)

T-GSI is designed to be adequately protected against misoperation due to electrical noise interference, provided the cables are pulled as recommended.

4.0 SPECIFICATIONS (Continued)

4.4.3 BAD SENSOR DETECTION AND INHIBITING

When an obviously inoperative T-GSI input sensor is detected, the data which would have appeared for that parameter will be replaced by the letters "OR" (for Out of Range). Once a sensor has been determined as bad, it is not included for any average or "hottest" calculations. Some sensor malfunctions are difficult to detect (such as a shorted TC, uncalibrated transducers, etc.) and are not considered in the T-GSI design. Obvious sensor malfunctions, such as those listed below, can be detected:

Open RTD
Shorted RTD
Open TC
Transducer output at maximum or minimum range

4.5 SCALES

The scale ratings are as follows:

vibration:	0 - 15 mils peak-to-peak displacement amplitude.
Eccentricity:	0 - 15 mils peak-to-peak for the bar graph display, with detector cat. No. 1257K28, located in the turbine standard.
Phase Angle:	0 - 360° on vibration phase angle.
Temperature:	All temperature ranges are 0 - 600°F, 0 - 1200°F, 0 - 300°C and 0 - 600°C.
Pressure:	Pressure ranges are 0 - 30 psia; 0 - 10, 0 - 100, 0 - 1000, 0 - 1500, 0 - 2500, 0 - 3000, 0 - 3500, 0 - 4000 and 0 - 5000 psig.
Expansion: Differential and Rotor	The scale range will match the range of the detector used The possible ranges with detectors presently in use are: 0 - 500 mils, 0 - 750 mils, 0 - 1 inch, 0 - 1.5 inches, 0 - 2 inches and 0 - 3 inches.
Shell Expansion:	The scale range will match the range of the detector used. The usual ranges with detectors presently in use are: 0 - 1 inch, 0 - 1.5 inches and 0 - 2 inches.
Speed:	0 - 5000 or 0 - 2500 rpm. Also applies for peak speed.
Control Valve Position:	0 - 100%, corresponding to 0 to -5V input from EHC equipment.

4.0 SPECIFICATIONS (Continued)

4.6 RECORDING INTERVAL

4.6.1 UPDATING (Response Time)

All data points are updated at least every three seconds; however, some critical inputs are updated as often as once per second.

4.6.2 LOG RECORDING INTERVAL

The time interval between each complete log of data is one of two values depending on conditions described below. The longer interval is 5 minutes; the shorter interval is 30 seconds. The system automatically goes to the 30 second interval, for 24 minutes, during the presence of a critical alarm condition. Alarm response time is one second. In addition, the change of state of selected inputs causes a complete log of data to be taken within one second, a snapshot action. If any of the following conditions are present, the longer interval is always used, regardless of the presence of an alarm condition:

Turning gear engaged
Rotor stopped
Historical memory system unavailable

Note: Snapshots are never inhibited.

4.7 MEMORY CAPACITY

4.7.1 TAPE

under normal circumstances, the 2400 foot reel of magnetic tape is large enough to store six weeks of data input to T-GSI.

4.7.2 DISK DRIVE

Under normal circumstances, the disk drive will contain all input to T-GSI for the last six weeks at any instance in time.

4.7.3 TREND DISPLAY

All inputs in the Trend Display have at least one hour of past history stored in memory at all times, at a sampling rate of once each minute for each channel.

4.0 SPECIFICATIONS (Continued)

4.8 I/O DEVICES

4.8.1 CRT DISPLAY

4.8.1.1 CRT

The CRT is a 19 inch color monitor capable of desk top or standard rack mounting.

4.8.1.2 CRT CONTROL CONSOLE

A matrix of suitable pushbuttons or keys are provided to select the CRT displays. The pushbuttons or keys are clearly marked with the names of the data pages and numerals 0-9. The keyboard is capable of rack or panel mounting.

4.8.2 TAPE RECORDER

The tape drive is in IBM compatible, 9 track, phase encoding unit. It is mounted in the master station control cabinet.

4.8.3 MAGNETIC DISK UNIT

A sealed Winchester type disk for historical scrolling is supplied. The disk is written to each time a log is taken.

4.8.4 TELEPRINTER PORT

A connection port is provided as standard for interfacing with a teleprinter. The port is in accordance with RS-232C and ASCII character set.

4.8.5 DATA LINK TO CUSTOMER COMPUTER

The data link complies with EIA RS-232C operated asynchronously at 9600 BAUD under link protocol Cat. 3.1/F1 of ANSI Std. X3.28.

5.0 INSTALLATION

5.1 GENERAL

The electronic equipment is for indoor, non-tropical installations. The four cabinets are NEMA type 12 which include air conditioning units and seals that inhibit dust from entering the cabinets when the doors are kept closed. The cabinets should be placed so both the front and rear doors can be opened fully for servicing.

The air conditioners are equipped with a 1/8-27 NPT drain at the center of the drain pan. In addition, under high termperature-humidity conditions condensation may appear on the cabinet walls. Therefore, the cabinets should be located such that any runoff is controlled.

Strip heaters are employed in the remote stations to warm the surface of the cabinets and reduce the times when condensation does form on the walls. These heaters must be manually operated when condesnsation does appear.

The master station must be located where it will not be subjected to water, moisture, or corrosive vapors and where it will be reasonably free from dust and dirt, i.e. the relay room or computer room.

****
* CAUTION *
****

THE TAPE AND DISK, HOUSED IN THE MASTER **STATION**, **MUST** NOT BE ALLOWED TO OPERATE IN AN ATMOSPHERE OF DUST OR LINT, AS DAMAGE TO THE RECORDING HEAD MAY RESULT.

The three remote cabinets are to be located in an area convenient to the type of inputs going to it: TSI in the relay room, Turbine remote near the turbine, and Generator remote near the generator. NOTE: The remote stations cannot be mounted on the turbine foundation (pedestal, island) due to vibration specifications.

The ambient temperature ranges for the cabinets are:

Master Station: 32 - 122° F when cabinet air conditioners are working properly.

50 - 90°F when cabinet air conditoners have failed, at which time THE CABINET DOORS SHOULD BE OPENED.

Remote Stations: 35 - 113°F. In the case of air conditioner failure, THE CABINET DOORS SHOULD BE OPENED, and repair of the air conditioners should be undertaken immediately.

5.0 INSTALLATION (continued)

5.1 GENERAL (continued)

The relative humidity ranges for the master station and the remote stations electronics are 20 - 80% without condensation, and 15 - 95% without condensation, respectively.

The cabinets must be located so the length of the wiring from the master station to the CRT will not exceed 300 feet.

It is recommended that the length of wiring from the detectors, on the turbine, to the TSI cabinet not exceed 650 feet, but in any case the loop resistance must not exceed 35 ohms. The maximum length of cable from the remote stations to the master cabinet shall not exceed 3000 feet.

The cabinets are designed with bottom access plates to permit wiring to be brought in from ducts or conduit under the floor. However, when preferable, wiring may be brought into the top of the cabinet, instead. Access plates may be removed for drilling or cutting of holes for cable or conduit entrance, then replace the plate and seal any holes.

The control console and CRT are to be installed in the control room in an easily accessible location. The CRT should be placed in a position where the screen can be easily read by the operator. The control console should be located adjacent to - and preferably below - the CRT.

The optional teleprinter and associated modem may be located in the control room, relay room, computer room, or other convenient and reasonably clean area.

During checkout and test, all simulated signals should be inputted at a point that most nearly checks the external wiring.

Note: Cabinet air conditioners must be running while operating with doors closed,

5.1.1 DIAGNOSTICS

To help in the check-out and maintenance of the TGSI system, each Controller has a monitor program with a variety of diagnostic tools. The diagnostics is available through a terminal (teleprinter or video terminal) connected via a DS3800HNMA Null Modem connector adaptor (supplied in the service kit) to the RS232C port on the HMPF printed circuit board. The default baud rate for the port is 300 but may be changed upwards to a max of 9600 baud via the terminal keyboard using the BAUDERATE command under diagnostics.

5.0 INSTALLATION (continued)

5.1.1 DIAGNOSTICS (continued)

The diagnostics port allows for a system self test (Confidence) and a current state-of-the-controller (status) display among others. Also provided is the ability to examine and modify system variables, and monitor and debug individual system programs while running ON LINE. Some of these functions, because of their nature, are available only after executing a privilege enabling sequence. They are intended for users that are intimately familiar with the TGSI system and are aware of the impact of the commands.

Privileges are enabled via the SETPRIVILEGE command. After typing the SETPRIVILEGE command the user will be prompted for the password (normally the last four digits of the turbine number). Entering anything other than the correct password will cause the privileges to be disabled. Hence, to disable the privileges, simply type CARRIAGE RETURN after the password prompt.

NOTE: Privileges will be automatically disabled after enabling if the terminal is not accessed for 24 hours.

The top level prompt of the monitor is the dollar sign($). To find out what commands are avilable to you, at your present level, type HELP. To get more information about a particular command, type HELP followed by the command. If the command is a sublevel process, more information will be available upon entering the sublevel. Unless otherwise noted, typing EXIT will return you to the top level ($) prompt.

When entering a command, you do not have to type the complete command name. One to four characters is all that is required and in general is denoted as such by use of () in the help summary display; i.e. H(ELP).

The SEL(ECT) command, available from the main Controller only, is a utility that transfers diagnostic control from the Main Controller to a user specified controller (i.e. DISP(LAY), GEN(ERATOR), HIST(ORICAL), ETC). After the controller is selected the diagnostic terminal is effectively attached to the specified controller's diagnostic port, without physically moving the diagnostic terminal.

After selection, if the specified controller is operational, diagnostic control is transferred. A prompt will then be echoed to the diagnostic terminal to indicate the selected controller's diagnostic utility is enabled. In general, this will be the top level "$" prompt.

To return to the Main Controller's SELECT utility the user must enter "[E".

If the controller selected is not available, an error message is outputted and control is returned to the Main Controller's SELECT utility.

5.0 INSTALLATION (continued)

5.1.1 DIAGNOSTICS (continued)

Note: All T-GSI diagnostics are single-user. However, in the case of the SELECT command it is possible to have multiple diagnostic terminals logically connected to the same controller. This situation can result in incoherent diagnostic printouts. It is the responsibility of the user to insure there is one and only one user accessing a controller's diagnostic at a time.

Below is a list of the top level unprotected and privileged commands for each processor in the system. A brief discription of each top level and sublevel command is available at the terminal and is not necessarily described elsewhere in this manual.

MAIN

Unprotected	Privileged
CAL	BAS(E)
CONF(IDENCE)	DSIM
ERR(ORLOG)	RSIM
LAMP	DIA(GNOSTICS)
RDTL	
SETP(RIVILEGE)	
SEL(ECT)	
STAT(US)	

DISPLAY

Unprotected	Privileged
CONF(IDENCE)	DIA(GNOSTICS)
CONS(OLE)	
ERR(OR)	
LAMP	
SETP(RIVILEGE)	
STAT(US)	

HISTORICAL

Unprotected	Privileged
CONF(FIDENCE)	DIA(GNOSTICS)
DRA(NGE)	DIN(IT)
LAB(EL)	DPO(WER)
LAM(P)	DRE(AD)
MON(ITOR)	ERR(ORCHECK)
SETP(RIVILEGE)	TCO(NTINUOUS)
STAT(US)	TDU(MP)
TQU(EUE)	SIM(ULATOR)

TSI, TURB., & GEN Remote

Unprotected	Privileged
CONF(IDENCE)	DIA(GNOSTIC)
ERR(OR)	SIM(ULATOR)
SETP(RIVILEGE)	
STAT(US)	

5.0 INSTALLATION (continued)

5.2 MASTER CABINET

5.2.1 EQUIPMENT INSTALLATION ,

After the double bay master cabinet is set in its desired location, as discussed in section 5.1 (GENERAL), it should be bolted in place.

Make connections to the T-GSI cabinet and associated equipment as shown on the customer connection diagram supplied for the system.

5.2.2 CHECKOUT AND TEST

5.2.2.1 POWER CHECKS

5.2.2.1.1 GROUND CHECK

****
* WARNING *
****

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED **GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.**

Due to the design of multiple grounds in the Master cabinet (and only in the Master Cabinet), no ground check is performed.

5.2.2.1.2 115 VAC POWER CHECK

Verify 115 VAC is supplied from an uninterruptible power supply and measure for correct voltages per the customer connection diagram.

Verify that the AC breaker on the back of the disk drive is off. Verify that the power switch on the tape deck is off. Verify that the power switch on the front of the display generator power supply (5560) panel is off. Reconnect AC power cords. (Power up of the disk drive and the tape deck is under item 9 in this section.)

5.2.2.1.3 230 VAC POWER CHECK

Vertify 230 VAC is supplied to cabinet. Measure for correct voltages per customer connection diagram.

Close breaker marked "AIR COND", located on the breaker panel and verify that the air conditioners supply cool air (Approx. 70° F).

5.0 INSTALLATION (continued)

5.2.2.1.3 230 VAC POWER CHECK (continued1

NOTE: AIR CONDITIONERS MUST BE RUNNING WHILE OPERATING WITH DOORS CLOSED.

5.2.2.1.4 LOADED POWER SUPPLY CHECK

TWO power supplies are located at the top of the Master station's right-hand bay; one in the front of the bay and one in the back. See Figures 3, 4, 11, and 14 for assistance in locating Master station components. Apply 115 VAC power to these power supplies by closing the breakers on the front of the supplies and two main breakers marked "MAIN DISCONNECT" and "MASTER", located on the breaker panel. Using the test rings at the top of each Medium Electronics Module (MEM), measure the following voltages:

Test Rings	Voltage	Adjustment (PS5A)	Adjustment (PS)
P5 to DCOM	+5 +/- 0.1 VDC	Vl	5V Adj.
P15 to ACOM	+15 +/- 0.3 VDC	v3	+15V Adj.
N15 to ACOM	-15 +/- 0.3 VDC	V4(top)	-15V Adj.
P28 to DCOM	+28 +/- 0.3 VDC	v2	+28v Adj.
ACOM to DCOM	0 +/- 0.1 VDC		

(Readjust as necessary)

5.2.2.2 DISPLAY (CRT) CONTROLLER VERIFICATION

A. Verify power is still applied to the Main/Display MEM..

B. Connect a teleprinter to the HMPF card in slot 1L of the MAIN-DISPLAY MEM with an RS-232C cable through a null modem connector adaptor (supplied in the service kit) to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

5.0 INSTALLATION (Continued)

5.2.2.2 DISPLAY (CRT) CONTROLLER VERIFICATION (continued)

C. Put the (Display) mode switch in the "Off-line" position and press the "reset" button. These are found on the Maintenance Module, the aluminum panel in the right hand bay of the Master Station, which contains lights and switches for the three controllers: Historical, Display, and Main. A drawing of the Maintenance Module is shown in Figure 4. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "F" which stands for pass or fail, respectively. If all the statuses are a "P", then proceed. If an "F" appears, or if the teleprinter does not begin printing, then refer to the Display Controller portion under TROUBLESHOOTING section 7.3.2. Figure 5 gives an example of the power up print out.

D. Check the LED's by typing "LAMP" on the teleprinter. (A dollar sign ($) prompt must be present before "LAMP" is typed.) At this time, the unlit LED'S on the Maintenance Module Display area should come on for approx 5 sec.

E. Check the status of the controller by typing "STAT" after the dollar sign ($) prompt. See Figure 6 for an example of this print out.

F. Next, the error logger should be checked. After the dollar sign ($) prompt, type "ERR". Then a "#" prompt should appear, at which time you should type "PRINT ALL". An example of this print out appears in Figure 7. If any errors are displayed, they should be corrected before proceeding (see trouble shooting section).

G. To finish this process, type "EX".

H. Put the (Display) mode switch in the "On-line" position.

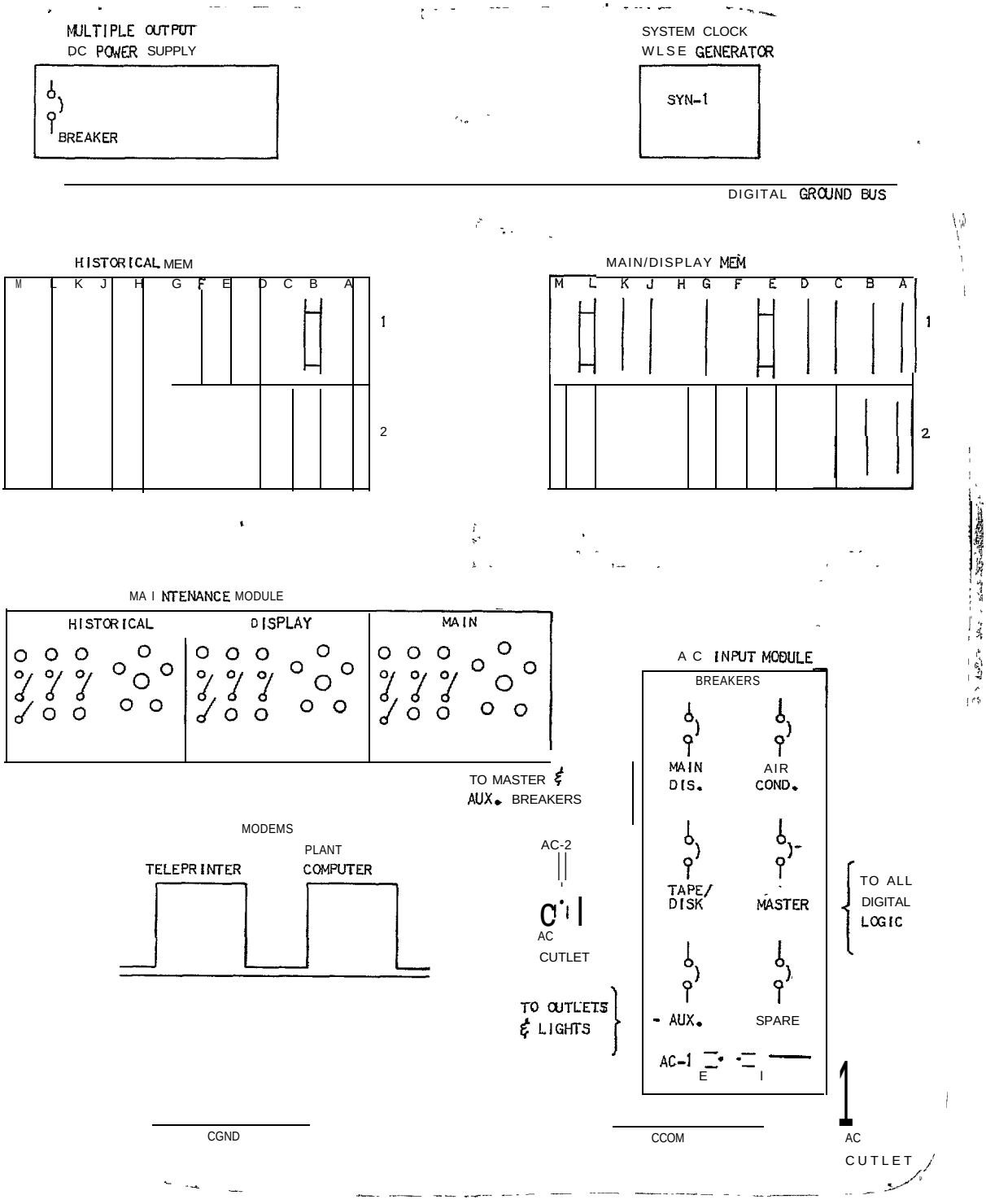


Figure 3: MASTER STATION - RIGHT BAY - FRONT

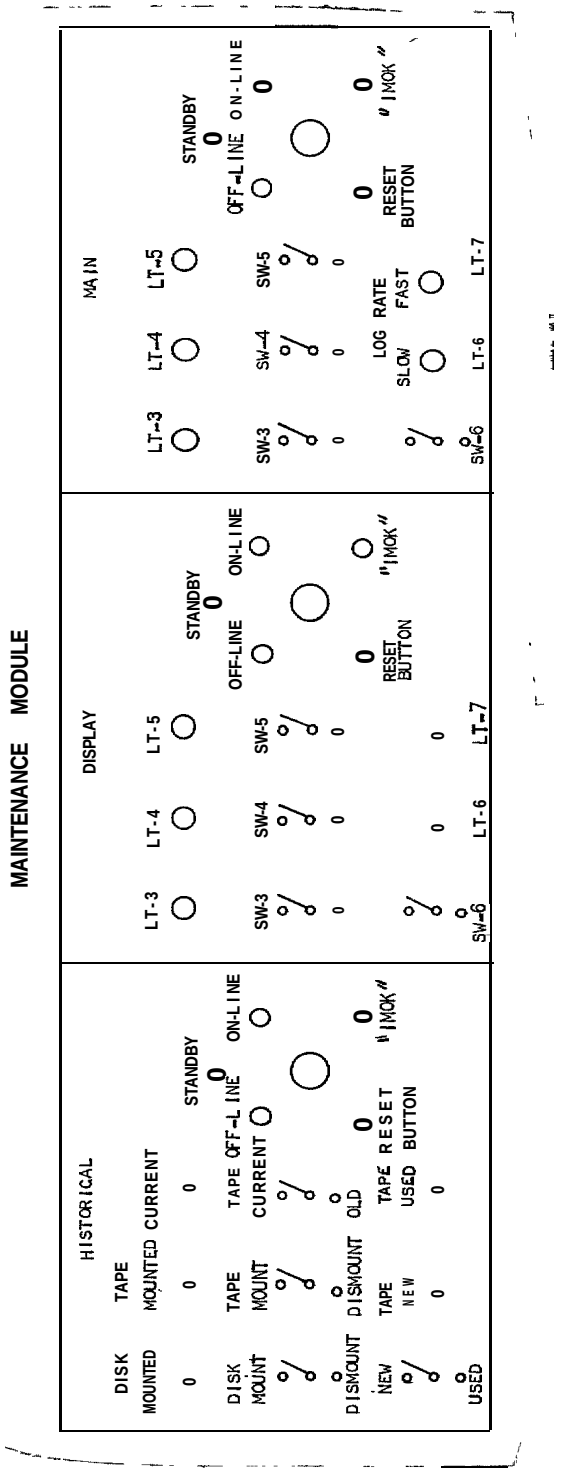


Figure 4: MAINTENANCE MODULE

ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS			EXP	TST
F0000-F1FFF	LO	P	9887	9887
F0000-F1FFF	HI	P	5299	5299
F2000-F3FFF	LO	P	4393	4393
F2000-F3FFF	HI	P	39C2	39C2
F4000-F5FFF	LO	P	99C8	99C8
F4000-F5FFF	HI	P	B4CC	B4CC
F6000-F7FFF	LO	P	4165	4165
F6000-F7FFF	HI	P	1390	1390
F8000-F9FFF	LO	P	7DD6	7DD6
F8000-F9FFF	HI	P	6B56	6B56
FA000-FBFFF	LO	P	3944	3944
FA000-FBFFF	HI	P	3006	3006
FC000-FDD8F	LO	P	7CB5	7CB5
FC000-FDF8F	HI	P	849D	849D
08000-09FFF	LO	P	17A7	17A7
08000-09FFF	HI	P	1449	1449
0A000-0BFFF	LO	P	A17D	A17D
0A000-0BFFF	HI	P	9AD8	9AD8
0C000-0DFFF	LO	p	0240	0240
0C000-0DFFF	HI	P	0223	0223
0E000-0FFFF	LO	p	E59F	E59F
0E000-0FFFF	HI	p	9787	9787

****

RAM CHECK

ADDRESS	
00000-06FFF	PASSED

****

ROM CHECKSUM TEST (P-PASS F=FAIL)

ADDRESS			EXP	TST
FE000-FFFDF	LO	P	6EA2	6EA2
FE000-FFFDF	HI	P	166E	166E

****

ROM CHECKSUM TEST (P=PASS F=FAIL

ADDRESS			EXP	TST
10000-11FFF	LO	P	7005	7005
10000-11FFF	HI	P	712D	712D
12000-13FFF	LO	P	F25C	F25C
12000-13FFF	HI	P	DDFE	DDFE
14000-15FFF	LO	P	543A	543A
14000-15FFF	HI	P	4B47	4B47
16000-17FAF	LO	P	0240	0240
16000-17FAF	HI	P	0223	0223

Figure 5: POWER UP PRINT OUT - DISPLAY

```
$STAT
*** DISPLAY CONTROLLER STATUS ***

TURBINE NUMBER - T138

TIME - 83/09/29 08:58:51 DST

CONTROLLER OPERATIONAL MODE - ON LINE

DIAGNOSTICS PRIVILEGES - DISABLED

IDLE TIME - 78 PERCENT
```

Figure 6: STATUS PRINT OUT - DISPLAY

The printout below contains errors only as an example. A printout without errors will include only the header information.

```
$ERR
#PRI ALL
```

DATE	TIME	TASK	ERROR
00/00/00	00:00:00	0	BUS HOG TIMEOUT
00/00/00	00:00:00	0	BUS HOG TIMEOUT
00/00/00	00:05:50	3	CRT LINK FAULT
00/00/00	00:05:52	3	CRT LINK FAULT
00/00/00	00:05:54	3	CRT LINK FAULT
00/00/00	00:05:56	3	CRT LINK FAULT
00/00/00	00:05:47	6	CONTROL CONSOLE OR CRT LINK TIMEOUT
00/00/00	00:05:47	6	CRT LINK FAULT
00/00/00	00:00:00	12	BUS HOG TIMEOUT
00/00/00	00:00:00	12	BUS HOG TIMEOUT

```
#EX
```

Figure 7: ERROR PRINT OUT - DISPLAY

5.0 INSTALLATION (Continued)

5.2.2.3 Main Controller Verification

A. Verify power is applied to the Main/Display MEM by confirming that 1) the breaker on the power supply in the FRONT of the MASTER STATION'S right-hand bay is closed, and 2) the breakers marked "MAIN DISCONNECT" and "MASTER" in the FRONT of the MASTER STATION'S right-hand bay are closed. See figure 3 for assistance.

B. Connect the teleprinter to the HMPF card in slot 1E of the MAIN-DISPLAY MEM with an RS-232 cable through a null modem connector adaptor to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

C. Put the (Main) mode switch in the "Off-line" position and press the "reset" button. These are found on the Maintenance Module, the aluminum panel in the right hand bay of the Master Station, shown in Figure 4. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "F" which stands for pass or fail, respectively. If all the statuses are a "P", then proceed. If an "F" appears, or if the teleprinter does not begin printing, then refer to the Main Controller portion under TROUBLESHOOTING section 7.3.2. Figure 8 gives an example of the power up print out.

D. Check the LED's by typing "LAMP" on the teleprinter. (A dollar sign ($) prompt must be present before "LAMP" is typed.) At this time, the unlit LED's on the Maintenance Module Display area should come on for approx. 5 seconds.

E. Check the status of the controller by typing "STAT" after the dollar sign ($) prompt. Fig. 9 gives an example of the status printout.

F. Next the error logger should be checked. After the dollar sign ($) prompt, type "ERR". Then a "#" prompt should appear, at which time you should type "PRINT ALL". An example of this print out appears in Figure 10. If any errors are displayed, they should be corrected before proceeding (see trouble shooting section).

G. To finish this process, type "EX".

H. Put the (Main) mode switch in the "On-line" position.

J. Set system clock via control console. Refer to TIME SET under operation section for control console.

```
ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS                  EXP      TST
F0000-F1FFF LO  P        5133     5133
F0000-F1FFF HI  P        15A4     15A4
F2000-F3FFF LO  P        F431     F431
F2000-F3FFF HI  P        F7B6     F7B6
F4000-F5FFF LO  P        E5EC     E5EC
F4000-F5FFF HI  P        D0C8     D0C8
F6000-F7FFF LO  P        8255     8255
F6000-F7FFF HI  P        9436     9436
F8000-F9FFF LO  P        D0EC     D0EC
F8000-F9FFF HI  P        9060     9060
FA000-FBFFF LO  P        0311     0311
FA000-FBFFF HI  P        C6A7     C6A7
FC000-FDF8F LO  P        F4C9     F4C9
FC000-FDF8F HI  P        7771     7771
08000-09FFF LO  P        76B4     76B4
08000-09FFF HI  P        6C05     6C05
0A000-0BFFF LO  P        071A     071A
0A000-0BFFF HI  P        4A31     4A31
0C000-0DFFF LO  P        0B9D     0B9D
0C000-0DFFF HI  P        3B9D     3B9D
****
RAM  CHECK

ADDRESS
00000-05FFF              PASSED
****

ROM  CHECKSUM  TEST  (P-PASS  F=FAIL)

ADDRESS                  EXP      TST
FE000-FFFDF LO  P        ED32     ED32
FE000-FFFDF HI  P        A05C     A05C
```

Figure 8: POWER UP PRINT OUT - MAIN

```
$STAT
*** MAIN CONTROLLER ***
83/09/29 07:52:19 T138

DATALINK          STS   ERR
TURBINE"          UP    000
GENERATOR         UP    000
TSI               UP    000
PLANT  COMPUTER   DWN   000

ALARM MEMORY STATUS

ON-LINE

%IDLE TIME = 53
  ("."=NA, "+"=ACTIVE, "*"=OUT OF RANGE, "<"=PRE-ALARM, ">"=ALARM)

                    1         2         3         4
START       01234567890123456789012345678901234567890123456789
CH.#0000    . . . . .>>>>>>>>>>.....++.++.+.>.+..+ . . . .
CH.#0050    ..............  ...  ..............................
CH.#0100    ........................  +...+.....*****.** .......
CH.#0150    >.............................................
CH.#0200    **.*********..********  ..... **.***..****...***..**
CH.#0250    ******.............................  *********  .....
CH.#0300    ..****  .... ***.*************  ....... ******..++++++++
CH.#0350    ++.....  ++.+++.++.....................................
CH.#0400    ..**.>.............  .**** ........................
CH.#0450    ..... ..................................................
CH.#0500    ++++++++.............  *>>+*+>,*.++,a**********
CH.#0550    ******************************  ....................
CH.#0600    ..........  ***  .............  ************************
CH.#0650    ******************  ..............................  **
CH.#0700    *.......................  **..***..******...*****..
CH.#0750    ...*.................................**..  ******..*.*.
CH.#0800    .....***.....**  ...................................
CH.#0850    ........  .++++++++++++++.+++++++.++..++++...  ..*..
CH.#0900    .*...*.*.*******....  **.*  ..........................
CH.#0950    ..................................................
```

Figure 9: STATUS PRINT OUT - MAIN

The printout below contains errors only as an example. A printout without errors will include only the header information.

```
$ERR
#PRI  ALL

   DATE      TIME        TASK                    ERROR

00/00/00  00:00:43        2      SCAN INHIBITED - DATALINK DOWN
83/09/29  07:53:46        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:53:51        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:53:55        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:53:56        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:54:00        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:54:05        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:54:09        3      SCAN ABORTED - LOST  CHANNEL  FROM  REMOTE
83/09/29  07:54:10        3      SCAN  ABORTED - LOST  CHANNEL  FROM  REMOTE
00/00/00  00:00:43        4      SCAN  INHIBITED - DATALINK DOWN
83/09/29  07:47:01        9      HISTORICAL  LOG  INHIBITED - HIST  SYS  OFF-LINE
83/09/29  07:47:07        9      HISTORICAL LOG INHIBITED - HIST SYS OFF-LINE
83/09/29  07:47:13        9      HISTORICAL  LOG  INHIBITED - HIST  SYS  OFF-LINE
83/09/29  07:47:19        9      HISTORICAL LOG INHIBITED - HIST SYS OFF-LINE
83/09/29  07:47:25        9      HISTORICAL  LOG  INHIBITED - HIST  SYS  OFF-LINE
83/09/29  07:47:31        9      HISTORICAL LOG INHIBITED - HIST SYS OFF-LINE
83/09/29  07:47:37        9      HISTORICAL  LOG  INHIBITED - HIST  SYS  OFF-LINE
83/09/29  07:47:43        9      HISTORICAL LOG INHIBITED - HIST SYS OFF-LINE
00/00/00  00:00:43       14      PLANT COMPUTER DATALINK NOT ESTB - SLAVE
00/00/00  00:00:43       14      PLANT COMPUTER DATALINK NOT ESTB - MASTER
#EX
```

Figure 10: ERROR PRINT OUT - MAIN (with errors)

5.0 INSTALLATION (Continued)

5.2.2.4 HISTORICAL CONTROLLER VERIFICATION

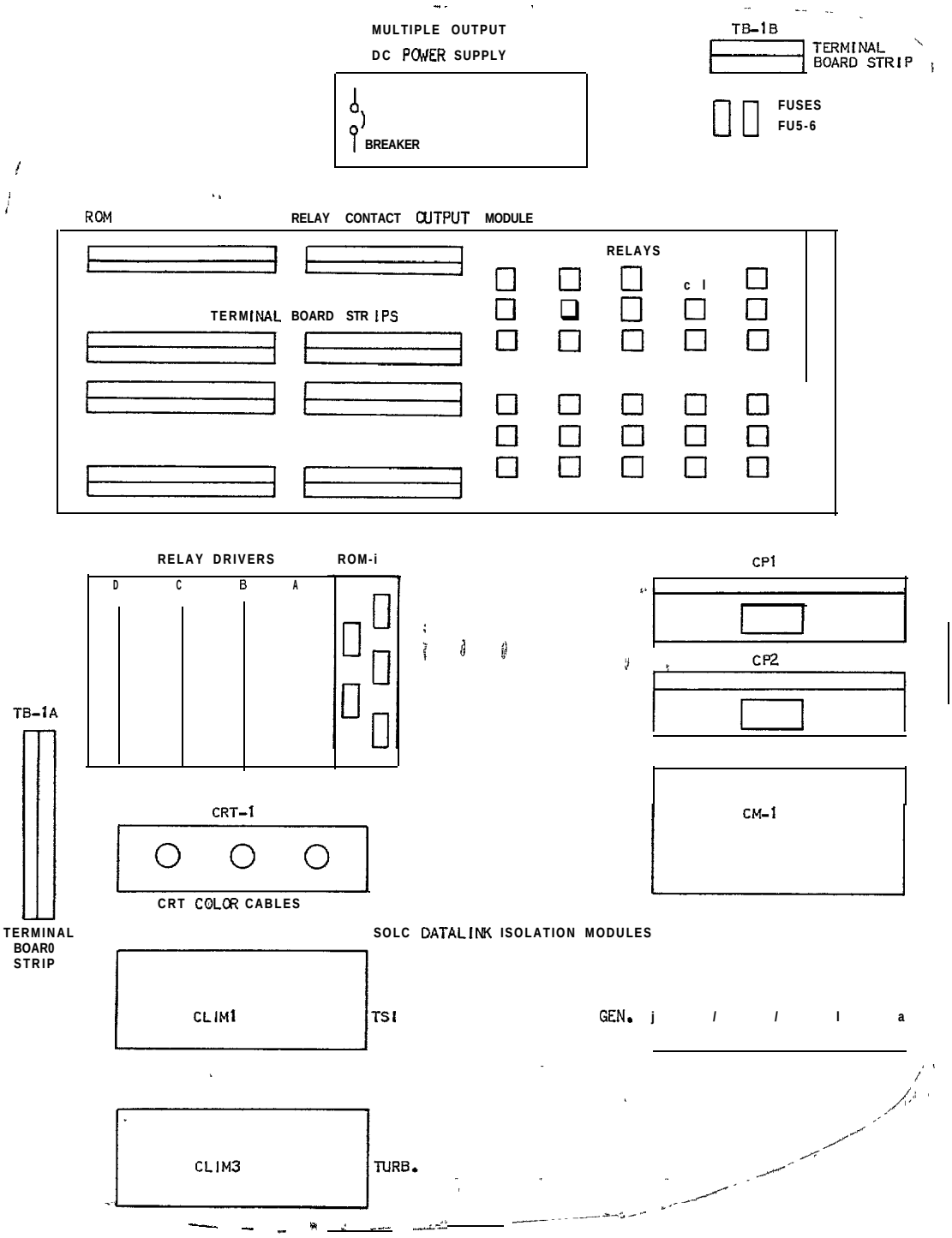
A. Verify power is applied to the Historical MEM by confirming that 1) the breaker on the power supply in the REAR of the MASTER STATION'S right-hand bay is closed, and 2) the breakers marked "MAIN DISCONNECT" and "MASTER" in the FRONT of the MASTER STATION'S right-hand bay are closed. See figures 3 and 11 for assistance.

B. Connect the teleprinter to the HMPF card in slot 1B of the HISTORICAL MEM with an RS-232 cable through a null modem adaptor to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

C. Put the (Historical) mode switch in the "Off-line" position and press the "reset" button. These are found on the Maintenance Module, the aluminum panel in the right hand bay of the Master Station, shown in Figure 4. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "F" which stands for pass or fail, respectively. If all the statuses are a "P", then proceed. If an "F" appears, or if the teleprinter does not begin printing, then refer to the Historical Controller portion under TROUBLESHOOTING section 7.3.2. Figure 12 gives an example of the power up print out.

D. check the LED's by typing "LAMP" on the teleprinter. (A dollar sign ($) prompt must be present before "LAMP" is typed.) At this time, the unlit LED's on the Maintenance Module Display area should come on for approx. 5 seconds.

E. check the status of the controller by typing "STAT" after the dollar sign ($) prompt. Figure 13 gives an example.



This figure is meant to be used as a guide in locating major components in the cabinets. Because components and their location may vary among units, refer to the customer connection diagram for information specific to your unit.

Figure 11: MASTER STATION - RIGHT BAY - REAR

```
**** HISTORICAL PROCESSOR CONFIDENCE CHECK ****

ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS          EXP   TST
EC000-EDFFF  LO  P  3CAE  3cAE
EC000-EDFFF  HI  P  070F  070F
EE000-EFFFF  LO' P  0AD4  oAD4
EE000-EFFFF, HI  P  32D5  32D5
F0000-F1FFF  LO  P  6730  6730
F0000-F1FFF  HI  P  559A  5598
F2000-F3FFF  LO  P  C176  C176
F2000-F3FFF  HI  P  A616  A616
F4000-F5FFF  LO  P  0904  0904
F4000-F5FFF  HI  P  E102  EL02
F6000-F7FFF  LO  P  B21C  B21C
F6000-F7FFF  HI  P  D9D2  D9D2
F8000-F9FFF  LO  P  C82F  C82F
F8000-F9FFF  HI  P  F349  F349
FA000-FBFFF  LO  P  23C2  23C2
FA000-FBFFF  HI  P  42B2  42B2
FC000-FDF9F  LO  P  DF27  DF27
FC000-FDF9F  HI. P  C683  C683

****

RAM CHECK

ADDRESS
00000-09FFE     PASSED

****

ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS          EXP   TST
FE000-FFFDF  LO  P  F5DF  F5DF
FE000-FFFDF  HI  P  8EF6  8EF6

****

ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS          EXP   TST
E0000-E1FFF  LO  P  3CCA  3CCA
E0000-E1FFF  HI  P  3663  3663
E2000-E3EFF  LO  P  0000  0000
E2000-E3EFF  HI  P  0000  0000
```

Figure 12: POWER UP PRINT OUT - HISTORICAL

```
$STAT
****** HISTORICAL CONTROLLER STATUS ************

83 09 29 08 03 25

DISK  POWERUP  INCOMPLETE

TAPE  READY

COUNT  OF  TAPE  RETURN  CODES

00001  LOAD  POINT

00017  HARD  ERROR

COUNT  OF  DISK  RETURN  CODES

00611 (HEX) BUSHOGS HAVE OCCURRED IN THE SYSTEM

D128  IS  THE  ADDRESS  OF  THE  BUSHOGS

87%  IDLE  TIME
```

Figure 13: STATUS PRINT OUT - HISTORICAL

5.0 INSTALLATION (Continued)

5.2.2.5 DISPLAY SECTION

5.2.2.5.1 HCMA CHECK (PTCA, PCPA)

Verify card "IMOK" LED is on for card in slot 2G (PTCA) and 2H (PCPA) of the Main-Display MEM. If LED is not lit, then refer to the "HCMA" heading under TROUBLESHOOTING section 7.3.2.

5.2.2.5.2 LED TEST ON CONTROL CONSOLE

Press "-" pushbutton. The response on the control console panel will be:
"LED TEST
"ENTER OR CANCEL"

Press the "ENTER" pushbutton, and the response on the control console should be:

"[[[[[[[[[[[[[[[[
[[[[[[[[[[[[[[[["
“111111111111111
111111111111111”
"1111111111111111
1111111111111111"
"LED TEST
COMPLETED"
"ENTER COMMAND"

5.2.2.6 COLOR CRT CHECK

5.2.2.6.1 POWER UP OF DISPLAY GENERATOR

1. Verify all power and I./O cables are connected properly.
2. Apply **115Vac** power to the TAPE/DISK BAY by closing the breaker marked "TAPE/DISK", located on the breaker panel.
3. Turn power switch on the front of the display power supply panel "ON".
4. Verify voltages at TB1 are correct as follows:

+ 5v +/- 0.05V
+12V +/- 0.24V
-12V +/- 0.24V

(Readjust as required)

5.0 INSTALLATION (Continued)

5.2.2.6.2 POWER UP OF COLOR CRT

1. Before connecting the monitor to the power source check the following:

   a. Determine that the line voltage and frequency are proper (120VAC +/-10%, 50/60 Hz).
   b. If the line voltage being used is not 120 VAC, remove cover and make certain the input voltage plug on power supply card is in the proper position for the input line voltage desired (100V, 110V, 220V, 240V). If the line voltage is not set properly, refer to the Operation and Maintenance Manual for the Color CRT.
   c. Plug AC line cord, supplied with unit into the power receptacle on rear panel.

2. connect monitor to power source.

3. set power switch to "ON".

4. After approximately a 30 second delay a raster should appear. If not, increase brightness control until the CRT image just appears.

5. Set the input impedance switch for each cable to the 75 ohm position.

6. connect the 3 system video cables (red, green, blue) from the Master Station to the CRT. (The connections in the Master Station are in the back of the right hand bay.)

7. A page containing a test pattern should appear. If not, reset the display controller. check that the words defining a certain color appear in that color. (The word "red" should be red in color.) If the colors do not match the words, recheck the connections of the 3 color video cables.

8. Adjust the brightness and video gain for the desired viewing level.

9. If necessary, adjust the sync and contrast controls.

10. If color purity or convergence are slightly out of adjustment, depress the "degauss" control, hold for 5 seconds and release. For additional information on the color CRT refer to the Operation and Maintenance Manual for the Color CRT.

11. Put the Display Controller "On-line".

5.0 **INSTALLATION** (Continued)

5.2.2.7 TELEPRINTER PORT CHECK

A. Verify that the Display mode switch is in the "On-line" position.

B. Connect the teleprinter modem in the cabinet to the remote modem per the customer connection diagram.

C. Set teleprinter baud rate to default rate of 1200 (300 on some units).

D. Set teleprinter for even parity.

E. Type a carriage return on the teleprinter. It should return with a "*" prompt. Type "H" followed by a carriage return. The teleprinter should respond with a help message followed by a "*" prompt again. If the port does not work properly, then refer to section 7.3.8 in the TROUBLESHOOTING section.

5.2.2.8 MAIN SECTION

5.2.2.8.1 HCMA CHECK (PCPA)

Verify card "IMOK" LED is on for card in slot 2A (PCPA) of the Main - Display MEM. If LED is not lit, then refer to the HCMA heading under TROUBLESHOOTING section 7.3.2.

5.2.2.8.2 HMHA CHECK (PDLA)

Verify PDLA card "IMOK" LED is on for card in slot 2B and 2C of the Main - Display MEM. If the LED is not lit, then refer to the HMHA heading under TROUBLESHOOTING section 7.3.2.

5.2.2.8.3 HARA CHECK

The HARA card in slot 1A of the Main-Display MEM, contains non-volatile memory in which is stored the changeable alarm limits for the TGSI system. The system was shipped with the factory settings programed into its memory. Verify that the "ENABLE" LED is ON when the switch on the front of the card is in the up position and OFF when down. The switch should normally be in the down (DISABLED) position except when it is desired to change an alarm setting (See Operation section).

NOTE: The IMOK light is never ON for this board.

5.2.2.8.4 MAIN/DISPLAY CONTROLLER ON-LINE CHECKOUT

When put into the On - Line mode, the Main Controller is enabled to log to the Historical Controller, Display Controller, and to the Plant Computer. At this point it is necessary for the Display controller to be On-Line and operational.

5.0 INSTALLATION (Continued1

5.2.2.8.4 MAIN/DISPLAY CONTROLLER ON-LINE CHECKOUT (Continued)

To verify logging to the Display Controller perform the following steps:

1. Insure that the Display and Main Controllers are On-Line by checking the Maintenance Module lights.
2. Verify the time on the system, is still correct. Refer to the procedure found in the Operator Control Console Instructions under OPERATION section 6.2.

****
*** CAUTION ***
****

WHEN **SETTING THE TIME, IT IS OF UTMOST IMPORTANCE THAT THE CORRECT TIME BE ENTERED. ALL RECORDS WILL BE REFERENCED TO A TIME RESULTING FROM THE TIME THAT IS ENTERED ON THE SYSTEM AT THIS POINT.**

3. Observe the system time display on the Color CRT. It should be incrementing.

5.2.2.9 HISTORICAL SECTION

Figure 14 contains a drawing that shows the location of the Historical Tape Drive, the Historical Disk Drive, and the CRT Electronics, which are located in the MASTER STATION'S left bay.

5.2.2.9.1 POWER UP OF DISK DRIVE -

****
*** CAUTION ***
****

CERTAIN MODELS MUST NOT BE SHIPPED OR MOVED ANY **SIGNIFICANT DISTANCE WITHOUT THE CARRIAGE(S) BEING SECURED.**

5.0 INSTALLATION (Continued)

5.2.2.9.1 POWER UP OF DISK DRIVE - (Continued)

1. Verify that all power and input/output (I/O) cables are connected properly.

2. Refer to the Operation and Maintenance Manual for the Disk Drive for the procedures describing unlocking of the carriage, if applicable, and powering up.

   If the disk drive is an Ampex unit, the upper and lower carriages must be unlocked before powering up. They must be re-secured before being moved or shipped any significant distance.

Lock/Unlock carriages

1. TO unlock the carriages, slide the drive forward out of the rack

2. Remove the two screws from the rear of the top cover, slide the cover back slightly and remove cover

3. Locate and remove the upper carriage-stopper screw from the shipping position and place screw in storage position.

4. Remove screws securing the small round inspection cover on the bottom of the unit.

5. Loosen the screw that holds the lower carriage-stopper shaft. Turn the shaft clockwise to unlock the carriage then re-tighten screw to secure the stopper shaft.

6. Replace the top and bottom covers.

7. To lock the carriages, reverse the above procedure.

5.0 INSTALLATION (Continued)

5.2.2.9.2 POWER UP OF TAPE DECK

Below is a general procedure for bringing the tape deck on line. For additional information or specification refer to the Operation and Maintenance Manual for the Tape Deck.

1. Verify all power and I/O cables are connected properly.

2. Clean tape path. Use a cotton swab or lint-free pad moistened with Genesolve D (an Allied Chemical Co. trademark), or Freon TF (a DuPont CO. trademark), or similar recommended cleaner. Never use carbon tetrachloride or vythene for cleaning magnetic tapes. When cleaning, wipe in the same direction as tape movement. For care and storage of magnetic tapes see GEK-83774.

3. Mount pre-erased 10 1/2" test tape on deck. Following posted diagram, thread tape and advance tape on take-up reel 2 or 3 turns or until tape no longer slips on reel.

4. With power switch on, verify that the "write enable" light on deck is on. If not, check for write ring on the back on tape reel.

5. Load the tape and verify the load point indicator.

6. Turn the tape deck "On-line". The deck is now ready to receive data.

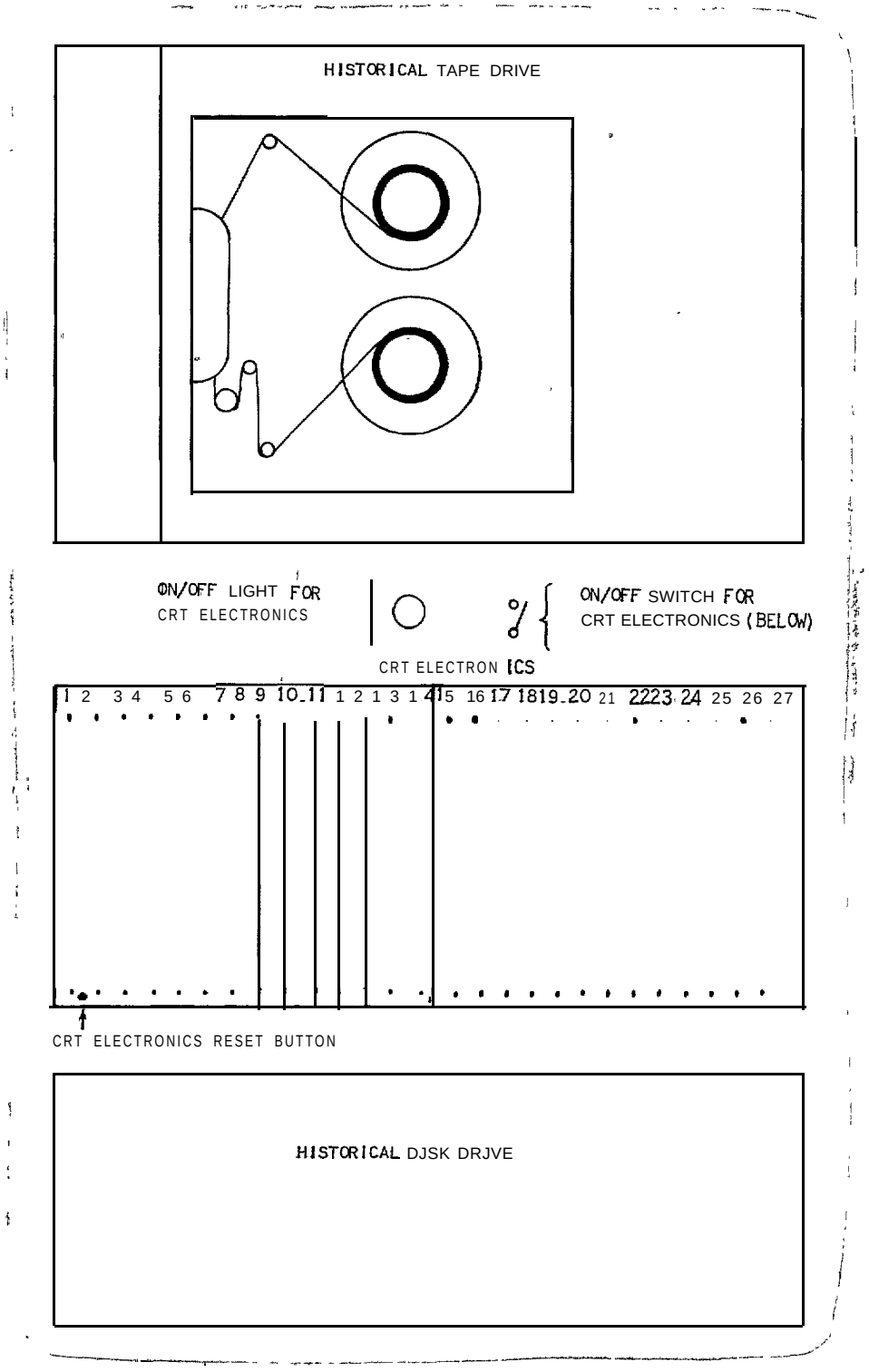


Figure 14: MASTER STATION - LEFT BAY - FRONT

5.0 INSTALLATION (Continued)

5.2.2.9.3 BRINGING HISTORICAL CONTROLLER ON LINE

1. Verify time has been entered on the operator's panel and is updating on the CRT.

****
* CAUTION *
****

IT IS OF UTMOST IMPORTANCE THAT THE CORRECT TIME HAS BEEN ENTERED. ALL RECORDS WILL BE IN RELATION TO THE TIME THAT IS SHOWN ON THE CRT.

IF THE CORRECT TIME IS BEING DISPLAYED, THEN PROCEED. IF NOT, THEN RESET THE CLOCK ACCORDING TO THE PROCEDURE LISTED IN THE OPERATOR CONTROL CONSOLE INSTRUCTIONS IN SECTION 6.2 UNDER OPERATION.

2. Verify Main and Display Controllers are "On-line".

3. verify E-bus is plugged into HRMA card in slot 1A of the Historical Mem. See figure 3 for assistance.

4. Set disk switch on the Maintenance Module to mount (closed). LED above switch should illuminate.

5. With the tape mounted and loaded on the drive, on the maintenance module, set the tape current and new switches up (closed). Then set tape mount switch to mount position (closed). LED above all three switches should illuminate. See figure 4 for switch locations on the Maintenance Module.

6. Turn the Historical mode switch on the Maintenance Module to the "On-line" position.

7. Observe CRT. The "HIST CNTLR OFFLINE" (historical controller offline) message should disappear and "DISK READY", "TAPE READY" should appear if the Main controller is logging properly.
NOTE: The above messages will appear only after each device has been written to.

8. If the above step is not completed, refer to the Historical portion under TROUBLESHOOTING section 7.3.2.

9. on the operator control console, enter a historical search request. A time near (earlier than) the current time would be appropriate to assure recording to disk is indeed operational. Refer to the Operator control console Instructions under OPERATION section 6.2 for advice.

5.0 INSTALLATION (Continued)

5.3 CONTROL CONSOLE

5.3.1 EQUIPMENT INSTALLATION

The control console is designed for panel or rack mounting with standard 19" rack spacing. For cutout dimensions, refer to the outline drawing supplied. Connections include two multi-wire cables that originate from the master station, and they are supplied.

5.3.2 CHECKOUT AND TEST

For checkout and test procedures on the Control console see item 5, "DISPLAY SECTION", in section 5.2.2, CHECKOUT AND TEST of the MASTER CABINET under INSTALLATION and section 7.3.6, CONTROL CONSOLE, under TROUBLESHOOTING.

5.4 CRT MONITOR

5.4.1 EQUIPMENT INSTALLATION

The monitor is designed for desk top placement, but it can be panel or rack mounted, as specified by the customer. The connections include an AC power cord, and three cables for color: red, green, and blue that connect to the master station.

5.4.2 CHECKOUT AND TEST

For checkout and test procedures on the CRT see item 6, "COLOR CRT CHECK", in section 5.2.2, CHECKOUT AND TEST of the MASTER CABINET under INSTALLATION and section 7.3.7, CRT MONITOR, under TROUBLESHOOTING.

5.5 TELEPRINTER (OPTIONAL)

5.5.1 EQUIPMENT INSTALLATION

Set the teleprinter in the desired location - no bolting is necessary. Two connections are required: an AC line to a convenient outlet, and the RS-232 interface to the remote modem, which is connected to the modem in the Master Station.

5.5.2 CHECKOUT AND TEST

For checkout and test on the teleprinter, see item 7, "TELEPRINTER PORT CHECK" in section 5.2.2, CHECKOUT AND TEST of the MASTER CABINET under INSTALLATION and section 7.3.8, TELEPRINTER (OPTIONAL), under TROUBLESHOOTING.

5.0 INSTALLATION (Continued)

5.6 TSI REMOTE CABINET

5.6.1 EQUIPMENT INSTALLATION

Position the TSI remote station in the desired location, bolt in place, and make the necessary connections according to standard electrical procedures and the information below.

5.6.1.1 CONNECTIONS

Make connections to the T-GSI cabinet, detectors, and associated equipment as shown on the customer connection diagram supplied for the system.

5.6.1.1.1 GROUNDING

****
* WARNING *
****

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.

5.6.1.1.2 DATA LINK

Special three-three wire shielded triads for transmitting signals from the TSI Remote Station to the Master Station is furnished with the equipment; the length specified at the time of order. The cable should be separated from all power and relay switching wires.

5.6.1.1.3 ALARM-INITIATION CONTACTS CONNECTED IN PARALLEL

Special instructions apply whenever a number of normally-open relay contacts of dual-alarm circuit boards (1589K29G70__) are connected in parallel. For details, refer to the INSTALLATION section of the publication for the dual-alarm board.

5.6.2 CHECKOUT AND TEST

5.6.2.1 POWER CHECKS

5.6.2.1.1 GROUND CHECK

After all external connections are made, lift the signal common cable to the EHC system and the CCOM bus bar to earth ground connection and verify an open circuit using an ohmmeter. Minimum resistance is 50 K ohms.

5.0 INSTALLATION (Continued)

5.6.2.1.2 115 VAC POWER CHECK

Verify 115 VAC is supplied. Measure for correct voltages per the customer connection diagram.

5.6.2.1.3 230 VAC POWER CHECK

Verify 230 VAC is supplied. Measure for correct voltages per the customer connection diagram.

5.6.2.2 ANALOG CHECK - TSI INPUTS

5.6.2.2.1 GENERAL PROCEDURE

Each system shall be tested with reference to the applicable Customer Connection and Elementary Diagrams. All connections to the cabinet terminal boards shall be made as specified on the Customer Connection Diagram.

Whenever measurements are performed on a TSI analog circuit board, the low side of the voltmeter or other test equipment is to be connected to the black test jack, unless otherwise specified.

5.6.2.2.2 POWER CHECKS

A. AC POWER CHECKS

1. Turn TSI Breaker (Bkr-16) on.
2. Turn the power switch on AC Power Supply on. The pilot light should light.

B. ADJUSTMENT OF TIME DELAY RELAY

Turn the TRIP switch to ON. The time delay relay should pick up 45 +/- 5 seconds after the TRIP switch is turned ON and drop out when the TRIP switch is switched to DISABLE.

C. ADJUSTING DC POWER SUPPLIES

1. Adjustment of Overvoltage Protectors
   If the vibration readings appear to go "dead", it could be due to a power supply trip. The +/- 15 VDC power supplies in the analog section of TSI have an overvoltage circuit that will cause the output to suddenly drop if the overvoltage setpoint is detected. If this occurs, consult the TROUBLESHOOTING section 7.3.3 for the above heading.

5.0 INSTALLATION (Continued)

5.6.2.2.2 POWER CHECKS (continued)

C. ADJUSTING DC POWER SUPPLIES (continued)

2. Adjustment of Power Supply Output Voltage
   Check the output of the positive power supply to 15.00 +/- 0.3 v, as measured with the digital voltmeter.
3. Repeat the above for the -15 V supply.

D. CHECKING REGULATED AC VOLTAGE

1. Ensure all circuit boards are installed into the modules.
2. Connect all detectors for the following inputs as required by the system:
   Expansion
   Eccentricity
   Position - LVDT's
3. Set expansion and position detectors to mid position values and eccentricity detector to 40 mils.
4. Measure the regulated AC voltage at terminals 5 and 6 of the AC Power Supply. This voltage should be 118 +/- 1 V.
5. If it is not within these limits, an autotransformer (XFMR-2) is provided

E. ADJUSTMENT OF 50V AC VOLTAGE

With all boards in place and detectors connected as indicated above, adjust variable resistor, R3006, in the AC Power Supply for 50 +/- 0.5 VAC at terminals 14 and 16 of the AC Power Supply.

5.6.2.2.3 ANALOG INPUTS

The various analog board/component publications are included in the T-GSI Composite Instruction Book. Each publication contains information on SPECIFICATIONS; RECEIVING, HANDLING, AND STORAGE; INSTALLATION; PRELIMINARY TEST AND ADJUSTMENTS; PRINCIPLES OF OPERATION; TROUBLESHOOTING; and REPAIR AND REPLACEMENT PARTS.

CHECKOUT AND TEST INSTRUCTIONS FOR THESE BOARDS AND COMPONENTS WILL NOT BE PROVIDED IN THIS SECTION ---THE TEST PROCEDURES OUTLINED IN THE SEPARATE ANALOG PUBLICATIONS ARE TO BE USED.

5.6.2.2.4 ANALOG ADJUSTMENTS

For analog adjustments, see the separate publications for each board or component in the T-GSI Composite Instruction book.

5.0 INSTALLATION (Continued)

5.6.2.3 LOADED POWER SUPPLY CHECK (Digital)

Apply power to all power supplies by closing the "MASTER" and "MAIN DISCONNECT" breakers. Using the test rings at the top of each SEM and MEM, measure the following voltages, and make sure they meet the specifications:

TEST RINGS	VOLTAGE	ADJUST
P5 to DCOM	+5 +/- 0.1 VDC	R1 (NPPB/C)
P15 to ACOM	t15 +/- 0.3 VDC	R6 (NPSK)
N15 to ACOM	-15 +/- 0.3 VDC	R7 (NPSK)
ACOM to DCOM	0 +/- 0.1 VDC	

Also check that the 105 VDC power supply on the contact input panel is within +/- 5 V. Measure at TB-C and adjust as necessary.

5.6.2.4 CONTROLLER VERIFICATION

NOTE: Open the "MASTER" breaker before performing "Controller Verification".

A. Connect the teleprinter to the HMPF card in slot 1A with an RS-232 cable through a null modem connector adaptor (supplied in the service kit) to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

B. Apply power to the Small Electronic Modules (SEMs) by closing the breaker marked "MASTER" on the breaker panel.

C. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "*F*", which stands for pass or fail, respectively. If all the statuses are a "P", then proceed to step D. If an "*F*" appears, then refer to the CONTROLLER VERIFICATION portion in section 7.3.3, TSI REMOTE CABINET under TROUBLESHOOTING. Figure 15 shows an example of this print out.

5.0 INSTALLATION (Continued)

5.6.2.4 CONTROLLER VERIFICATION (continued)

D. Check the status of the controller by typing "STATUS" after the dollar sign ($) prompt. An example of this print out is shown in Figures 16 and 17.

```
ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS              EXP   TST
F0000-F1FFF LO P     EA4A  EA4A
F0000-F1FFF HI P     AA0B  AA0B
F2000-F3FFF LO P     51E1  51E1
F2000-F3FFF HI P     4C6E  4C6E
F4000-F5FFF LO P     92CA  92CA
F4000-F5FFF HI P     94CF  94CF
F6000-F7FFF LO P     CD35  CD35
F6000-F7FFF HI P     A585  A585
F8000-F9FFF LO P     5820  5820
F8000-F9FFF HI P     6396  6396
FA000-FBFFF LO P     332A  332A
FA000-FBFFF HI P     0498  0498
FC000-FDF9F LO P     258F  258F
FC000-FDF9F HI P     B173  B173

****

RAM CHECK

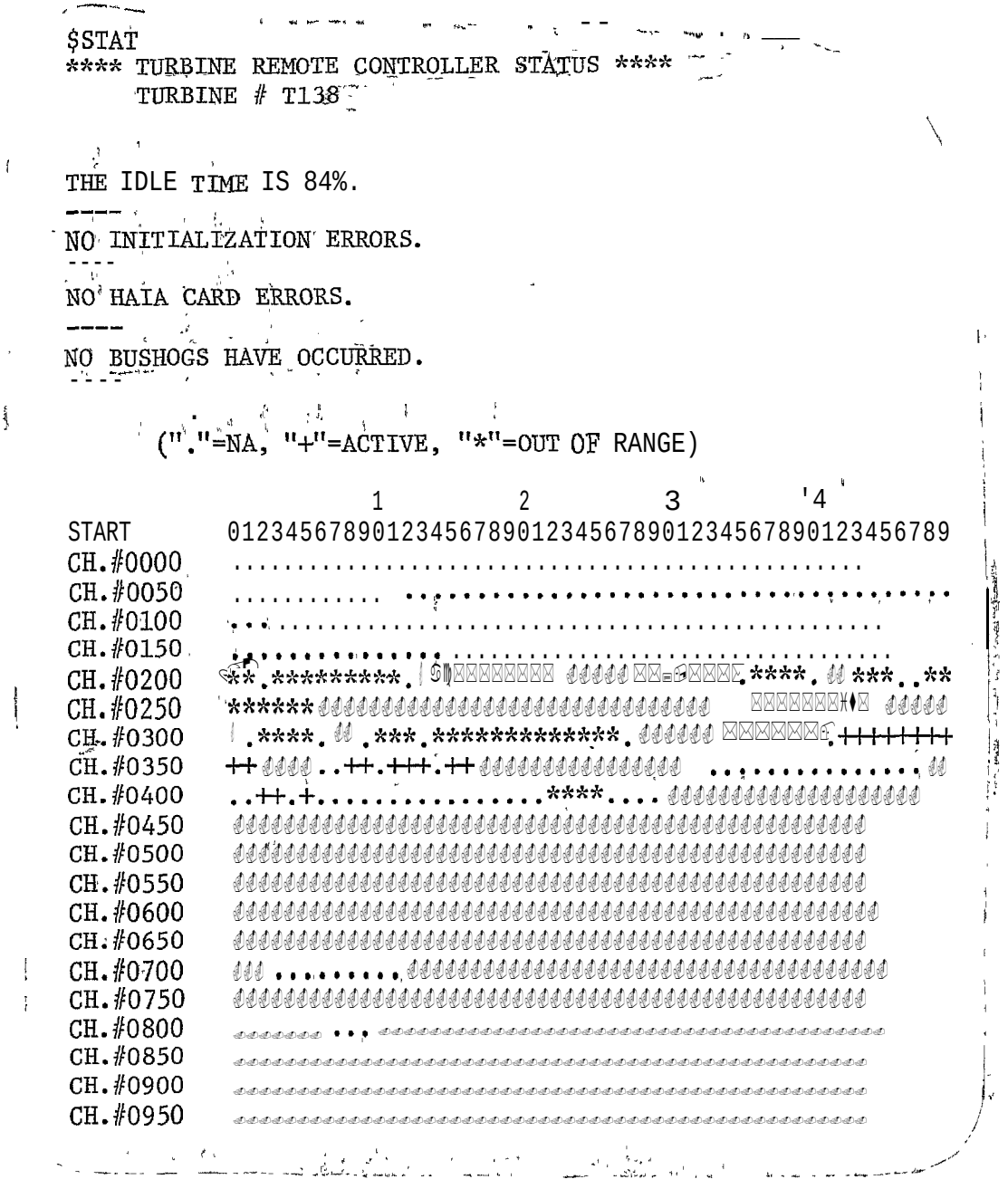
ADDRESS
00000-02FFF      PASSED

****

ROM CHECKSUM TEST (P=PASS F=FAIL)

ADDRESS              EXP   TST
FE000-FFFDF LO P     6EA2  6EA2
FE000-FFFDF HI P     166E  166E
```

Figure 15: POWER UP PRINTOUT - REMOTE CONTROLLER



```
$STAT
**** TURBINE REMOTE CONTROLLER STATUS ****
     TURBINE # T138


THE IDLE TIME IS 84%.
----
NO INITIALIZATION ERRORS.
----
NO HAIA CARD ERRORS.
----
NO BUSHOGS HAVE OCCURRED.
----

      ("."=NA, "+"=ACTIVE, "*"=OUT OF RANGE)

                    1         2         3         4
START       01234567890123456789012345678901234567890123456789
CH.#0000    ..................................................
CH.#0050    ..........  ..,.....................................
CH.#0100    ...  ..............................................
CH.#0150    ,.............,.....................................
CH.#0200    **.*********.  ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0250    ******,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0300    .****.,,,.***.**************.,,,,,,,,,,,,,,,,,,,,,,
CH.#0350    ++,,,,,..++.+++.++,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0400    ..++.+.................****....,,,,,,,,,,,,,,,,,,,,
CH.#0450    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0500    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0550    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0600    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0650    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0700    ,,,.........,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0750    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0800    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0850    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0900    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
CH.#0950    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
```

Figure 16: STATUS PRINT OUT (with no errors) - REMOTE CONTROLLER

```
$STAT
**** GENERATOR REMOTE CONTROLLER STATUS ****
     TURBINE # T138

THE IDLE TIME IS 77%.

THERE ARE 0002T INITIALIZATION ERRORS. 'THE FIRST 02T ERRORS ARE
LISTED BELOW:

01) RAW DATA RECORD #0113T TRIED TO USE HIOD #2, WHICH DOES NOT
    EXIST.

02) RAW DATA RECORD #0114T TRIED TO USE HIOD #2, WHICH DOES NOT
    EXIST.

NO HAIA CARD ERRORS.

NO BUSHOGS HAVE OCCURRED,

      ("."=NA, "+"=ACTIVE, "*"=OUT OF RANGE)

                    1    2    3    4
 START      01234567890123456789012345678901234567890123456789
  CH.#0000  ..... ..................................
  CH.#0050  .....  .... .......... .... .....
  CH.#0100  ..... I  ..........................
  CH.#0150  ..................................................
  CH.#0200  ..................................  ............
  CH.#0250  ..................................................
CH.#0300    ..................................................
  CH.#0350  ..................................................  I
  CH.#0400  ..................................................
  CH.#0450  .....  ..........................
  CH.#0500  ++++++I..... ++++++++.++..  .++.**********
: CH.#0550  ****************************.....................
  CH.#0600  ..... ***.....  ************************
  CH.#0650  ********************.............................  **
  CH.#0700  *.....  ** .. ***. .******. .. ******..
  CH.#0750  ...*.....  **.....  ******..*.*.
  CH.#0800  ..... ***..... **.....  ......  .....
  CH.#0850  ..... ++++++++++++++.+++++++.++..++++.......
  CH.#0900  ..................................................
  CH.#0950  ..................................................
```

Figure 17: STATUS PRINT OUT (with errors) - REMOTE CONTROLLER

5.0 INSTALLATION (Continued)

5.6.2.5 INPUT/OUTPUT CHECK

For the following tests the remote diagnostics need to be in the "error" sublevel. To get to the "error" sublevel, type "ERR" followed by a carriage return on the teleprinter. YOU will know you are in the "error" sublevel because you will now have a "#" prompt. (NOTE: If you already have the "#" prompt, you do not need to do the above.)

5.6.2.5.1 CONTACTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
135 - 150	234A - 235A

Using the elementary diagram and the customer connection diagram, test each of the above inputs using the procedure below. some of the inputs may not be in the system, these should not be checked.

Type "VAL" on the teleprinter followed by the input number. The value should be either 0 or 8191. change state of the contact input. Type "VAL" on the teleprinter followed by the input number again. This time the value should be the opposite of what it was the first time.

5.6.2.5.2 SPEED

INPUT POINTS	ELEMENTARY SHEET NUMBER	INPUT FREQUENCY
125 - 126	232A	4800 +/- 1 Hz
127 - 128	232A	3600 +/- 1 Hz

Using the elementary diagram and customer connection diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

With no frequency input, type "VAL" on the teleprinter followed by the input number. The value should be 0 exactly. Apply the frequency in the table above to the input. This input should be a 30 - 70 V square wave. Type "VAL" on the teleprinter followed by the input number. The value should be 3600 +/- 2 counts.

5.6.2.5.3 THERMOCOUPLE INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
79 - 92	231A
94 - 107	231B

5.0 INSTALLATION (Continued)

5.6.2.5 INPUT/OUTPUT CHECK (continued)

5.6.2.5.3 THERMOCOUPLE INPUTS [continued)

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Open the input and type "VAL" on the teleprinter followed by the input number. Check for "OR" - the Out of Range indication to insure that the thermocouple being tested is the one being "called up" on the system.

Short the input and type "VAL" on the teleprinter followed by the input number. (The units for this value can be found on the Customer Connection Diagram.) Measure the ambient temperature near the Thermocouple Input Module in the cabinet. This value should equal the above value +/- 4.63°F or +/- 2.57°C. Record this Cold Junction value.

For spot checking the calibration of the various thermocouples, follow the procedure below: In the table that follows, find the column for the appropriate type of thermocouple. Using a book of thermocouple output voltages, look up the cold junction value measured above and find the voltage that corresponds to it. subtract this voltage from the voltage in the table below and apply it to the input terminals. Type "VAL" on the teleprinter followed by the input number. This value should equal the value in the table +/- 11°F or +/- 6.1°C. Repeat this paragraph for the other two input voltages in the table.

VOLTAGE INPUT (mv)

(J) IRON CONSTANTAN	(K) CHROMEL ALUMEL	(E) CHROMEL CONSTANTAN	(T) COPPER CONSTANTAN	TEMPERATURE DEG F	DEG C
-3.49	-2.70	-3.98	-2.58	-100	-73.3
14.11	10.56	17.94	12.57	500	260.0
29.52	22.25	40.06	N/A	1000	537.8

5.6.2.5.4 4 - 20 MA INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
45 - 52	230B
62 - 69	230B

5.8 INSTALLATION (Continued)

5.6.2.5.4 4 - 20 MA INPUTS (continued)

Using the Elementary Diagram and the customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Apply 12.00 ma to the input. Type "VAL" on the teleprinter followed by the input number. This value should be half way between the low and the high value on the Customer connection Diagram +/- 2 counts.

5.6.2.5.5 VOLTAGE INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBERS
11 - 26	211X & 212X
28 - 29	214X
31 - 44	214X, 215X & 232A
53 - 60	232A & 216X
70 - 77	232A & 218X

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system and should not be checked.

Apply the proper voltage to the input to get a half scale reading. Type "VAL" on the teleprinter followed by the input number. This value should be equal to the reading calculated above ± 2 counts.

5.7 TURBINE REMOTE CABINET

5.7.1 EQUIPMENT INSTALLATION

Position the turbine remote station near the turbine, BUT NOT ON THE TURBINE FOUNDATION (PEDESTAL), and secure.

Make the necessary connections following standard electrical procedures, the notes below, and the customer connection diagram.

5.7.1.1 GROUNDING

****
* *
* WARNING *
* *
****

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED **GROUNDING PRACTICES AND PROCEDURES MUST BE** ADHERED TO.

5.0 INSTALLATION (Continued)

5.7.1.2 CONNECTIONS TO TRANSDUCERS

As shown on the customer connection diagram, make the signal connection to the transducer with shielded cable (2 wires within a shield, with plastic covering over the shield).

If the 4-20 ma transmitters are shared with multiple users, the transmitters should be grounded at the power supply for it. In the case of multiple transmitters powered by the same power supply, grounding only the power supply will insure that multiple grounds do not exist. The ground will be in the remote station and the T-GSI input module must be the first input TAP after the power supply ground, if the power supply is internal to the T-GSI remote station. If an external power supply is used, the customer must insure that the input to the T-GSI input module is kept electrically within 5 volts of the remote station's internal ground. Figure 18-A shows an example for wiring when the transmitter is not shared and Figure 18-B shows an example for wiring when the transmitter is shared.

5.7.1.3 DATA LINK

Special three-three wire shielded triads for transmitting signals from the Turbine Remote Station to the Master Station is furnished with the equipment; the length specified at the time of order. The cable should be separated from all power and relay switching circuits.

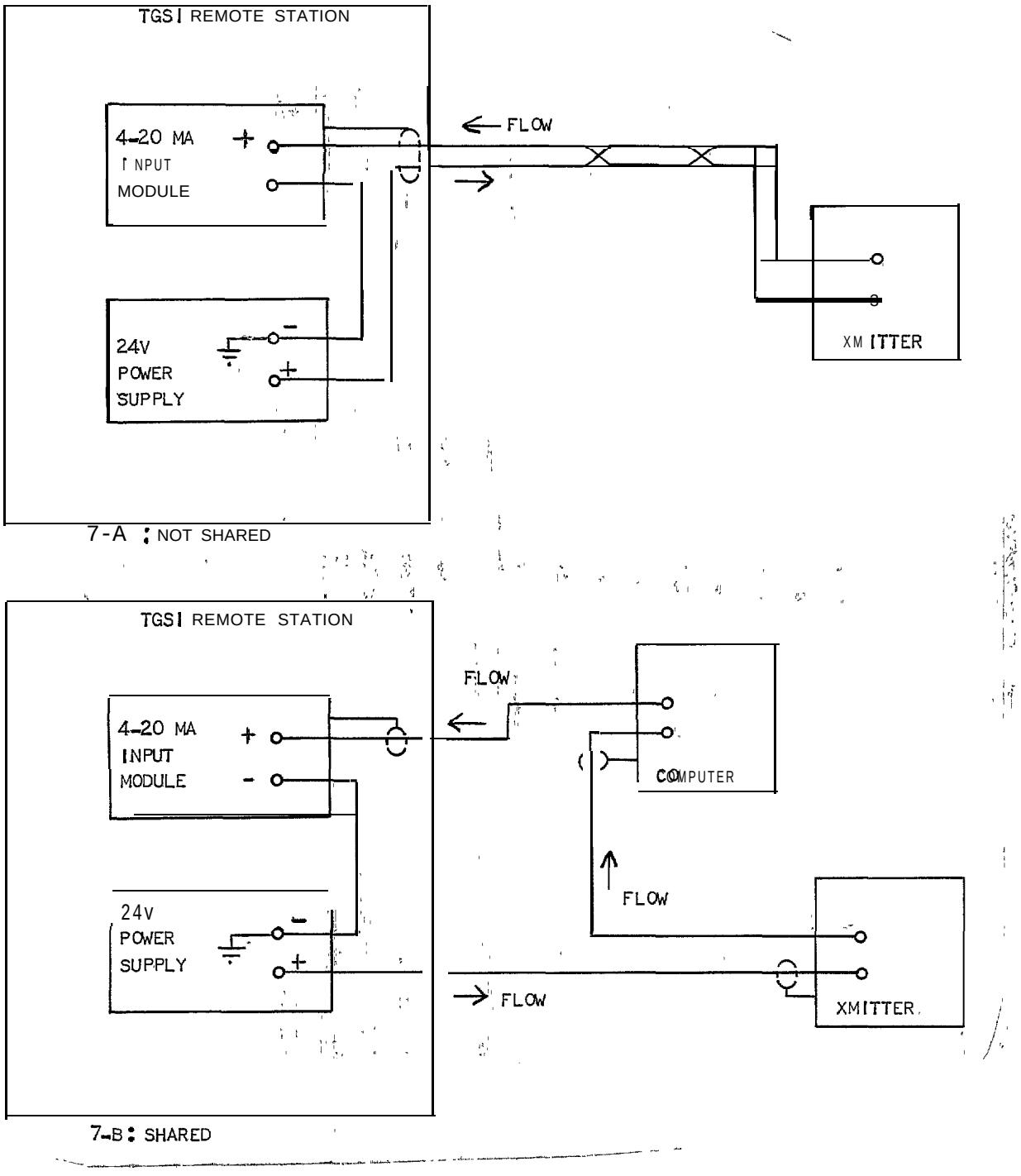


Figure 18: TRANSDUCER WIRING

5.0 INSTALLATION (Continued)

5.7.2 CHECKOUT AND TEST

5.7.2.1 POWER CHECKS

5.7.2.1.1 GROUND CHECK

After all external connections are made, lift the CCOM bus bar to earth ground connection and verify an open circuit using an ohmmeter. Minimum resistance is 50 K ohms.

5.7.2.1.2 115 VAC POWER CHECK

Verify 115 VAC is supplied from the uninterruptible power supply and measure for correct voltages per the customer connection diagram.

5.7.2.1.3 230 VAC POWER CHECK

Verify 230 VAC is supplied and measure for correct voltages per the customer connection diagram.

5.7.2.1.4 LOADED POWER SUPPLY CHECK

Apply power to all power supplies. close the "MASTER" and "MAIN DISCONNECT" breakers. Using the test rings at the top of each SEM and MEM, measure the following voltages, and make sure they meet the specifications:

TEST RINGS	VOLTAGE	ADJUST
P5 to DCOM	+5 +/- 0.1 VDC	R1 (NPPB/C
P15 to ACOM	t15 +/- 0.3 VDC	R6 (NPSK)
N15 to ACOM	-15 +/- 0.3 VDC	R7 (NPSK)
ACOM to DCOM	0 +/- 0.1 VDC	

When applicable, check that the 105 VDC power supply on the contact input panel is within +/- 5 V. Measure at TB-C and adjust as necessary.

5.0 INSTALLATION (continued)

5.7.2.2 CONTROLLER VERIFICATION

NOTE: Open the "MASTER" breaker before performing "Controller Verification".

A. connect the teleprinter to the HMPF card in slot 1A with an RS-232 cable through a null modem connector adaptor (supplied in the service kit) to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

B. Apply power to the Small Electronic Modules (SEMs) by closing the breaker marked "MASTER" on the breaker panel in the remote cabinet.

C. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "*F*", which stands for pass or fail, respectively. If all the statuses are a "P", then proceed to step D. If an "*F*" appears, then refer to the CONTROLLER VERIFICATION portion in section 7.3.4 TURBINE REMOTE CABINET under TROUBLESHOOTING. Figure 15 shows an example of this print out.

D. Check the status of the controller by typing "STAT" after the dollar sign ($) prompt. An example of this print out is shown in Figures 16 and 17.

5.7.2.3 INPUT/OUTPUT CHECK

For the following tests the remote diagnostics need to be in the "error" sublevel. TO get to the "error" sublevel, type "ERR" followed by a carriage return on the teleprinter. You will know you are in the "error" sublevel because you will now have a "#" prompt. (NOTE: If you already have the "#" prompt, you do not need to do the above.)

5.7.2.3.1 CONTACTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
390 - 421	232A - 232D

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Type "VAL" on the teleprinter followed by the input number. The value should be either 0 or 8191. Change state of the contact input. Type "VAL" on the teleprinter followed by the input number again. This time the value should be the opposite of what it was the first time.

5.0 INSTALLATION (Continued1

5.7.2.3.2 THERMOCOUPLE INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
200 - 227	331A - 331B
244 - 313	331C - 331G
440 - 443	331G

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Open the input and type "VAL" on the teleprinter followed by the input number. Check for "OR" - the Out of Range indication to insure that the thermocouple being tested is the one being "called up" on the system.

Short the input and type "VAL" on the teleprinter followed by the input number. (The units for this value can be found on the Customer Connection Diagram.) Measure the ambient temperature near the Thermocouple Input Module in the cabinet. This value should equal the above value +/- 4.63°F or +/- 2.57°C. Record this cold Junction value.

For spot checking the calibration of the various thermocouples, follow the procedure below: In the table that follows, find the column for the appropriate type of thermocouple. Using a book of thermocouple output voltages, look up the cold junction value measured above and find the voltage that corresponds to it. Subtract this voltage from the voltage in the table below and apply it to the input terminals. Type "VAL" on the teleprinter followed by the input number. This value should equal the value in the table +/- 11°F or +/- 6.1°C. Repeat this paragraph for the other two input voltages in the table.

VOLTAGE INPUT (mv)

(J) IRON CONSTANTAN	(K) CHROMEL ALUMEL	(E) CHROMEL CONSTANTAN	(T) COPPER CONSTANTAN	TEMPERATURE DEG F	TEMPERATURE DEG C
-3.49	-2.70	-3.98	-2.58	-100	-73.3
14.11	10.56	17.94	12.57	500	260.0
29.52	22.25	40.06	N/A	**1000**	537.8

5.0 INSTALLATION (Continued)

5.7.2.3.3 4 - 20 MA INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
342 - 373	330A & 330C

Using the Elementary Diagram and the customer connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Apply 12.00 ma to the input. Type "VAL" on the teleprinter followed by the input number. This value should be half way between the low and the high value on the Customer connection Diagram +/- 2 counts.

5.7.2.3.4 0 - 10 VOLT INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
374 - 389	330B

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system, these should not be checked.

Apply 5.00 V to the input. Type "VAL" on the teleprinter followed by the input number. This value should be half way between the low and the high value on the Customer connection Diagram +/- 2 counts.

5.8 GENERATOR REMOTE CABINET

5.8.1 EQUIPMENT INSTALLATION

Place the generator remote station in the desired location, NOT ON THE TURBINE FOUNDATION (PEDESTAL), and secure in place.

Make the necessary connections following standard electrical practices, the notes below, and the customer connection diagram.

5.8.1.1 GROUNDING

****
* WARNING *
****

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED **GROUNDING PRACTICES AND PROCEDURES** MUST BE ADHERED TO.

5.0 INSTALLATION (Continued)

5.8 GENERATOR REMOTE CABINET (continued)

5.8.1.2 CONNECTIONS TO TRANSDUCERS

As shown on the customer connection diagram, make the signal connection to the transducer with shielded cable (2 wires within a shield, with plastic covering over the shield),

If the 4 - 20 ma transmitters are shared with multiple users, the transmitters should be grounded at the power supply for it. In the case of multiple transmitters powered by the same power supply, grounding only the power supply will insure that multiple grounds do not exist. The ground will be in the remote station and the T-GSI input module must be the first input TAP after the power supply ground, if the power supply is internal to the T-GSI remote station. If an external power supply is used, the customer must insure that the input to the T-GSI input module is kept electrically within 5 volts of the remote station's internal ground. Figure 10-A shows an example for wiring when the transmitter is not shared and Figure 10-B shows an example for wiring when the transmitter is shared.

5.8.1.3 DATA LINK

Special three-three wire shielded triads for transmitting signals from the Generator Remote Station to the Master Station is furnished with the equipment; the length specified at the time of order. The cable should be separated from all power and relay switching cables.

5.8.2 CHECKOUT AND TEST

5.8.2.1 POWER CHECKS

5.8.2.1.1 GROUND CHECK

After all external connections are made, lift the CCOM bus bar to earth ground connection and verify an open circuit using an ohmmeter. Minimum resistance is 50 K ohms.

5.8.2.1.2 115 VAC POWER CHECK

Verify 115 VAC is supplied from the uninterruptible power supply and measure for correct voltages per the customer connection diagram.

5.8.2.1.3 230 VAC POWER CHECK

Verify 230 VAC is supplied and measure for correct voltages per the customer connection diagram.

5.0 INSTALLATION (Continued)

5.8.2.1.4 LOADED POWER SUPPLY CHECK

Apply power to all power supplies. Close the "MASTER" and "MAIN DISCONNECT" breakers. Using the test rings at the top of each SEM and MEM, measure the following voltages, and make sure they meet the specifications:

TEST RINGS	VOLTAGE	ADJUST
P5 to DCOM	+5 +/- 0.1 VDC	R1 (NPPB/C)
P15 to ACOM	+15 +/- 0.3 VDC	R6 (NPSK)
N15 to ACOM	-15 +/- 0.3 VDC	R7 (NPSK)
P28 to DCOM	+28 +/- 0.3 VDC	R1 (NPPB/C)
ACOM to DCOM	0 +/- 0.1 VDC	

Also check that the 105 VDC power supply on the contact input panel is within +/- 5 V. Measure at TB-C. Adjust as necessary.

5.8.2.2 CONTROLLER VERIFICATION

NOTE: Open the "MASTER" breaker before performing controller verification.

A. Connect the teleprinter to the HMPF card in slot 1A with an RS-232 cable through a null modem connector adaptor (supplied in the service kit) to be located at the card. The adaptor is marked for proper orientation. The power cord for the teleprinter can be connected at the receptacle on the power distribution module.

B. Apply power to the Small Electronic Modules (SEMs) by closing the breaker marked "MASTER" on the breaker panel in the remote cabinet.

C. In a couple of seconds, the teleprinter should start printing out addresses followed by a "P" or an "*F*", which stands for pass or fail, respectively. If all the statuses are a "P", then proceed to step D. If an "*F*" appears, then refer to the CONTROLLER VERIFICATION portion in section 7.3.5 GENERATOR REMOTE CABINET under TROUBLESHOOTING. Figure 15 shows an example of this print out.

D. Check the status of the controller by typing "STAT" after the dollar sign ($) prompt. An example of this print out is shown in Figures 16 and 17.

5.8.2.3 INPUT/OUTPUT CHECK

For the following tests the remote diagnostics need to be in the "error" sublevel. To get to the "error" sublevel, type "ERR" followed by a carriage return on the teleprinter. You will know you are in the "error" sublevel because you will now have a "#" prompt. (NOTE: If you already have the "#" prompt, you do not need to do the above.)

5.0 INSTALLATION (Continued)

5.8.2.3 INPUT/OUTPUT CHECK (continued)

5.8.2.3.1 CONTACTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
860 - 923	435A - 435H

Using the Elementary Diagram and the customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system.

Type "VAL" on the teleprinter followed by the input number. The value should be either 0 or 8191. Change state of the contact input. Type "VAL" on the teleprinter followed by the input number again. This time the value should be the opposite of what it was the first time.

5.8.2.3.2 THERMOCOUPLE INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
538 - 609	430A - 430F
613 - 624	430F
782 - 795	430G
799 - 812	430H
815 - 842	4305 - 430K

Using the Elementary Diagram and the customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system.

Type "VAL" on the teleprinter followed by the input number. Open the input and check for "OR" - the Out of Range indication to ensure that the thermocouple being tested is the one being "called up" on the system.

Short the input and type "VAL" on the teleprinter followed by the input number. (The units for this value can be found on the Customer Connection Diagram.) Measure the ambient temperature near the Thermocouple Input Module in the cabinet. This value should equal the above value +/- 4.63°F or +/- 2.57°C. Record this Cold Junction value.

For spotchecking the calibration of the various thermocouples, follow the procedure below: In the table that follows, find the column for the appropriate type of thermocouple. Using a book of thermocouple output voltages, look up the cold junction value measured above and find the voltage that corresponds to it. Subtract this voltage from the voltage in the table below and apply it to the input terminals. Type "VAL" on the teleprinter followed by the input number. This value should equal the value in the table +/- 11°F or +/- 6.1°C. Repeat this procedure for the other two input voltages in the table.

5.0 INSTALLATION (Continued)

5.8.2.3.2 THERMOCOUPLE INPUTS (continued)

VOLTAGE INPUT (mv)

(J) IRON CONSTANTAN	(K) CHROMEL ALUMEL	(E) CHROMEL CONSTANTAN	(T) COPPER CONSTANTAN	TEMPERATURE DEG F	TEMPERATURE DEG C
-3.49	-2.70	-3.98	-2.58	-100	-73.3
14.11	10.56	17.94	12.57	500	260.0
29.52	22.25	40.06	N/A	1000	537.8

5.8.2.3.3 4 - 20 MA INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
522 - 537	431A

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system.

For linear inputs, apply 12.00 ma to the input. Type "VAL" on the teleprinter followed by the input number. This value should be half way between the low and the high value on the Customer Connection Diagram -I-/-2 counts.

For non-linear inputs such as hydrogen purity and conductivity apply 12.00 ma to the input. Type "VAL" on the teleprinter followed by the input number. Check the calibration curve for the particular transducer to see if the value obtained corresponds to the appropriate value on the curve.

5.8.2.3.4 0 - 10 VOLT INPUTS

INPUT POINTS	ELEMENTARY SHEET NUMBER
843 - 858	431B

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system.

5.0 INSTALLATION (Continued)

5.8.2.3.4 0 - 10 VOLT INPUTS (continued)

Apply 5.00 V to the input. Type "VAL" on the teleprinter followed by the input number. This value should be half way between the low and the high value on the Customer Connection Diagram +/- 2 counts.

5.8.2.3.5 RTDS

INPUT POINTS	ELEMENTARY SHEET NUMBER
626 - 697	434D - 434H
701 - 708	4348
725 - 728	434A
730 - 733	434A
735 - 742	434A
745 - 752	434B
754 - 769	434B - 434C
773 - 780	434c

Using the Elementary Diagram and the Customer Connection Diagram, test each of the above inputs using the procedure below. Some of the inputs may not be in the system.

Note: All unused or open RTD inputs must be shorted and grounded before calibration checks are attempted.

If this system does not use grounded RTDS, the C inputs must be grounded to ACOM per the customer Connection Diagram. Disconnect the RTD at the generator junction box. Type "VAL" on the teleprinter followed by the input point number. The value printed out should be "OR" for Out of Range. Reconnect the RTD. The print out should now be the ambient temperature at the RTD.

For spotchecking the calibration of the various RTDs the following procedure may be used:

Pick a resistor within the range of the RTD this input uses. Connect this resistor from input terminal A to input terminals B and C.

Note: Lead C must also be connected to ACOM.

Type "VAL" on the teleprinter followed by the input point number. The value printed should be correct within + 1% of full scale.

If you shorted any of the C leads above to ACOM, this short must be removed after the RTD is checked.

5.0 INSTALLATION (Continued)

5.8.2.3.6 3-PHASE POWER & FREQUENCY

The following tests are to be made after the Generator is started up.

Using the channels and the formulas below, type "VAL" on the teleprinter followed by the channel number. The formulas give the values that should be printed.

CHANNEL NUMBER	ELEMENTARY SHEET	SIGNAL	FORMULA
505 - 507	432A	Voltage (KV)	KV = (Input Volts * PT Ratio)/1000
502 - 504	432A	Current (KA)	KA = (Input Amps * CT Ratio)/1000
500	432A	Real Power (MW)	MW = (KV * KA) COS(Phase Angle)
501	432A	Reactive Power (MVAR)	MVAR = (KV * KA) SIN(Phase Angle) Note: The value is negative for leading PF.
521	432A	Power Factor (PF)	PF = COS(Phase Angle)

Measure the generator frequency. Type "VAL 520" on the teleprinter. The two values should agree within ± 0.05 Hz.

5.9 SYSTEM CHECKS

5.9.1 EXERCISING REMOTES AND MASTER

With the Main Controller in the Standby mode, the controller attempts to communicate with all remote stations in the system, whether powered up or not. An indication of this is the blinking (approximately 2 Hz rate) of each remote's associated "ACTIVE" LED on the PDLA (HMHA) cards in slots 2B & 2C of MEM - B1S.

No logging to the Display Controller, Historical Controller or Plant Computer takes place in the Standby mode.

To printout the remote station datalink status using the Main Controller's status function, enter via the diagnostic teleprinter, the command "Status DTL". The operational status of each datalink should be "UP". If not, ensure that the datalink cables are in place and connectors at the PDLA cards are properly seated.

5.9.2 DATA RETRIEVAL

Retrieve present and historical data logs via the control console and teleprinter. Ensure retrieved data agrees with the input data. Randomly check that the CRT Page data is also correct.

5.0 INSTALLATION (Continued)

5.9.2 DATA RETRIEVAL (continued)

After a few days of recording data on tape, dismount and remount the tape as an old tape. Then perform a tape read function and examine the disk reserve directory using the (DRA(NGE) Command to ensure that a proper tape read was accomplished. (One day on each side of the requested day written to reserve directory.)

Remount the tape as a used tape and ensure the tape repositions at the end of data and continues to update. (Message on CRT should be "Tape READY".)

6.0 OPERATION

6.1 INTRODUCTION

The OPERATION section contains the information necessary for communicating with the T-GSI system. Included are various component instructions and descriptions. In the paragraphs below is a brief description of the material covered in the succeeding subsections.

A. The CONTROL CONSOLE KEY DESCRIPTION gives a brief summary of the pushbuttons on the console and their function.

B. DATA and DISPLAY SPECIFICATIONS describe the data pages and their contents; and the calculated values and the algorithms used in their calculation.

C. The OPERATOR'S CONTROL CONSOLE INSTRUCTIONS describe general operation of the control console. They give the procedures for obtaining current operating data and recent* historical data, the most common mode of operation. They detail the sequence and type of pushbuttons to press for each type of information desired. They also describe what messages will be displayed when certain areas of the system aren't working or if improper pushbuttons are selected.

D. The OPERATOR TELEPRINTER INSTRUCTIONS are comprised of operating instructions and the various command descriptions for use on the teleprinter. The last page contains a list of the input point categories available and how to address them.

E. The HISTORICAL DATA STORAGE INSTRUCTIONS contain information on mounting and dismounting of tapes, reading old tapes, explanation of tape and disk CRT messages, and explanation of diagnostic commands.

F. The ALARM SET POINT INSTRUCTIONS direct the operator on how to change the pre-alarm and alarm settings for certain inputs. The inputs with changeable alarm limit setpoints are marked as such on the Recorded Data List in the Customer Connection Diagram, pages 02A-02T.

6.2 CONTROL CONSOLE KEY DESCRIPTION

A. GENERAL DATA Causes display on the CRT of the General Data Page.

B. ALARM SUMMARY Causes display on the CRT of the Alarm Summary Page showing all the channels which are in the alarm condition.

C. COMPOSITE DATA Causes display on the CRT of the Composite Data Page.

* Recent historical data includes approximately 45 days of data, which is the amount of information that can be stored on the disk.

6.0 OPERATION (Continued)

6.2 CONTROL CONSOLE KEY DESCRIPTION (continued1

D. DATA GROUPS — starts the input sequence to select the specific data group page and causes the specific page to be displayed on the CRT.

E. SELECTIVE DATA — Causes display on the CRT of the Selective Data Page showing the inputs which were previously selected.

F. READ TAPE — Starts the sequence to read data from the magnetic tape to disk.

G. TIME SET — Starts the input sequence to change the clock.

H. TREND DIRECTORY — Causes the display on the CRT of a directory of inputs available for trending.

I. TREND DISPLAY — Causes display of the Trend Display Page on the CRT showing previously selected trend inputs.

J. CURRENT LOG — Causes display on the CRT of all new data inputs in consecutive order, including descriptive labels, values and measurement units.

K. SCAN LOG FWD — Scrolls the log on the CRT forward or stops scrolling. The log may be either current data or historic data.

L. SCAN LOG REV — Scrolls the log on the CRT in reverse order or stops scrolling.

M. FAST — Accelerates scrolling of the log displayed on the CRT, or if already scrolling, fast then scrolls slow. Log may be either current or historic data.

N. HIST LOG SEARCH — Starts sequence to select a time frame for retrieval of an historic log. The selected log is then displayed on the CRT.

0. HIST LOG NEXT — Starts the search for the next historic log in time sequence.

P. ENTER — Causes number or command appearing on console display to be entered. Key signifies completion of command sequence.

Q. CANCEL — Aborts command or data appearing on console display.

6.0 OPERATION (Continued)

6.2 CONTROL CONSOLE KEY DESCRIPTION (continued)

R. ALARM ACK — Allows operator acknowledgement of an alarm when alarm summary page is displayed on CRT and stops flashing of alarm message on CRT.

S. DELETE CHANNEL — Deletes selected data channel (input) from either "TREND DISPLAY" or "SELECTIVE DATA."

6.3 DATA And DISPLAY DESCRIPTIONS

6.3.1 CRT DISPLAY - GENERAL

The color CRT monitor provides graphic and alpha-numeric data indication for the system. Data pages are selectable via a pushbutton control console. Several pages have individual pushbuttons for selection, whereas so called "Data Group" pages are selected in two steps. (Refer to the Operator control Console Instructions for procedure.) The monitor is configured in a matrix format of 72 positions/line by 48 lines. The page data may use all but the bottom four lines of each page, which are reserved for system messages.

Various pages displayed on the monitor have data displayed in bar graph form. Most bar graphs have normal video (dark background) but some use inverse video (white background). When inverse video is employed, the dark background that appears in varying lengths (Max of one character space) at the tip of the bars as the data changes should be ignored. Therefore, the value is read at the tip of the main bar not the dark background.

6.3.2 COMPOSITE GRAPH DATA PAGE

The composite display page provides data that is critical to the safe and continuous operation of the turbine-generator. The composite display page will be the primary display during turbine-generator start-ups and shutdowns. The composite display is divided into steam turbine data and generator data.

The bar graphs of the composite display are considered summaries and have scales of absolute value. Most depict the maximum value (closest to the limit) parameter of each group. I.E., of the 16 points of vibration only one point is determined to have the greatest value or the value closest to a pre-defined limit). Other parameters may be singles. Examples of parameters displayed on this page are vibration, eccentricity, water detection temperature, shell metal temperature, P-Bar liquid, stator slot, and cold gas temperatures.

6.0 OPERATION (Continued)

6.3.3 GENERAL DATA PAGE

The General Data page provides current value data on operating conditions, i.e., speed, control valve postion, load, etc., and system status data, i.e., status of circuit breaker, turning gear, rotor, etc., of the steam turbine, generator , and auxiliary equipment, if applicable.

6.3.4 DATA GROUP PAGES

The T-GSI contains up to twelve pages under the "Data Group" heading. Five pages are available for turbine data, six pages for generator data and one page for auxiliary equipment, e.g., boiler feed pump turbines, if applicable. Pages included in a particular system are listed on the front of the Control Console. A description of some of the typical pages follows.

6.3.4.1 Bearing Data Page

This page provides current value data on the bearing span. Numerical values of journal bearing vibration, temperature, and thrust bearing temperature are updated once every three seconds or less. Bearing symbols and values change color when alarm set points are exceeded.

6.3.4.2 Turbine Temperature Data Page

This page provides current value data in a bar graph format describing the status of water detection temperatures and shell metal temperatures. Bar values are updated once every three seconds or less. Most bars of the two graphs are distinct temperature pairs. Bar pairs change color and blink when alarm set points are exceeded.

6.3.4.3 LP Turbine Data Page

The LP Turbine Data Page provides current value data in a bar graph format describing the status of the LP turbine temperatures. Bar values are updated once every three seconds or less.

6.3.4.4 Cooling Gas Temperatures Page

This page shows a plan view of the generator, and collector and exciter components. Outlined in the plan view are the various components associated with the cooling system.

6.0 OPERATION (Continued)

6.3.4.5 Stator RTD And TC Temperatures Page

This page shows a listing of stator liquid (TC) and bar (RTD) temperatures associated with the generator. This listing changes depending on the particular generator that it reflects. Three main variables are considered: 1) If GENERREX* Control, then P-bar liquid temperatures are shown, 2) The number of slots in the generator, and 3) if the unit has liquid cooled bushings or liquid cooled connection rings.

When a temperature approaches an alarm value, the background around the temperature will change to yellow. When the alarm value is reached, the background will change to red but no flashing occurs.

6.3.4.6 Stator Water System Data Page

The Stator Water System Data page shows a schematic type diagram of the stator water pumping system. The numerical and status data is normally green and changes to yellow and red appropriately. There are two pumping units, and either may be operated as the primary with the other as a backup. The pump which is selected and in operation will be displayed in red. If the primary pump stops and the'backup pump starts, then the symbols will change colors and the words "Reserve Pump Running" will appear in red. If stator winding cooling water is lost, and a turbine runback is initiated, "SCW Protection - Trip" will appear in red at the left side of the screen, as opposed to "SCW Protection - Normal".

6.3.4.7 Excitation System Data Page

This page shows data relating to the excitation system, such as generator field voltage and current, generator field breaker status, power system stabilizer, etc.

6.3.5 TREND DISPLAY

A mix of turbine, generator, and if applicable, boiler feed pump data points, are available for display under the Trend Display Page. up to three parameters from the "bank" are capable of being displayed at one time. Data points are color coded if in the alarm area. All parameters listed under Trend Display have at least one hour's past history stored in memory at all times at a sampling rate of once each minute for each point. Upon making a trend display selection, the entire past hour's trend-curve will come onto the CRT without any appreciable waiting. A legend of parameters available for access and display on the CRT is identified in the Trend Channel Directory Page. Trendable points are also identified as such in the Recorded Data List of the Customer Connection Diagram (pages 02A - 02).

* GENERREX is a trademark of General Electric Company, U. S. A.

6.0 OPERATION (Continued)

6.3.6 TREND CHANNEL DIRECTORY PAGE

The Trend Channel Directory Page contains a listing of all trend points available.

6.3.7 SELECTIVE DATA PAGE

Point data is written on the selective Data Page as accessed from the "Data Table" when selected by the operator through the CONTROL console. Any and all points are displayed upon command on a selectable basis with current value data being displayed. Previously recorded data on the disk is also accessible upon command on a selectable basis. Cancellation of the Data Page is enabled through the "CANCEL" pushbutton. Any displayed point may be cancelled by entering the point number via the key panel, followed by the "DELETE CHANNEL" pushbutton.

6.3.8 ALARM SUMMARY PAGE

This page shows all alarmed parameters which are accessed automatically and displayed on the Alarm Summary Page as they enter the alarm state. The time of the alarm occurrence is recorded and included on the Alarm Page. On those parameters having alarm set points, when the parameter enters the alarm condition, the corresponding set point will also be displayed. Alarm set points are listed in the Manufacturing Instructions, included in the T-GSI COMPOSITE INSTRUCTION BOOK. .

6.3.9 DATA TABLE (LOG) DISPLAY

The points of the Data Table Log are available for review by first selecting either a current or historical log via the control console, then scanning the log at the operator's command. At any point along the table, the operator has the capability of stopping the scan to display the data presently on the CRT. The operator also has the capability to scan in reverse and to scan the table at a fast or slow rate. The pushbuttons for these commands are on the control Console.

6.3.10 ALARM AND TRIP PROVISIONS

Alarm provisions consist of both audible and visual indication. Audible indications are via switch contacts provided in the T-GSI to be wired to the customer's annunciator board and/or operator's console. A summary of these contacts are included in the customer Connection Diagram. One of these contacts, "Non-Acknowledged T-GSI Alarm", is for new alarms and another "Alarm Present in T-GSI System" remains as long as any alarm is present. Visual indication of an alarm state are provided in the following manner:

6.0 OPERATION (Continued)

6.3.10 ALARM AND TRIP PROVISIONS (continued)

1. On the CRT display, the value, symbol, or bar turns yellow in pre-alarm and red on alarm. Simultaneously while turning red, they also flash at a 1/sec. rate until the alarm is acknowledged. The flashing then ceases, but the red color remains until the parameter's value returns to the non alarm condition.

2. On the CRT "Composite Display Page" the word, "ALARM", is written vertically down the CRT page and flashed at a 1/sec. rate.

3. The message, "SEE ALARM PAGE", is written within the last four lines of the CRT page and flashed at a 1/sec. rate.

4. Last, on the "Alarm Summary Page" a statement is written describing the alarmed point.

In reference to the alarm logic, an alarm acknowledge (ACK) button is provided on the Control Console for cancelling the statement "SEE ALARM PAGE"; and for cancelling the flash of the alarm message on the composite page, once the operator has displayed the alarm message page for review and acknowledged the alarm. Additional alarms will still be added to the alarm summary page as they occur.

Tripping provisions consist of one contact which may be actuated by certain alarm conditions specified at the time of ordering. The conditions to be alarmed and which may be specified for tripping are analog vibration and differential expansion.

6.3.11 ALARM LOGIC: TURBINE-GENERATOR SHAFT VIBRATION

In T-GSI, separate alarm contacts for shaft vibration are supplied from the analog and digital sections. The analog setting is set higher to act as a backup to the digital alarm, which is the primary alarm. The analog setting is fixed, whereas the digital setting is a function of speed. The alarm setpoints are given in the Manufacturing Instructions.

6.3.12 SYSTEM MESSAGES

Messages are displayed within the bottom four line spaces of all pages to alert the operator to the following conditions.

6.3.12.1 Display Controller Messages

1. SEE ALARM PAGE
2. CONTROL CONSOLE LINK BAD
3. OUT OF RANGE INPUT

6.0 OPERATION (Continued)

6.3.12.2 Main Controller Messages

1. DTL DOWN = TUR GEN TSI
2. REMOTE SIMULATOR ON

6.3.12.3 Historical Controller Messages - Disk

1. DISK FAULT
2. DISK WRITE PROTECT ON
3. OVERLAP IN TIME
4. DISK DISMOUNTED
5. DISK READY
6. DISK UNIT NOT READY
7. DISK POWER UP INCOMPLETE
8. INVALID HIST. LOG DATE
9. HISTORICAL OFFLINE
10. OVERLAP RECOVERY

6.3.12.4 Historical controller Messages - Tape

1. END OF TAPE NEAR
2. TAPE FULL
3. TAPE REWINDING
4. TAPE HARDWARE ERROR
5. MOUNTED TAPE FAULTY
6. TAPE CONTROLLER ERROR
7. WRITE PROTECTED TAPE
8. TAPE OFFLINE
9. TAPE DISMOUNTED
10. TAPE READY
11. TAPE RECORDING FROM DISK
12. DISK RECORDING FROM TAPE
13. OVERLAP RECOVERY
14. EOT NOT MARKED
15. TAPE DMA OVERRUN
16. MOUNT A LATER TAPE
17. MOUNT AN EARLIER TAPE

6.3.12.5 Lights

If any of the three processors in the Master Station stalls, the operator is alerted by lights on the control console as follows:

1. Display Alert
2. Main Alert
3. Logging Alert

6.0 OPERATION (Continued)

6.3.13 JOURNAL BEARING TEMPERATURE SPIKE MONITOR

The journal bearing temperatures, up to 16 per turbine, and the inlet oil temperature are scanned every 15 seconds when turbine speeds are below 2000 rpm. After each scan, the previous five data readings, which are saved, are examined for a continuous temperature rise,and alarmed if above a limit. Journal bearing temperature data is displayed on the Bearing Group Display Page. The absolute temperature for each bearing is displayed. All bearing temperatures are recorded.

6.3.14 TAPERED LAND THRUST BEARING TEMPERATURE MONITOR

The thrust bearings are segregated between turbine and generator sides. The bearing metal temperature and the inlet oil temperature are then scanned routinely. The highest thrust bearing temperature is established and a thrust bearing temperature difference is calculated for both the turbine and generator sides. The numerical quantities of each are displayed on the Bearing Group Data Page. Appropriate alarms are output and displayed on the Alarm Summary Page. In case of a system alarm, this page displays the word "ALARM" in red, vertically in the center of the screen. The word flashes until the alarm is acknowledged, then remains displayed at reduced intensity until the parameter is again within limits. The four possible alarms are:

Thrust Bearing - Turb. side	Maximum	Abnormal
Thrust Bearing - Turb. side	Differential	Abnormal
Thrust Bearing - Gen. side	Maximum	Abnormal
Thrust Bearing - Gen. side	Differential	Abnormal

6.3.15 RATE OF MAIN STEAM AND REHEAT STEAM TEMPERATURE CHANGE ALARM

Data inputs of main and hot reheat steam (second reheat also, if provided), temperatures are scanned and stored once every thirty seconds. At each scanning period, the temperature decrease, if any, over the past fifteen minutes will be calculated for both inputs. These calculated rates of change, if found to be negative and in magnitude exceeding preset limits, will trigger an alarm relay whose contacts are available to the customer for annunciation. The preset alarm limits are listed in the Manufacturing Instructions.

6.3.16 GENERATOR STATOR TEMPERATURE VARIABLE ALARM LIMIT

For generator operation at light loads, it is desirable to have reduced stator temperature, by bar RTD's and liquid header TC'S, alarm limits from those required at full load operation. To do this, a method has been developed to group similar temperature readings together and give an alarm as a function of both inlet water temperature and armature current.

6.0 OPERATION (Continued)

6.4 OPERATOR'S CONTROL CONSOLE INSTRUCTIONS

6.4.1 INTRODUCTION

This is an operator's user guide with step-by-step instructions of the operations which can be performed using the operator's control console. Following are some general rules for any function. At the beginning of all sequences the console will display the prompt, "ENTER COMMAND". At this point only, the following function keys may be entered:

General Data
Alarm Summary
Composite Data
Data Groups
Read Tape
Selective Data
Trend Directory
Trend Display
Time Set
Current Log
Historical Log Search
Alarm Acknowledge

"Historical Log Next" is valid at this point as well as when historical log is currently being displayed. Alarm Acknowledge is valid ONLY when the Alarm Summary Page is displayed.

6.4.2 CANCELLING A SEQUENCE

A sequence may be cancelled at any point. The cancel may be executed during any time the operator is prompted for the next key or is entering numeric data by simply pressing the cancel button. If cancel is pushed during the time enter sequence, the data already entered up to that point will be ignored. In addition, the cancel button can be used as an abort during a tape read if the operator decides he cannot wait for the process to complete.

6.4.3 ENTERING NUMERIC DATA

Numeric data is requested during several command sequences. There is no character delete key, but the operator can correct mistakes by simply continuing the entry. The extra characters will roll off the display. When the enter key is pressed, the number appearing on the display is entered,

6.0 **OPERATION** (Continued)

6.4.4 OPERATOR ERRORS

Any time a button is pushed out of sequence, the operator control console will display the message: "INVALID PUSHBUTTON". The key is ignored and the last prompt, before the key was pressed, is restored.

6.4.5 PUSHBUTTON OPERATIONS

One button should be pressed at a time and released. If two buttons are pressed simultaneously, they are both ignored. A button must be released before it can be accepted again. Each properly pushed button will cause a new message to appear on the panel, or in the case of numeric data entry, an echo of the data entered.

6.4.6 GENERAL RULES FOR USING THE FOLLOWING SEQUENCES

1. The following sequences can only be started when the Panel Message displayed is "ENTER COMMAND".

2. If "ENTER COMMAND" is not on the display, the user may hit the cancel button to get back to the "ENTER COMMAND" mode.

3. To follow a sequence, start with step 1 and continue to go to the next step unless indicated to go to another step. In some cases, there is a choice of steps to go to.

4. When the response on the panel is "ENTER COMMAND", the sequence is complete, and any other sequence may be started.

5. At certain points, messages will briefly appear on the panel to give the operator the status of the data entered. Therefore, it is important to pay attention to the panel at all times during a sequence. If the operator accidentally pushes the wrong button, the nature of the status message will help him to recognize his mistake.

6.4.7 SEQUENCE TO DISPLAY A GENERAL DATA PAGE

STEP	ACTIONS
1.	Press "GENERAL DATA" pushbutton - Response on Panel - "GENERAL DATA ENTER OR CANCEL" - Go to 2a or 2b
2a.	Press "ENTER" pushbutton - Response on Panel - "GENERAL DATA ENTERED" "ENTER COMMAND" - Response on CRT - General Data Page Displayed

6.0 OPERATION (Continued)

6.4.7 SEQUENCE TO DISPLAY A GENERAL DATA PAGE (continued)

STEP	ACTIONS
OR	
2b.	Press "CANCEL" pushbutton
	- Response on Panel - "GENERAL DATA CANCELLED" "ENTER COMMAND"

6.4.8 SEQUENCE TO DISPLAY AN ALARM SUMMARY PAGE

STEP	ACTIONS
1.	Press "ALARM SUMMARY" pushbutton
	- Response on Panel - "ALARM SUMMARY ENTER OR CANCEL"
	- Go to 2a or 2b
2a.	Press "ENTER" pushbutton
	- Response on Panel - "ALARM SUMMARY ENTERED"
	- Response on CRT - "ENTER COMMAND" Alarm Summary Page Displayed
OR	
2b.	Press "CANCEL" pushbutton
	- Response on Panel - "ALARM SUMMARY CANCELLED" "ENTER COMMAND"

6.4.9 SEQUENCE TO DISPLAY COMPOSITE DATA PAGE

STEP	ACTIONS
1.	Press "COMPOSITE DATA" pushbutton
	- Response on Panel - "COMPOSITE DATA ENTER OR CANCEL"
	- Go to 2a or 2b
2a.	Press "ENTER" pushbutton
	- Response on Panel - "COMPOSITE DATA ENTERED" "ENTER COMMAND"
	- Response on CRT - Composite Data Page Displayed
OR	
2b.	Press "CANCEL" pushbutton
	- Response on Panel - "COMPOSITE DATA CANCELLED" "ENTER COMMAND"

6.0 OPERATION (Continued)

6.4.10 SEQUENCE TO DISPLAY DATA GROUP

STEP	ACTIONS
1.	Press "DATA GROUPS" pushbutton
	- Response on panel - "ENTER DATA GROUP PAGE NO."
	- Go to 2a or 2b
2a.	Press "CANCEL" pushbutton
	- Response on panel - "DATA GROUPS CANCELLED" "ENTER COMMAND"
OR	
2b.	Key in page number "XYZ"
	- Response on panel - data is echoed as keyed in
3.	Press "ENTER" pushbutton
	- Response on panel - "DATA GROUP XYZ ENTERED" "ENTER COMMAND"
	- Response on CRT - Data group "XYZ" displayed

6.4.11 SEQUENCE TO ENTER A TIME SET

****
* *
* CAUTION *
* *
****

WHEN SETTING THE TIME, IT IS OF UTMOST IMPORTANCE THAT THE CORRECT **TIME** BE **ENTERED. ALL** RECORDS **WILL BE REFERENCED TO A TIME RESULTING** FROM THE TIME THAT IS ENTERED **ON** THE SYSTEM AT THIS POINT.

STEP	ACTIONS
1.	Press "TIME SET" pushbutton
	If the main is not on line, time cannot be set:
	- Response on panel - "MAIN PROCESSOR OFFLINE/STANDBY" "TIME SET CANCELLED"
OR	
	- Response on panel - "TIME SET ENTER OR CANCEL"
	- **Go to 2a** or 2b

6.0 OPERATION (Continued)

6.4.11 SEQUENCE TO ENTER A TIME SET (continued)

STEP	ACTIONS (continued)
2a.	Press "CANCEL" pushbutton
	- Response on Panel - "TIME SET CANCELLED" "ENTER COMMAND"
OR	
2b.	Press "ENTER" pushbutton
	- Response on panel - "TIME CODES: DST=0 STD=1" "TIME CODE = X ENTER OR CHANGE"
	- Go to Step 3a, 3b or 3c
3a.	Key in Code "0" (To change from standard to daylight time)
	- Response on panel - Data is echoed as keyed in
3b.	Key in Code "1" (To change from daylight to standard time)
	- Response on panel - Data is echoed as keyed in
OR	
4a.	Press "ENTER" pushbutton
	- Response on panel - "DAYLIGHT TIME" "ENTER COMMAND"
4b.	Press "ENTER" pushbutton
	- Response on panel - "STANDARD TIME" "ENTER COMMAND"
OR	
3c.	Press "ENTER" pushbutton
	- Response on panel - "STANDARD TIME" "YEAR = XX ENTER OR CHANGE"
OR	
	- Response on panel - "DAYLIGHT TIME" "YEAR = XX ENTER OR CHANGE"
5.	Key in Year or leave data as is
	- Response on panel- Data is echoed as keyed in
6.	Press "ENTER" pushbutton
	- Response on panel - "MONTH = XX ENTER OR CHANGE"

6.0 **OPERATION** (Continued)

6.4.11 SEQUENCE TO ENTER A TIME SET (continued)

STEP	ACTIONS (continued)	
7.	Key in month or leave data as is	
	- Response on panel -	Data is echoed as keyed in
8.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL DATE ENTERED" "MONTH = XX ENTER OR CHANGE"
	- Go to step 7	
OR		
	- Response on panel -	"DAY = XX ENTER OR CHANGE"
9.	Key in day or leave data as is	
	- Response on panel -	Data is echoed as keyed in
10.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL DATE ENTERED" "DAY = XX ENTER OR CHANGE"
	- Go to Step 9	
OR		
	- Response on Panel -	"HOUR = XX ENTER OR CHANGE"
11.	Key in Hour or leave data as is	
	- Response on Panel -	Data is echoed as keyed in
12.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "HOUR = XX ENTER OR CHANGE"
	- Go to step 11	
OR		
	- Response on panel -	"MIN = XX ENTER OR CANCEL"
13.	Key in minute or leave data as is	
	- Response on panel -	echo data as keyed in
14.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "MIN = XX ENTER OR CHANGE"
	- Go to step 13	
OR		

6.0 OPERATION (Continued)

6.4.11 SEQUENCE TO ENTER A TIME SET (continued)

STEP ACTIONS (continued)

- Response on panel - "SEC = XX ENTER OR CHANGE"

15. Key in seconds or leave data as is
- Response on panel - data is echoed as keyed in

16. Press "ENTER" pushbutton
- Response on panel - "ILLEGAL TIME ENTERED"
"SEC = XX ENTER OR CHANGE"
- Go to step 15

OR

- Response on panel - "DATE= YY/MM/DD TIME= HH/MM/SS"
"ENTER COMMAND"

OR

- Response on panel - "WILL CAUSE HIST OVERLAP"
"UNSEQUENCED LOGS WILL BE DELETED"
"TIME SET ENTER OR CANCEL"

NOTE: since logs are stored on the disk and tape in chronological order, it is very important not to set time back. However, if it is necessary to do this, the Historical controller will execute a recovery to delete files from the disk which are no longer in order. Also, the tape will rewind and search forward to the new log date, and continue recording.

17a. Press "ENTER" pushbutton
- Response on panel - "DATE = YY/MM/DD TIME = HH/MM/SS"
"ENTER COMMAND"

OR

17b. Press "CANCEL" pushbutton
- Response on panel - "TIME SET CANCELLED"
"ENTER COMMAND"

6.4.12 SEQUENCE TO'ALARM ACKNOWLEDGE

STEP ACTIONS

1. check to see Alarm Summary Page Displayed on CRT **(if not,** go to Sequence to Display Alarm Summary).

6.0 **OPERATION** (Continued)

6.4.12 SEQUENCE TO ALARM ACKNOWLEDGE (continued)

STEP	ACTIONS (continued)	
2.	Press "ALARM ACK" pushbutton	
	- Response on Panel	- "ALARM ACK ENTER OR CANCEL"
	- Go to 3a or 3b	
OR		
	- (If failed to do step 1)	
	- Response on Panel	- "ALARM SUMMARY NOT DISPLAYED" "ENTER COMMAND"
	- Go to step 1	
3a.	- Press "CANCEL" pushbutton	
	- Response on Panel	- "ALARM ACK CANCELLED" "ENTER COMMAND"
OR		
3b.	Press "ENTER" pushbutton	
	- Response on Pane1	- "ALARM ACK ENTERED" "ENTER COMMAND"

6.4.13 SEQUENCE TO DISPLAY TREND DIRECTORY

STEP	ACTIONS	
1.	Press "TREND DIRECTORY" pushbutton	
	- Response on Panel	- "TREND DIRECTORY ENTER OR CANCEL"
	- Go to 2a or 2b	
2a.	Press "ENTER" pushbutton	
	- Response on Panel	- "TREND DIRECTORY ENTERED" "ENTER COMMAND"
	- Response on CRT	- Trend directory displayed
OR		
2b.	Press "CANCEL" pushbutton	
	- Response on Pane1	- "TREND DIRECTORY CANCELLED" "ENTER COMMAND"

6.0 **OPERATION** (Continued)

6.4.14 SEQUENCE TO DISPLAY TREND GRAPH

STEP ACTIONS

1. Press "TREND DISPLAY" pushbutton
   - Response on Panel - "TREND DISPLAY ENTER OR CANCEL"
   - Go to 2a or 2b

2a. Press "CANCEL" pushbutton
   - Response on Panel - "TREND DISPLAY INPUT CANCELLED"
   "ENTER COMMAND"

OR

2b. Press "ENTER" pushbutton
   - Response on Panel - "TYPE CHAN NO. ENTER OR DELETE"
   - Response on CRT - Trend Graph already entered appears
   - Go to 3a or 3b

3a. Enter "CANCEL" pushbutton
(Normal termination when no changes in trend graph are desired.)
   - Response on Panel - "TREND DISPLAY INPUT CANCELLED"
   "ENTER COMMAND"

OR

3b. Enter channel number "XYZ"
   - Response on panel - echoes data entered in
   - Go to 4a, 4b, or 4c

4a. Press "ENTER" pushbutton
   - Response on Panel - "CHAN NO. XYZ ENTERED"
   "TYPE CHAN NO. ENTER OR DELETE"
   - Response on CRT - Trend graph for channel XYZ appears
   - Go to Step 3b or 4c

OR

   - Response on Panel - "MAXIMUM CHANNELS ENTERED"
   "TYPE CHAN NO. ENTER OR DELETE"
   - Go to Step 3b or 4c

OR

   - Response on Panel - "CHAN NO. XYZ ILLEGAL"
   "TYPE CHAN NO. ENTER OR DELETE"
   - Go to Step 3b or 4c

6.0 OPERATION (Continued)

6.4.14 SEQUENCE TO DISPLAY TREND GRAPH (continued)

STEP	ACTIONS (continued)	
4b.	Press "DELETE CHANNEL" pushbutton	
	- Response on Panel -	"CHAN NO. XYZ DELETED" "TYPE CHAN NO. ENTER OR DELETE"
	- Response on CRT -	Trend Graph for Channel XYZ Disappears
	- Go to Step 3b or 4c	
OR		
	- Response on Panel -	"CHANNEL NO. XYZ NOT DISPLAYED" "TYPE CHAN NO. ENTER OR DELETE"
	- Go to Step 3b or 4c	
OR		
	- Response on Panel -	"CHANNEL NO. XYZ ILLEGAL" "TYPE CHAN NO. ENTER OR DELETE"
	- Go to Step 3b or 4c	
OR		
4c.	Press "CANCEL" pushbutton	
	- Response on Panel -	"TREND DISPLAY INPUT CANCELLED" "ENTER COMMAND"

6.4.15 SEQUENCE TO DISPLAY SELECTIVE DATA

STEP	ACTIONS	
1.	Press "SELECTIVE DATA" pushbutton	
	- Response on Panel -	"SELECTIVE DATA HIST. OR CURRENT"
	- Go to 2a or 2b	
2a.	Press "CURRENT LOG" pushbutton	
	- Response on Panel -	"CURRENT SEL. DATA ENTER OR CANCEL"
	- Go to 3a or 3b	
OR		
2b.	Press "HISTORICAL LOG SEARCH" pushbutton	
	- Response on Panel -	"HIST. SEL. DATA ENTER OR CANCEL"
	- Go to 3a or 3b	
3a.	Press "CANCEL" pushbutton	
	- Response on Panel -	"SELECTIVE DATA INPUT CANCELLED" "ENTER COMMAND"

6.0 OPERATION (Continued)

6.4.15 SEQUENCE TO DISPLAY SELECTIVE DATA (continued1

OR

3b. Press "ENTER" pushbutton
- Response on Panel - "TYPE CHAN NO. ENTER OR DELETE"
- Response on CRT - Selective data already entered appears
- go to 4a or 4b

4a. Enter "CANCEL" pushbutton
(Normal termination when no changes in selective data are desired.)
- Response on Panel - "SELECTIVE DATA INPUT CANCELLED"
"ENTER COMMAND"

OR

4b. Enter channel number XYZ
- Response on Panel - echoes data entered in
- Go to 5a, 5b, or 5c

5a. Press "ENTER" pushbutton
- Response on Panel - "CHANNEL NO. XYZ ENTERED"
"TYPE CHAN NO. ENTER OR DELETE"
- Response on CRT - Selective data for channel XYZ appears
- Go to Step 4b or 5c

OR

- "CHANNEL NO. XYZ ALREADY ENTERED"
- "TYPE CHAN NO. ENTER OR DELETE"

OR

- Response on Panel - "MAXIMUM CHANNELS ENTERED"
"TYPE CHAN NO. ENTER OR DELETE"
- Go to Step 4b or 5c

OR

- Response on Panel - "CHANNEL NO. XYZ ILLEGAL"
"TYPE CHAN NO. ENTER OR DELETE"
- Go to Step 4b or 5c

6.0 OPERATION (Continued)

6.4.15 SEQUENCE TO DISPLAY SELECTIVE DATA (continued)

STEP	ACTIONS (continued)
	OR
5b.	Press "DELETE CHANNEL" pushbutton
	- Response on Panel - "CHANNEL NO. XYZ DELETED" "TYPE CHAN NO. ENTER OR DELETE"
	- Response on CRT - Selective data for channel XYZ disappears
	- Go to step 4b or 5c
	OR
	- Response on Panel - "CHANNEL NO. XYZ NOT DISPLAYED" "TYPE CHAN NO. ENTER OR DELETE"
	- Go to Step 4b or 5c
	OR
	- Response on Panel - "CHANNEL NO. XYZ ILLEGAL" "TYPE CHAN NO. ENTER OR DELETE"
	- Go to Step 4b or 5c
5c.	Press "CANCEL" pushbutton
	- Response on Panel - "SELECTIVE DATA INPUT CANCELLED" "ENTER COMMAND"

6.4.16 SEQUENCE TO DISPLAY CURRENT LOG

STEP	ACTIONS
1.	Press "CURRENT LOG" pushbutton
	- Response on Panel - "CURRENT LOG ENTER OR CANCEL"
	- Go to 2a or 2b
2a.	Press "CANCEL" pushbutton
	- Response on Panel - "CURRENT LOG CANCELLED" "ENTER COMMAND"
	OR
2b.	Press "ENTER" pushbutton
	- Response on Panel - "ENTER SCAN COMMANDS"
	- Response on CRT - Current Log appears
	- Go to 3a, 3b, 3c or 3d

6.0 OPERATION (Continued)

6.4.16 SEQUENCE TO DISPLAY CURRENT LOG (continued)

STEP	ACTIONS (continued)
3a.	Press "SCAN LOG FWD" pushbutton
- Response on CRT	- Log Scrolls forward or, if already scrolling, stops
- Go to Step 3a, 3b, 3c or 3d	
OR	
3b.	Press "SCAN LOG REV" pushbutton
- Response on CRT	- Log scrolls in reverse or, if already scrolling, stops
- Go to step 3a, 3b, 3c or 3d	
OR	
3c.	Press "FAST" pushbutton
- Response on CRT	- Log scrolls fast or, if already scrolling fast, scrolls slow
- Go to step 3a, 3b, 3c or 3d	
OR	
3d.	Press "CANCEL" pushbutton
- Response on Panel	- "CURRENT LOG CANCELLED" "ENTER COMMAND"
- Response on CRT	- If log is scrolling, it stops

6.4.17 SEQUENCE TO DISPLAY AN HISTORICAL LOG

STEP	ACTIONS
1.	Press "HIST LOG SEARCH" pushbutton
- Response on Panel	- "HIST LOG SEARCH ENTER OR CANCEL"
- Go to 2a or 2b	
OR	
- Response on Panel	- "HISTORICAL OFFLINE/STANDBY" "HIST LOG SEARCH INPUT CANCELLED" "ENTER COMMAND"
2a.	Press "CANCEL" pushbutton
- Response on Panel	- "HIST LOG SEARCH INPUT CANCELLED" "ENTER COMMAND"

6.0 OPERATION (Continued)

6.4.17 SEQUENCE TO DISPLAY AN HISTORICAL LOG (continued)

STEP	ACTIONS (continued)
OR	
2b.	Press "ENTER" pushbutton
	- Response on panel - "TIME CODES: DST=0 STD=1" "TIME CODE = X ENTER OR CHANGE"
	- Go to Step 3a or 3b
3a.	Key in Code "0"
	- Response on panel - Data is echoed as keyed in
4a.	Press "ENTER" pushbutton
	- Response on panel - "DAYLIGHT TIME" "YEAR = XX ENTER OR CHANGE"
OR	
3b.	Key in Code "1"
	- Response on panel - Data is echoed as keyed in
4b.	Press "ENTER" pushbutton
	- Response on panel - "STANDARD TIME" "YEAR = XX ENTER OR CHANGE"
5.	Key in Year or leave data as is
	- Response on panel - Data is echoed as keyed in
6.	Press "ENTER" pushbutton
	- Response on panel - "MONTH = XX ENTER OR CHANGE"
7.	Key in month or leave data as is
	- Response on panel - Data is echoed as keyed in
8.	Press "ENTER" pushbutton
	- Response on panel - "ILLEGAL DATE ENTERED" "MONTH = XX ENTER OR CHANGE"
	- Go to step 7
OR	
	- Response on panel - "DAY = XX ENTER OR CHANGE"
9.	Key in day or leave data as is
	- Response on panel - Data is echoed as keyed in

6.0 OPERATION (Continued)

6.4.17 SEQUENCE TO DISPLAY AN HISTORICAL LOG (continued)

STEP	ACTIONS (continued)	
10.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL DATE ENTERED" "DAY = XX ENTER OR CHANGE"
	- Go to Step 9	
OR		
	- Response on Panel -	"HOUR = XX ENTER OR CHANGE"
11.	Key in Hour or leave data as is	
	- Response on Panel -	Data is echoed as keyed in
12.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "HOUR = XX ENTER OR CHANGE"
	- Go to step 11	
OR		
	- Response on panel -	"MIN = XX ENTER OR CANCEL"
13.	Key in minute or leave data as is	
	- Response on panel -	echo data as keyed in
14.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "MIN = XX ENTER OR CHANGE"
	- Go to step 13	
OR		
	- Response on panel -	"SEC = XX ENTER OR CHANGE"
15.	Key in seconds or leave data as is	
	- Response on panel	data is echoed as keyed in
16.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "SEC = XX ENTER OR CHANGE"
	- Go to step 15	
OR		

6.0 OPERATION (Continued)

6.4.17 SEQUENCE TO DISPLAY AN HISTORICAL LOG (continued)

STEP ACTIONS (continued)

- Response on panel - "DATE= YY/MM/DD TIME= HH/MM/SS"
  "SEARCH IN PROGRESS"

17. Wait for Log Search

- Response on Panel - "LOG FOUND"
  "HIST LOG SEARCH ENTER NEXT/SCAN"
- Response on CRT - Hist log displayed
- Go to 2a, 18a, 18b, 18c or 18d

OR

- Response on Panel - "LOG NOT FOUND"
  "HIST LOG SEARCH ENTER NEXT/SCAN"
- Response on CRT - Remains the same.
- Go to 2a, 18a, 18b, 18c or 18d

OR

- Response on Panel - "LOG SEARCH INOPERATIVE"
  "HIST LOG SEARCH INPUT CANCELLED"
  "ENTER COMMAND"

OR

- Response on Panel - "TIME OR DATE NOT ON DISK"
  "CURRENT SEARCH WINDOW IS:"
  "YY/MM/DDHH:MM:SS YY/MM/DDHH:MM:SS"
  "YY/MM/DDHH:MM:SS YY/MM/DDHH:MM:SS"
  'HIST LOG SEARCH INPUT CANCELLED"
  "ENTER COMMAND"

18a. Press "HIST LOG NEXT" pushbutton

- Response on Panel - "SEARCH IN PROGRESS"
- Go to 17

18b. Press "SCAN LOG FWD" pushbutton

- Response on CRT - Log scrolls forwards or if already scrolling stops.
- Go to 18a, 18b, 18c, 18d or 18e

18c. Press "FAST" pushbutton

- Response on CRT - Log scrolls fast or if already scrolling fast, scrolls slow
- Go to 18a, 18b, 18c, 18d or 18e

6.0 OPERATION (Continued)

6.4.17 SEQUENCE TO DISPLAY AN HISTORICAL LOG (continued)

STEP ACTIONS (continued)

18d. Press "SCAN LOG REV" pushbutton
- Response on CRT - Log Scrolls in reverse or if already scrolling, stops.
- Go to 18a, 18b, 18c, 18d or 18e

18e. Press "CANCEL" pushbutton
- Response on CRT - Log stops scrolling
- Response on panel - "HIST LOG SEARCH INPUT CANCELLED"
"ENTER COMMAND"

6.4.18 SEQUENCE TO READ TAPE

NOTE: REFER TO THE HISTORICAL DATA STORAGE INSTRUCTIONS FOR THE SEQUENCE TO MOUNT A TAPE TO READ.

STEP ACTIONS

1. Press "TAPE READ" pushbutton
- Response on Panel - "TAPE READ ENTER OR CANCEL"
- Go to 2a or 2b

OR

- Response on Panel - "HISTORICAL OFFLINE"
"TAPE READ CANCELLED"
"ENTER COMMAND"

2a. Press "CANCEL" pushbutton
- Response on Panel - "TAPE READ CANCELLED"
"ENTER COMMAND"

2b. Press "ENTER" pushbutton
- Response on panel - "TIME CODES: DST=0 STD=1"
"TIME CODE = X ENTER OR CHANGE"
- Go to Step 3a or 3b

3a. Key in Code "0"
- Response on panel - Data is echoed as keyed in

4a. Press "ENTER" pushbutton
- Response on panel - "DAYLIGHT TIME"
"YEAR = XX ENTER OR CHANGE"

OR

6.0 **OPERATION** (Continued)

6.4.18 SEQUENCE TO READ TAPE (continued)

STEP	ACTIONS (continued)	
3b.	Key in Code "1"	
	- Response on panel -	Data is echoed as keyed in
4b.	Press "ENTER" pushbutton	
	- Response on panel -	"STANDARD TIME" "YEAR = XX ENTER OR CHANGE"
5.	Key in Year of leave data as is	
	- Response on panel -	Data echoed as keyed in
6.	Press "ENTER" pushbutton	
	- Response on panel -	"MONTH = XX ENTER OR CHANGE"
7.	Key in month or leave data as is	
	- Response on panel -	Data is echoed as keyed in
8.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL DATE ENTERED" "MONTH = XX ENTER OR CHANGE"
	- Go to step 7	
OR		
	- Response on panel -	"DAY = XX ENTER OR CHANGE"
9.	Key in day or leave data as is	
	- Response on panel -	Data is echoed as keyed in
10.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL DATE ENTERED" "DAY = XX ENTER OR CHANGE"
	- Go to Step 9	
OR		
	- Response on Panel -	"HOUR = XX ENTER OR CHANGE"
11.	Key in Hour or leave data as is	
	- Response on Panel -	Data is echoed as keyed in
12.	Press "ENTER" pushbutton	
	- Response on panel -	"ILLEGAL TIME ENTERED" "HOUR = XX ENTER OR CHANGE"
	- Go to step 11	

6.0 OPERATION (Continued)
6.4.18 SEQUENCE TO READ TAPE (continued)

OR

STEP ACTIONS (continued)

- Response on panel - "MIN = XX ENTER OR CANCEL"

13. Key in minute or leave data as is
- Response on panel - Data echoed as keyed in

14. Press "ENTER" pushbutton
- Response on panel - "ILLEGAL TIME ENTERED"
"MIN = XX ENTER OR CHANGE"
- Go to step 13

OR

- Response on panel - "SEC = XX ENTER OR CHANGE"

15. Key in seconds or leave data as is
- Response on panel - data is echoed as keyed in

16. Press "ENTER" pushbutton
- Response on panel - "ILLEGAL TIME ENTERED"
"SEC = XX ENTER OR CHANGE"
- Go to step 15

OR

- Response on panel - "DATE = YY/MM/DD TIME = HH/MM/SS"
- "TAPE READ IN PROGRESS"
- GO to 17a or 17b

17a. Wait for tape read
- Response on Panel - "TAPE READ COMPLETE"
- "ENTER COMMAND"

OR

- Response on Panel - "TAPE INOPERATIVE"
- "TAPE READ ABORTED"
- "ENTER COMMAND"

OR

- Response on Panel - "TAPE INOPERATIVE"
- "CHECK DIAGNOSTIC PANEL SWITCHES"
- "TAPE READ ABORTED"
- "ENTER COMMAND"

6.0 OPERATION (Continued)

6.4.18 SEQUENCE TO READ TAPE (continued)

STEP	ACTIONS (continued)	
OR		
	- Response on Panel -	"DISK INOPERATIVE" "TAPE READ ABORTED" "ENTER COMMAND"
OR		
	- Response on Panel -	"DATES ON TAPE EARLIER THAN REQ" "TAPE READ ABORTED "ENTER COMMAND"
OR		
	- Response on Panel -	"DATES ON TAPE LATER THAN REQ" "TAPE READ ABORTED "ENTER COMMAND"
OR		
	- Response on Panel -	"MORE LOGS MAY BE ON PREVIOUS TAPE" "TAPE READ COMPLETE" "ENTER COMMAND"
OR		
	- Response on Panel -	"DISK DIRECTORY FULL" "TAPE READ COMPLETE" "ENTER COMMAND"
OR		
17b.	Press "CANCEL" pushbutton	
	- Response on Panel -	"TAPE READ CANCELLED" "ENTER COMMAND"

6.4.19 SEQUENCE TO DISPLAY NEXT HISTORICAL LOG

STEP	ACTIONS	
1.	Press "HIST LOG NEXT" pushbutton	
	- Response on Panel -	"HIST LOG NEXT ENTER OR CANCEL"
	- Go to 2a or 2b	

6.0 **OPERATION** (Continued)

6.4.19 SEQUENCE **To** DISPLAY NEXT HISTORICAL LOG (continued)

STEP ACTIONS (continued)

OR

- Response on Panel - "HISTORICAL OFFLINE/STANDBY"
  "HIST LOG NEXT INPUT CANCELLED"
  "ENTER COMMAND"

2a. Press "CANCEL" pushbutton
- Response on Panel - "HIST LOG NEXT INPUT CANCELLED"
  "ENTER COMMAND"

OR

2b. Press "ENTER" pushbutton
- Response on Panel - "SEARCH IN PROGRESS"

3. Wait for Log Search
- Response on Panel - "LOG NOT FOUND"
  "HIST LOG SEARCH ENTER NEXT/SCAN"
- Go to 2a, 4a, 4b, 4c or 4d

OR

- Response on Panel - "LOG SEARCH INOPERATIVE"
  "HIST LOG CANCELLED"
  "ENTER COMMAND"

OR

- Response on Panel - "LOG FOUND"
  "HIST LOG SEARCH ENTER NEXT/SCAN"
- Response on CRT - Hist log displayed
- Go to 2a, 4a, 4b, 4c or 4d

4a. Press "HIST LOG NEXT" pushbutton
- Response on Panel - "SEARCH IN PROGRESS"
- Go to 3

4b. Press "SCAN LOG FWD" pushbutton
- Response on CRT - Log Scrolls forward or if already scrolling, stops
- Go to 4a, 4b, 4c, 4d or 4e

6.0 OPERATION (Continued)

6.4.19 SEQUENCE TO DISPLAY NEXT HISTORICAL LOG (continued)

STEP ACTIONS (continued)

4c. Press "SCAN LOG REV" pushbutton
- Response on CRT - Log scrolls in reverse or, if already scrolling stops
- Go to 4a, 4b, 4c, 4d or 4e

4d. Press "FAST" pushbutton
- Response on CRT - Log scrolls fast or, if already scrolling fast, scrolls slow
- Go to 4a, 4b, 4c, 4d or 4e

4e. Press "CANCEL" pushbutton
- Response on CRT - Log stops scrolling
- Response on Panel - "HIST LOG SEARCH INPUT CANCELLED"
- "ENTER COMMAND"

6.4.20 SEQUENCE TO PERFORM LED TEST ON PANEL

STEP ACTIONS

1. Press "-" pushbutton
- Response on Panel - "LED TEST ENTER OR CANCEL"
- Go to 2a or 2b

2a. Press "ENTER" pushbutton
- Response on Panel - "[[[

6.0 OPERATION (Continued)

6.5 TELEPRINTER INSTRUCTIONS

6.5.1 INTRODUCTION

T-GSI is equipped for the use of a terminal, referred to here as the teleprinter, at which system logs may be listed for examination or archiving. The teleprinter is an auxiliary to T-GSI's primary information source, the color CRT, and may be used while T-GSI is on-line, concurrent with tasks that are being performed by the Control console. For example, while current scan information is being displayed on the CRT, tape reads and historical searches may be executed using the teleprinter, or vice versa. Also, both devices may display either current or historical information at the same time.

The teleprinter port is intended for a hard copy device, but any buffered terminal compatible with the RS232C communications interface will suffice; for example, a video terminal. For connection procedure details, refer to the installation section of this book and the Customer Connection Diagram supplied for the system.

6.5.2 OPERATING INSTRUCTIONS

6.5.2.1 General Information

The T-GSI teleprinter is an interactive device; system logs are obtained ONLY by entering a request at the terminal itself. In other words, the teleprinter does not automatically or periodically log data; neither is it instructed to do so from the control console or any other part of the system. After a prompt (*) is received, the user may enter commands at the teleprinter keyboard to request current, historical or alarm listings as described in the following sections. If the prompt is not present on the terminal display, hitting a carriage return or CONTROL-Y should revive it. If not, consult the troubleshooting guide for the Display Controller.

Note: A "*" prompt must be present just prior to entering a valid input command or a command error will occur.

6.5.2.2 Edit Functions

The teleprinter provides text editing functions for correcting errors entered within a command string. Each function is listed below with a brief description.

NOTE: In the examples below, the asterisk symbol (*) is the teleprinter prompt (not typed by the user) and the symbol [CUR] represents the cursor position at the end of sequence shown.

6.0 OPERATION (Continued)

6.5.2.2 Edit Functions (continued)

A. DEL/RUBOUT

Removes the previous character from the message and echoes the deleted character to the terminal.

Example:

```
Typed   - * HEK(DEL)LP
Echoed  - * HEKRLP [CUR]
```

B. CONTROL-R

Causes all characters currently in the message to be transmitted to the terminal on a new line. This function is convenient after using DEL/RUBOUT because it shows the state of the message after the editing.

Example:

```
Typed   - * HEK(DEL)LP(CONTROL-R)
Echoed  -   * HEKKLP
              HELP [CUR]
```

C. CONTROL-X

Deletes the entire message that has been entered so far.

NOTE: This has the same effect as using the DEL/RUBOUT function to remove all the characters from the message.

Example:

```
Typed   -  * HEKL(CONTROL-X)HELP
Echoed  -  * HERL
             HELP [CUR]
```

D. CONTROL-Y

Aborts the current command sequence and returns the teleprinter prompt.

NOTE: CONTROL-Y may be entered at any time to abort the current process - even during the listing of a log at the terminal.

Example:

```
Typed   - * HE(CONTROL-Y)
Echoed  -   * HE ^ Y
        - * [CUR]
```

6.0 OPERATION (continued)

6.5.2.3 command Format

The general teleprinter command formats are:

(Hist. Log and Tape Read)
(mnemonic) (date or date/time) [(subclass)]

(Other Log commands)
(mnemonic) [(subclass)]

(Set command)
SET (parameter) (number)

(Alarm command)
AL(ARM)

(Channel commands)

ADD/DELETE [(channel no.),(channel no.),...]

(Help command)
H(ELP) [(command mnemonic)]

A description of each command and its particular syntax is included in section 3.

The (subclass) parameter, when appended to a log request command, will cause only the T-GSI channels included in that subclass to be printed at the terminal. (See the Appendix for a list of the available subclass groupings.) Also, once a subclass has been specified, that same subclass will be assumed on subsequent requests until a different one is specified. When no subclass has yet been specified, for example, after a system reset, subclass "A" is assumed.

For Historical Log requests, the date and time of the log desired must be entered as part of the command string. The format is shown below:

(date/time) :== (year)/(month)/(day) (hour):(minute):(second)

For Tape Read requests, only the date is required. The format for that is:

(date) :== (year)/(month)/(day)

6.0 OPERATION (Continued)

6.5.3 COMMAND DESCRIPTIONS

6.5.3.1 Current Log - CL [(subclass)]

The current state of the T-GSI inputs is captured in a 'snapshot' fashion and printed at the terminal in a tabular format. The subclass parameter is optional.

6.5.3.2 Historical Log - HL (date/time) [(subclass) 1

This command sequence will initiate a search of the information currently stored on the T-GSI disk. The log returned and subsequently printed out will be the log CLOSEST to the time and date requested. If the exact log requested exists, of course, it will be returned.

If the historical search is unsuccessful, a message will be printed at the teleprinter terminal indicating the nature of the problem. An error in the search sequence could be caused by one of the following conditions:

1. The time and date requested is not within the range of the logs stored on the disk.

2. The Historical Controller interface is currently being used by the T-GSI control console for a Tape Read or Historical Search.

3. The historical disk hardware is not functioning properly.

4. The Historical Controller did not respond to the request for a search.

6.5.3.3 Next Log - NL [(subclass11

This command will fetch and print out the log on the disk immediately following the log most recently fetched by the Historical Log command. The time difference between the two logs depends on the historical logging rate at the time the logs were recorded. This rate is variable and depends on system alarm conditions. The same rules and conditions apply to the **Next** Log command as listed for the Historical Log command above.

6.5.3.4 Previous Log - PL [(subclass)]

To fetch and print the log immediately previous to the log most recently fetched by the Historical Log command, this command should be used. The time difference between the two logs depends on the historical logging rate at the time the logs were recorded. This rate is variable and depends on system alarm conditions. The same rules and conditions apply to the Previous Log command as listed for the Historical Log command above.

6.0 OPERATION (Continued)

6.5.3.5 Same Log - SL [(subclass)]

If another listing is desired from the log most recently accessed, whether it be current or historical, the Same Log command may be used. This function is especially useful when two or more different subclass listings are needed from the same log. Execution of this command will neither initiate a historical search nor take a new snapshot of the current scan information; it simply uses the data already in memory from the last time a log was requested.

6.5.3.6 Tape Read - TR (date)

The user may also initiate a 'tape read to disk' sequence from the teleprinter. When the command is executed, the tape is searched for the (date) entered, and then historical logs for approximately three days, the date specified plus one day on either side, will be read from the tape to the disk. This information will then be available for access by the historical log commands.

Tape Read, of course, is intended to be used to read in archived tapes, since any information on the tape currently being recorded should also be in the current disk window, unless the disk had been down for maintenance. Before executing the Tape Read command, then, the archive tape, if applicable, must be mounted on the drive according to the procedures in the T-GSI Historical Data Storage Instructions. After dumping the desired information to the disk for access by the teleprinter, the old tape may be removed and the current tape replaced on the drive so that it may continue recording.

NOTE: Tape information is placed on the disk in an area reserved for this purpose. This is the same area used by the Tape Read command run from the Control Console. Hence, only the tape information from the most recent Tape Read will be available on the disk. Previous information is overwritten. However, the teleprinter's historical log fetch commands may be used to search for and print information placed on the disk by a Control Console initiated Tape Read: and conversely, information read from the tape by the teleprinter may be placed on the CRT historical data displays.

6.5.3.7 Set - SET BAUD/COLUMNS/LINES (number)

SET may be used to modify one of the teleprinter's variable parameters - namely printer baud rate, the number of columns in the listing, or the number of lines between headers in the listing. Each of these is described briefly below.

6.0 OPERATION (Continued1

6.5.3.7 Set - SET BAUD/COLUMNS/LINES (number) (Continued)

SET B(AUD) (number)

Resets the serial data transmission rate of the communication link between T-GSI and the teleprinter terminal. The parameter (number) may be any decimal number up to 9600. The default value, at power-up or after a system reset, is either 300 or 1200 baud. To change the terminal baud rate, or to use another terminal with a different rate, enter the SET BAUD command with the terminal set to the default baud rate and then change the terminal baud rate setting or the terminal.

SET C(OLUMNS) (number)

Changes the number of columns in the output listing from its present value to (number), where number = 1,2,3 or 4. This facilitates the use of a variety of terminals, from very narrow to very wide carriage types. The default number of columns is two.

SET L(INES) (number)

Changes the number of lines printed between column headers in the output listing from its present value to (number), where number = 1,2,3,...255). This flexibility, along with the variable columns described above, allows the user to choose the appearance of the output listing. The default number of lines between headers is ten.

6.5.3.8 Alarm - AL(ARM)

A list of all channels currently being retained by T-GSI as unacknowledged or uncleared alarms may be listed at the teleprinter. The listing format is like that of the Alarm Summary display on the CRT, including the time of the alarm's occurrence, the channel's identifier, the current input value and the alarm setpoint.

6.5.3.9 Add Channel - AC [(channel No.),(channel No.),...]

T-GSI channels may be added to subclass Z using this command. The maximum number of channels in subclass Z is 34.

6.5.3.10 Delete Channel - DC [(channel No.),(channel No.),...]

T-GSI channels may be deleted from subclass Z using this command. If an '*' is typed as one of the channel numbers, then all channels will be deleted from subclass Z.

6.0 OPERATION (Continued)

6.5.3.11 Help - H(ELP) [(mnemonic)]

As an aid to the user, a brief description of each teleprinter command, along with its correct syntax, may be obtained at the terminal by entering a HELP request. Typing Only HELP will print out a list of available commands; to obtain specific information on a command, the string HELP (mnemonic) may be entered, where mnemonic is any valid teleprinter command or the word SUBCLASSES. A brief message describing the command will then be printed at the terminal. For HELP SUBCLASSES, the list of available subclasses will be printed,

6.0 OPERATION (Continued)

APPENDIX

INFORMATION SUBCLASSES

A. ALL CHANNELS IN ORDER

B. ALL THE BELOW SUBCLASSES (C-P) IN ONE COMPOSITE LOG

C. VIBRATION CHANNELS

D. EXHAUST HOOD TEMPERATURES AND PRESSURES

E. WATER DETECTION TEMPERATURES, LP TURBINE WET REVERSE FLOW CONTACTS AND LP TURBINE WATER INDUCTION CONTACTS

F. BEARING METAL, JOURNAL AND THRUST PLATE TEMPERATURES

G. SHELL METAL, STEAM AND STEAM SEAL TEMPERATURES

H. ECCENTRICITIES, EXPANSIONS, SPEEDS AND VALVE POSITIONS

I. TURBINE STATUS CONTACTS

J. GENERATOR LOAD, VOLTAGE, CURRENT AND FREQUENCY - CORE MONITOR LEVEL, FIELD VOLTAGE, CURRENT AND TEMPERATURE, HYDROGEN PRESSURE AND PURITY, AND WATER CONDUCTIVITY

K. GAS TEMPERATURES - COMMON COLD GAS, H2 COOLER COLD AND HOT AND GENERREX TRANSFORMER AND REACTOR COLD

L. GENERATOR LIQUID HEADER OUTLET TEMPERATURES

M. GENERATOR STATOR SLOT AND STATOR COOLING WATER TEMPERATURES

N. GENERATOR STATUS CONTACTS

0. LIQUID COOLED BUSHING TEMPERATURES AND EXCITER TEMPERATURES - RECTIFIER COOLING WATER, ALTERNATOR AND COLLECTOR AIR, ALTERNATOR SLOTS, GENERREX TRANSFORMER AND REACTOR CORES, AND P BAR WATER

P. EXCITER STATUS CONTACTS

Y. ALL CHANNELS ON THE CURRENT SELECTIVE DATA PAGE ON THE CRT

Z. ALL CHANNELS IN THE TELEPRINTER SELECTIVE DATA LIST

6.0 OPERATION (Continued)

6.6 HISTORICAL DATA STORAGE INSTRUCTIONS

6.6.1 HISTORICAL CONTROLLER

6.6.1.1 POWERFAIL PROCEDURE

IF THE SYSTEM POWER FAILS DURING RECORDING OF A TAPE THERE IS NO NEED TO MOUNT A NEW TAPE. LEAVE THE MAINTENANCE MODULE SWITCHES AS THEY ARE. THE TAPE DRIVE WILL RETURN TO ON-LINE AND THE TAPE SOFTWARE WILL DETERMINE WHERE RECORDING LEFT OFF BEFORE THE POWERFAIL, AND CONTINUE FROM THAT POINT.

6.6.2 TAPE FUNCTIONS

6.6.2.1 DISMOUNTING A FULL TAPE

After a tape has recorded 13,500 logs a message will appear on the CRT "END OF TAPE NEAR". This gives the operator a warning so he can find a new tape and remount it before the tape becomes full and stops recording logs. The operator may either dismount the tape now or wait until the tape is completely full.

TO DISMOUNT A FULL TAPE:

Simply wait and observe the CRT message "FULL TAPE". At this point the tape will automatically rewind. The operator will observe that the rewinding is slower than a normal rewind in order to prevent the tape from being wound too tightly for archival storage.

On the maintenance module, turn the tape mount switch to "DISMOUNT".

Refer to the TAPE TRANSPORT MAINTENANCE AND TEST MANUAL for specific instructions concerning the tape unloading procedure.

Remove the tape.

Secure end of tape.

Remove write-protect ring and place in archival storage.

Label tape for later reference. The beginning and ending date can be found by using the label command in the diagnostics prior to remounting a tape.

6.6.2.2 DISMOUNTING A PARTIALLY RECORDED TAPE

The operator may decide to dismount the tape in order to do a tape read, other maintenance to tape drive, or for normal archival storage before the tape is completely full.

6.0 OPERATION (Continued)

6.6.2.2 DISMOUNTING A PARTIALLY RECORDED TAPE (continued)

On the maintenance module turn the tape mount switch to "DISMOUNT". The tape will finish the current log, then write file markers to mark the place in the tape so it may be found again. Now the tape will automatically and slowly rewind.

When the load point indicator comes on, unload the tape.

Remove the tape.

If the operator plans to reuse the tape he may set it aside. It should be marked for easy identification, the write-protect ring should be removed, and it should be placed in archival storage.

6.6.2.3 PERFORMING A TAPE READ ON AN HISTORICAL TAPE

When an operator needs to view a past log that is no longer on the main directory on the fixed disk, he must mount an historical tape, and dump those logs onto a second directory on the disk. NOTE: THE DATA ALREADY ON THAT SECOND DIRECTORY WILL BE DELETED.

Make sure the disk mount switch is in "MOUNT" position in order to be able to record to the disk. If not, "DISK FAULT" will be returned; dismount the current tape.

Find the historical tape which contains the date desired to be dumped.

Mount this tape onto the tape drive. NOTE: REMOVE WRITE ENABLE RING IF PRESENT.

Refer to the tape transport maintenance and test manual for the tape load procedure.

On the maintenance module turn the tape switch to "OLD TAPE".

Turn the tape mount switch to "MOUNT".

Enter the sequence on the operator control console or teleprinter to request a tape read of the desired date. The actual dump will contain the day requested and the day before and after the requested day.

Observe **status** messages on CRT and operator control console or teleprinter. If the requested date is found, the CRT will display "DISK RECORDING FROM TAPE". The control console or teleprinter will display "TAPE READ IN PROGRESS". When a dump is completed, the response will be "TAPE READ COMPLETE". If this is being done via teleprinter, the letter **"W"** is output every 4 seconds during the tape read sequence. If the tape

6.0 OPERATION (Continued)

6.6.2.3 PERFORMING A TAPE READ ON AN HISTORICAL TAPE (continued)

read is not completed within 32 minutes, a message "HISTORICAL DID NOT RESPOND" is output and tape read is aborted.

If the switches have not been set to indicate it is an old (historical) tape, the response to the operator will be "CHECK DIAGNOSTIC PANEL SWITCHES".

If all of the requested data is not on the tape mounted, the response will be "MORE LOGS MAY BE ON PREVIOUS TAPE" or "MORE LOGS MAY BE ON NEXT TAPE".

If the operator has mounted the wrong tape, the response will be "DATES ON TAPE LATER THAN REQ." or "DATES ON TAPE EARLIER THAN REQ".

If the tape or disk become dismounted or offline, the response will be "TAPE INOPERATIVE" or "DISK INOPERATIVE", respectively.

When the tape read is complete, either successfully or unsuccessfully, the tape will rewind. The operator may then request another tape read or dismount the tape.

6.6.2.4 TO DISMOUNT AN HISTORICAL TAPE

Turn maintenance module "TAPE MOUNT" switch to "DISMOUNT".

Refer to TAPE TRANSPORT MAINTENANCE AND TEST MANUAL for tape load procedure.

Remove tape and return to archival storage.

If the operator is through performing tape reads, as soon as possible, he should remount the current tape he previously removed.

6.6.2.5 REMOUNTING THE PARTIALLY RECORDED CURRENT TAPE

To prevent the tape from missing very many logs which will need to be recorded from the disk, the current tape should be remounted as soon as possible.

Mount the proper tape, set aside earlier, on the tape drive.

Set maintenance module "TAPE SELECT" switches to "CURRENT", "USED". NOTE: IT IS VERY IMPORTANT TO TURN THE "NEW/USED" SWITCH TO "USED". IF THIS IS NOT DONE, THE TAPE WILL BE TREATED AS A NEW TAPE, AND LOGS OVERWRITTEN.

6.0 OPERATION (continued)

6.6.2.5 REMOUNTING THE PARTIALLY RECORDED CURRENT TAPE (continued)

Set tape mount switch to mount.

Observe the tape search for the end of file markers to find the last log previously recorded. Now the tape will record back logs from disk, then start recording current logs once again.

6.7 ALARM SET POINT INSTRUCTIONS

6.7.1 ALARM LIMIT OPERATION

Integral to the T-GSI system is the ability to modify/display designated alarm limits. The command, called changeable Alarm Limit (CAL), features are:

- Field changeable alarm limits.
- Non-volatile retention of alarm limits.
- On-line modification of alarm limits.
- Elimination of alarm limit switches and pots.

Using a teleprinter connected to the diagnostic port on the HMPF card in the main controller, designated alarm limits can be modified/displayed on-site by entering commands to the T-GSI system. New limits are stored into Electrically Alterable Read Only Memory (EAROM) to ensure limits are saved - even in the event of a power loss. The CAL command sequence eliminates the need for adjusting pots or setting switches to modify/display alarm limits. Moreover, the changing of alarm limits is performed with the T-GSI powered up and on-line. Secondary features of the system allow:

- Displaying of the current channel value and alarm status;
- Resetting of the alarm limits to the factory original settings; and
- Displaying of built-in "HELP" messages to explain all CAL commands.

6.7.2 SET UP CONDITIONS

To be able to modify/display an alarm limit the following conditions must exist:

- The EAROM card (HARA) enabled switch must be "up" and enable LED must be lit.
- A teleprinter must be connected to the Main controller diagnostic port (HMPF card).
- The user must turn on the CAL function by typing in "CAL". The system will respond with the CAL "-" prompt.

6.0 OPERATION (Continued)

6.7.2 SET UP CONDITIONS (continued)

After all of the above has been satisfied the system is ready to accept commands to display/modify alarm limits.

6.7.3 ALARM LIMIT IDENTIFIERS

Different algorithms are used for alarming T-GSI channels. Furthermore, a given alarm method can have multiple limits and a given channel can be checked with more than one alarm method. To uniquely distinguish specific alarm method and limit type, alarm limit identifiers are used. The identifiers are:

LIMIT ID	DESCRIPTION
TSH/R	Two Stage High Alarm/Red Limit
TSH/Y	Two Stage High Alarm/Yellow Limit
TSL/R	Two Stage Low Alarm/Red Limit
TSL/Y	Two Stage Low Alarm/Yellow Limit
JBS/R	Journal Bearing Spike Monitor/Red Limit
JBS/Y	Journal Bearing Spike Monitor/Yellow Limit
VDP/R	variable Differential Pair Alarm/Red Limit
VDP/Y	variable Differential Pair Alarm/Yellow Limit
VLA/R	Variable Limit Alarm/Red Limit
VLA/Y	variable Limit Alarm/Yellow Limit
GLH/R	Generator Load Hold Alarm/Red Limit
GLH/Y	Generator Load Hold Alarm/Yellow Limit
TBP	Turbine By-Pass Alarm
SVB/HR	Shaft Vibration Alarm/High Range-Red Limit
SVB/HY	Shaft vibration Alarm/High Range-Yellow Limit
SVB/LR	Shaft vibration Alarm/Low Range-Red Limit
SVB/LY	shaft Vibration Alarm/Low Range-Yellow Limit

6.7.4 DISPLAY ALARM LIMIT SETTINGS

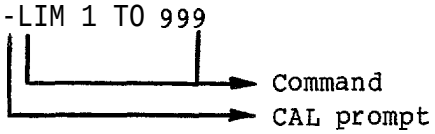
There are three commands available to display alarm limits. They are:

COMMAND	DESCRIPTION
LIM XXX	Display alarm limits for channel XXX.
LIM XXX LEN AA	Display AA channel limits - starting at channel XXX.
LIM XXX TO YYY	Display alarm limits for channel XXX to channel YYY.

As an example, to display all T-GSI limits use the following command:

6.0 OPERATION (Continued)

6.7.4 DISPLAY ALARM LIMIT SETTINGS (continued)

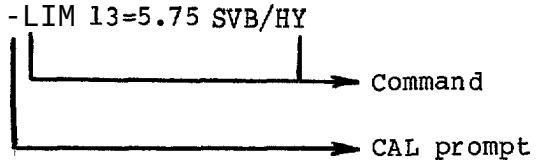


6.7.5 MODIFY ALARM LIMIT SETTINGS

There is a single command to modify a channel alarm limit. Only one limit type for one channel can be changed per command. The general form of the command is:

COMMAND	DESCRIPTION
LIM XX=SAAAA. (ID)	Modify channel xx to the value SAAAA for limit type (ID), where SAAAA., is a signed decimal number and (ID) defines the specific alarm limit to be modified. (See ALARM LIMIT IDENTIFIERS)

As an example, the command used to change the Shaft Vibration High Range yellow alarm limit setting to 5.75 mils for channel 13 is shown below.



NOTE: The HARA card's Enable LED must be on (Front edge switch in up poisition) before emtering a change command. The card's "active" LED WILL BLINK WHEN THE LIMIT IS CHANGED. If the card is inoperative an error message will be returned.

6.7.6 RESETTING ALARM LIMIT SETTINGS

The original factory setting for ALL alarm settings can be regenerated with the following command:

CAUTION: This operation must be performed with the Main Controller in the OFF LINE mode.

NOTE: This command will destroy all field set alarm limits. The command takes approximately four minutes to complete.

COMMAND	DESCRIPTION
RESET	Resets all alarm limits to factory settings

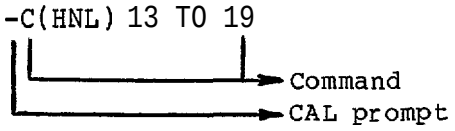
6.0 OPERATION (Continued)

6.7.7 DISPLAY CHANNEL VALUE

The current value and status of a channel(s) can be displayed with the following commands:

COMMAND	DESCRIPTION
C(HNL) XXX	Display channel XXX value and status
C(HNL) XXX LEN AA	Display AA channel limits and status starting with channel XXX.
C(HNL) XXX TO YYY	Display channel XXX to channel YYY value and status.

As an example, the command to display the value and status of channels 13 to 19 is displayed below.



6.7.8 MISCELLANEOUS ALARM LIMITS COMMANDS

The operation of the CAL system is internally documented and is available by a series of "help" messages. The CAL system is turned off with the CAL EXIT command. The HELP and EXIT commands are listed below.

COMMAND	DESCRIPTION
EX(IT)	Exits the CAL system
H(ELP)	Lists all CAL commands only
H(ELP) LIM	Describes the limit modify and display command
H(ELP) C(HNL)	Describes the channel display command
H(ELP) RESET	Describes the CAL RESET command
H(ELP) EX(IT)	Describes the CAL EXIT command

7.0 MAINTENANCE

7.1 SYSTEM OVERVIEW

The flow of information through the T-GSI begins with the Remote Station continuously scanning its inputs. The inputs are transferred to the Master Station as a result of the Main controller scanning the Remote Station. When the last input is received the Main Controller alarm checks the received values and tags the buffer with the current system time/date. The contents of the buffer, called a "system log", is transferred to the Display controller for operator display, and passed to the Historical Controller at predetermined intervals for permanent storage. Upon request from an external device, e.g., a plant computer, a copy of a system log can be obtained via a Plant Computer datalink.

Block Diagrams of the Master and Remotes are given in Figures 19 through 22. These should be helpful when reading through the signal flow paragraphs below,

7.1.1 SIGNAL FLOW

A. Remote Data Flow

The inputs to a remote station can be divided into three categories: contact signals, analog signals, and the speed signal.

Contact signals are fed into a contact input module (CIM). This module applies 105 VDC to the contacts and reads if the contact is open or closed. The state of the contact is then converted to a TTL signal and sent to the HIOD card. The remote controller then reads the HIOD and puts the value of each input in the data log.

There are five types of analog signals used by the remote stations. Not all types are used in each remote. voltage signals are one type. They are input into the NAIB card. Vibration signals are an example of a voltage signal. A second type are current signals. These signals are input into the NAIA cards which convert them into voltage signals. An example of a current input is the signal from the main steam pressure transducer. A third type are thermocouple inputs. These signals are input into one of four card types: NTCA, NTCB, NTCC, or NTCD, depending on the type of thermocouple used. The NTC cards amplify the low voltage signals from the thermocouple to approximately 10V at full scale. A fourth type of input are RTD's (resistance temperature detectors). Depending on the range and type of RTD, this signal is input to one of four cards: NRTA, NRTB, NRTC or NRTD. These cards convert the resistance of the RTD to a voltage signal. All the above voltage signals are converted to digital values by the HAIA cards. This card is read by the remote controller to get these digital values. It then scales and linearizes these values and puts them in the data log. The last type of analog input is in the generator remote station, the complex power module,

7.0 MAINTENANCE (Continued)

7.1.1 SIGNAL FLOW (continued1

A. Remote Data Flow (continued)

This module converts the PT and CT, power and current transformer signals into voltage signals for generator voltage, current, real power and reactive power. These signals are also converted to digital values by the HAIA card. It is then read and scaled by the controller and put into the data log.

The speed signal is generated by teeth on a wheel passing a sensor. This signal is counted by the HPRA card which is then read by the processor. This value is then converted to actual speed in rpm and entered into the data log.

The above data is held in the data log until the Master Controller requests it. Upon a request, the data is moved to the HMHA card which transmits it to the Master Controller.

B. Main-Remote Interface

The creation of a new system log begins with the Main Controller scanning each of the Remote Stations for the current state and value of its respective system inputs. The communication channel used by the Main Controller to scan the Remotes is an SDLC datalink. The datalink, which operates at a link speed of 48K baud, is controlled by an HMHA card. (See Main Controller slot 2B and 2C). An HMHA is required at each of the datalinks. Datalink isolation modules are inserted into the datalinks to provide voltage and noise isolation.

To scan the Remote Stations, the Main Controller simultaneously transmits command messages to all the Remote Stations in the system. The Remote Stations respond by transmitting back a buffer containing the state of its respective process inputs. Several transmissions may be required to obtain all the process inputs. Upon the last response from the slowest Remote Station, i.e., the remote with the most channels, the Main controller tags the system log with the current system time/date and alarm checks the system log per requisition requirements.

New requests are not issued until the current log is completed. The resulting scanning rate is approximately 2 HZ.

C. Main-Display Controller Interface

The operator display information is updated each time the Main Controller transfers a system log to the Display Controller. The Main Controller "refreshes" the Display controller with the latest system status, immediately after the completion of a system log. The resulting logging rate is approximately 2 HZ.

7.0 **MAINTENANCE** (Continued)

7.1.1 SIGNAL FLOW (continued)

C. Main-Display Controller Interface (continued)

The communication channel between the Main Controller and The Display Controller is by a dual ported RAM, implemented with an HRMA card (Main controller slot 1D). The Main Controller accesses the shared memory through the Main Controller's L-bus and the Display Controller **uses** the Main-Display-Historical E-bus.

In addition to the system log buffers, a message buffer is passed to the Display Controller. The message buffer contains an operator message describing the operating state of the Remote Station datalinks and the operating mode of the Main controller.

D. Main-Historical Controller Interface

The Historical Controller permanently stores the state of the Turbine-Generator to tape and disk in response to the transfer of a system log from the Main Controller to the Historical Controller. The system logs are transferred at one of two rates, except for immediate logs. The logging rate, governed by the Main controller, is determined by the alarm state of the system. In a "normal" state, a system log will be transferred to the Historical Controller once every 5 minutes. Under "abnormal", i.e., certain alarm conditions, a system log will be transferred once every 30 seconds. The faster logging rate is maintained for a duration of 24 minutes from the last detected alarm. Refer to MI (Manufacturing Instructions) supplied with the equipment. The dual rate timer is derived from the system clock.

Immediate logs are logs taken immediately upon detection of any one of the following conditions: (AS specified at time of order)

1. Certain alarm conditions.

2. Change of state of certain **System status** inputs.

NOTE: An immediate log starts the "Fast Log" time period and repeats with each reset (new) occurence.

The communication channel between the Main Controller and the Historical Controller is a dual ported RAM, implemented with an HRMA card. (See Historical Controller slot 1A). The Main Controller accesses the shared memory though the Main-Display-Historical E-bus and the Historical Controller **uses** the L-bus.

Unlike the Main-Display interface, an additional level of security to Historical logs is provided by the passing of a system log checksum word to the Historical Controller. If the Historical Controller detects a checksum error the Historical log is discarded. An error is logged by the main controller and a message displayed.

7.0 MAINTENANCE (Continued)

7.1.1 SIGNAL FLOW (continued1

E. Main Controller-Plant Computer Interface

A Plant Computer can obtain a copy of the T-GSI system log via the Plant computer datalink interface. The system log is sent only upon request from the Plant Computer. The Main Controller services and responds to the request on a time available basis.

The Plant Computer's port to the T-GSI system is an RS-232C serial datalink. Using an HCMA card, (see Main Controller slot 2C), system logs are passed from the Main Controller to the Plant Computer using a standard datalink protocol, ANSI Standard X3.28 subcategory 3.1/F1. The Plant Computer obtains a system log by issuing a command for a copy of the latest system log to the datalink. The Main controller responds by transmitting a copy of the current system log. A modem is inserted into the datalink to extend the range of the datalink.

F. Main Controller-Diagnostic Interface

The Main Controller is accessible to maintenance/engineering personnel for system diagnostic purposes via the Main Controller's HMPF RS-232C port. Using a dumb terminal, the user has the ability to perform system wide diagnostics, inspect the main Controller's status and many other functions.

G. Main Controller Relay Contact Output Interface

The alarm state of the Turbine-Generator system is available to the customer through a set of relay contact outputs. The relay contacts are energized and de-energized as a result of the system log alarming.

The Main Controller operates the T-GSI relay contact outputs through a set of logic level parallel I/O signals driven by an HIOD card, (see Main Controller slot 2E), to the Relay Driver Module, (see RDM). The Relay Driver Module, in turn, energizes/de-energizes the relays in the Relay Output Module, (see ROM). The relay contacts are brought out to terminal strips for connection to customer devices.

H. Main Controller Maintenance Module Interface

The operational state of the Main Controller can be controlled by the Maintenance Module. The modes, OFF-LINE, STANDBY and ON-LINE, permit engineering/maintenance personnel to vary the degree of Main Controller interaction with the T-GSI system.

7.0 MAINTENANCE (Continued)

7.1.1 SIGNAL FLOW (continued)

H. Main Controller Maintenance Module Interface (continued)

The maintenance mode is determined by a key switch on the Maintenance Panel. The switch is directly connected to the Main Controller's HXPC card. (See Main Controller slot 2F). When a switch position change is detected the Main Controller puts itself into the selected mode and turns on a mode indicator on the Maintenance Module. The Main Controller turns on/off the mode indicators by an HIOD-Relay Driver-Mode Indicator arrangement. The HIOD and Relay Driver is shared with the T-GSI relay contact output sub-system.

I. Main Controller System clock Interface

The T-GSI system time is based on a 60 HZ square wave derived from the customer's AC input to the T-GSI equipment.

Using a 60 HZ AC Transducer Module, (see SYN.), connected to the AC line, a logic level 60 HZ pulse is generated and inputted to the Main Controller's HXPC counter. (See Main controller slot 2F). The counter is set-up as a divided by 60 counter to produce a 1HZ interrupt. This interrupt is used to increment the system time/date value.

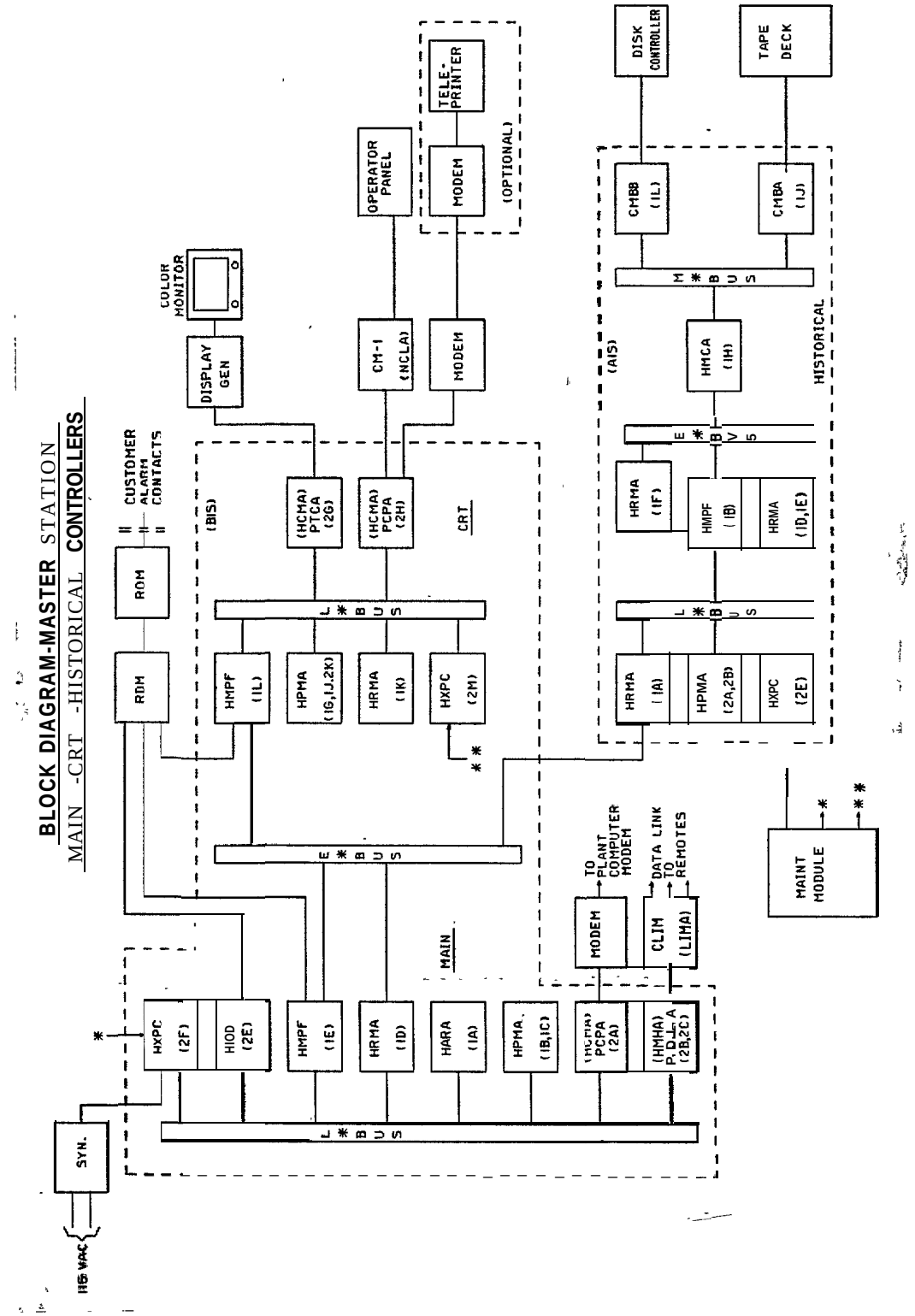


Figure 19. BLOCK DIAGRAM -MASTER STATION WITH PUSHBUTTON STYLE CONTROL CONSOLE.

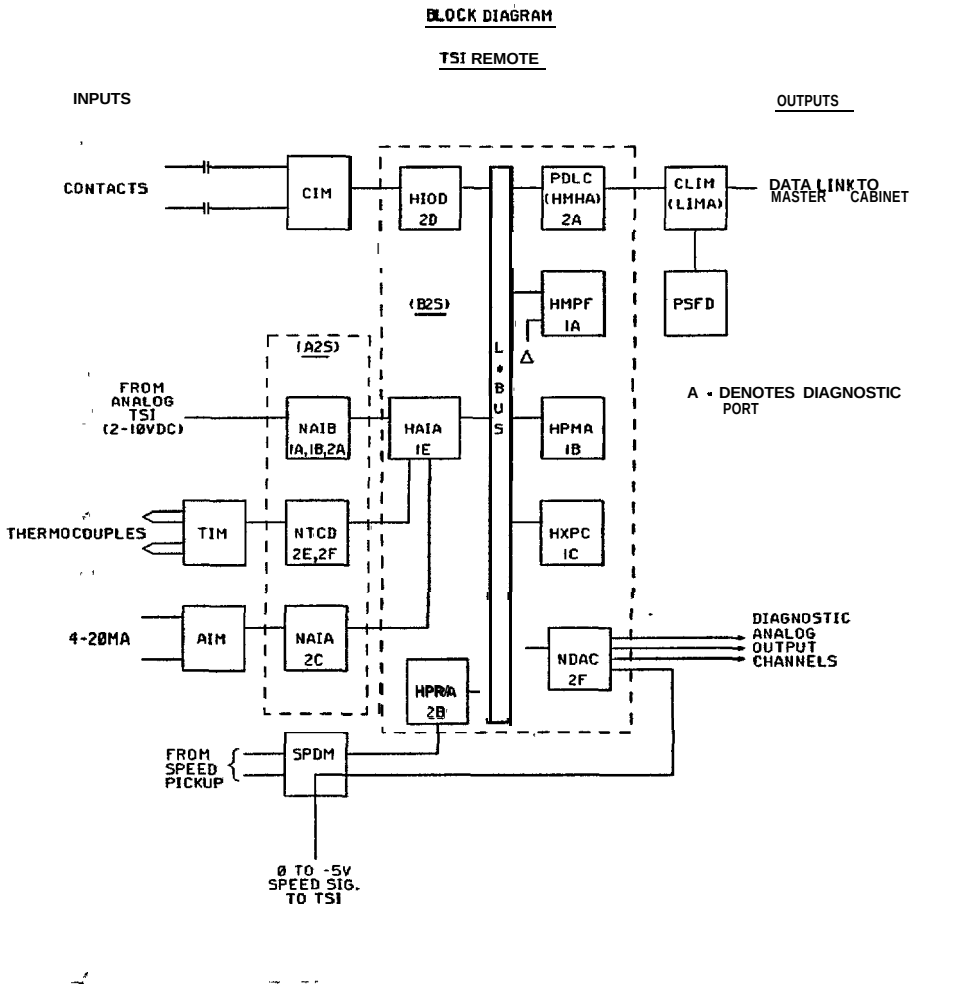


Figure 20: - BLOCK DIAGRAM - TSI REMOTE

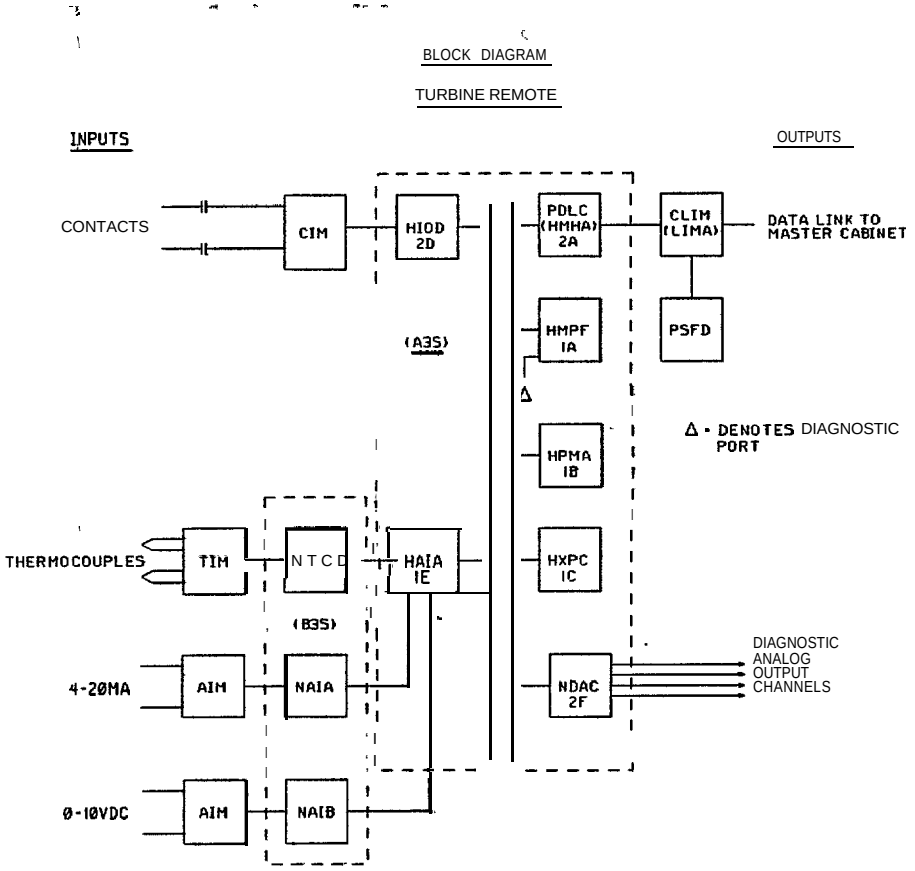


Figure 21: BLOCK DIAGRAM - TURBINE REMOTE

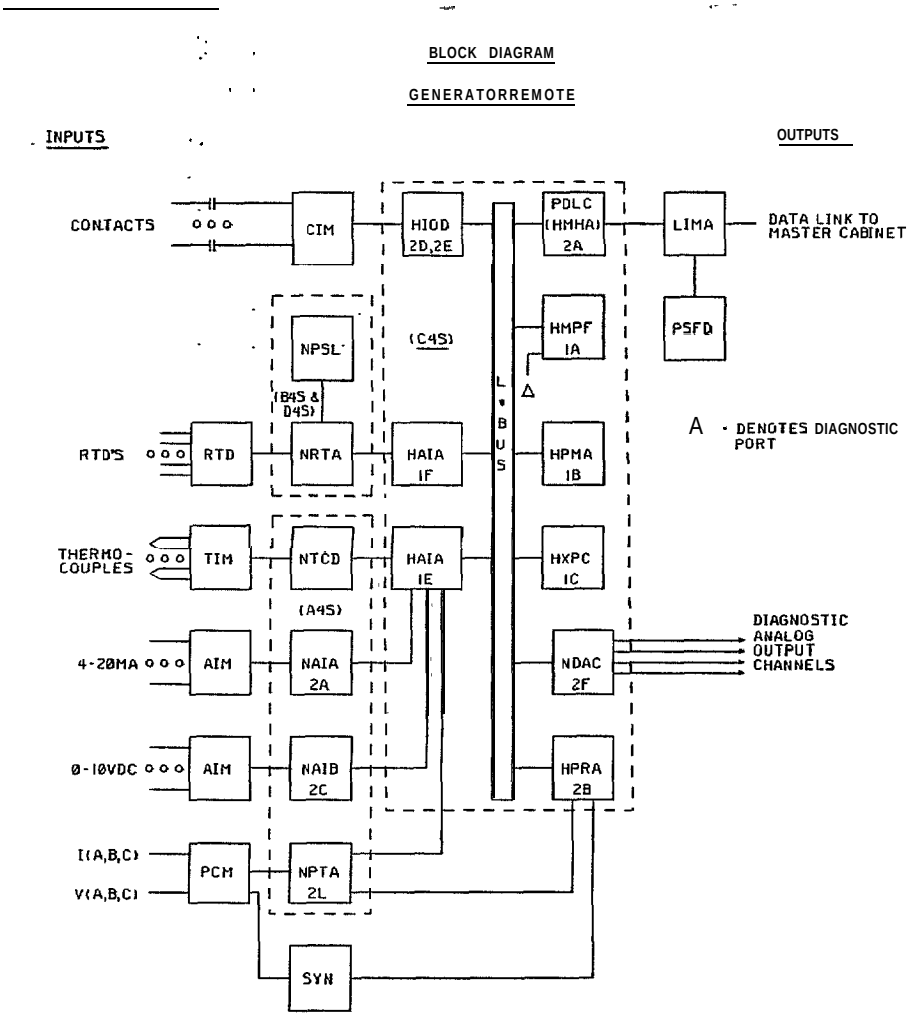


Figure 22: - BLOCK DIAGRAM - GENERATOR REMOTE

7.0 MAINTENANCE (Continued)

7.2 ROUTINE MAINTENANCE

A recommended maintenance schedule is given below. For further information on servicing the major components of the system, refer to their separate publications which are listed in the system TABLE OF CONTENTS in the T-GSI COMPOSITE INSTRUCTION BOOK.

NOTE: The cabinets should be opened only when necessary for service or inspection and then only by qualified personnel. The surrounding area should be kept as clean as possible, especially during intervals when the doors must be open for servicing.

7.2.1 MONTHLY

Air filters, (Air Conditioners): Inspect for dirt and dust build up. It is anticipated that the filters on the Turbine and Generator Remote Stations will need changing more often than on the Master and the TSI.

7.2.2 QUARTERLY

shaft Vibration Detectors

Inspect for wear of rider tip. Oil upper probe bearing with several drops of light machine oil. (No maintenance required for bracket mounted detectors.)

7.2.3 MAINTENANCE RECORDS

Most users develop a procedure for keeping records showing maintenance routines including dates of calibration, lubrication, replacement, details of down time, results of various checks and information concerning service calls. Such records are strongly recommended.

7.3 TROUBLESHOOTING

7.3.1 GENERAL

In many instances during the troubleshooting process, the replacement of a card is called for. When this is done, compare the number and location of Berg jumpers (little blue plastic pieces) on the replacement board against the original.

7.0 MAINTENANCE (Continued)

7.3.2 MASTER CABINET

7.3.2.1 MAIN CONTROLLER

Effective troubleshooting of the Main Controller requires:

- Terminal connected to the Main Controller Diagnostic port
- Customer Connection Diagram
- Common sense

Using the above, card LED indicators, the maintenance module and this document, the majority of Main Controller faults can be identified and corrected. For problems that are not addressed by the Troubleshooting guide, the user is advised to use the "Error Logger" and "STATUS" facilities found in the Main controller's diagnostics port. Both functions are documented with "HELP" commands.

A. MAIN CONTROLLER IS DEAD

Symptoms: Subsequent to power-up reset:

- HMHA and HCMA IMOK LEDS come on
- No PROM check printout to diagnostic port
- HMPF IMOK LED does not come on

Diagnosis: Controller has not executed the initialization code in response to the reset condition. The controller is in a "hung" state.

Remedy: Try the following:

- Verify EPROM's are properly seated (In particular look for bent IC pins or incorrect IC orientation).
- Verify EPROM part number to IC socket.

B. CONTROLLER DIES DURING INITIALIZATION

Symptoms: controller dies during initialization printout.

Diagnosis: controller is attempting to initialize the system, but has detected a hardware fault. The initialization printout will document detected fault as:

- CPU test failure
- PROM checksum failure
- RAM check failure

If the LED on HMPF goes out, comes back on after about 2 seconds, and no PROM check occurs, CPU failure is indicated.

7.0 MAINTENANCE (Continued)

7.3.2.1 MAIN CONTROLLER (continued)

B. CONTROLLER DIES DURING INITIALIZATION (continued)

Remedy: - Perform corrective action per initialization printout.

C. CONTROLLER DIES AFTER INITIALIZATION

Symptoms: - System passes initialization and runs in the Off-Line mode.

- System crashes when: put into the Standby or On-line mode from the Off-line mode; or Maintenance panel lights go on then off, followed by the HMPF IMOK light going on then off when powered-up directly into Standby or On-line modes.
- The diagnostic port is dead.

Diagnosis: The system has passed initialization but has crashed during the remote station scanning. The alarm database on the HARA card has been contaminated.

Remedy: Reprogram the HARA card by:

- Generating a power reset from the maintenance panel with the system in the Off-Line mode
- Using the diagnostic port, execute the Changeable Alarm Limit facility
- Run the "RESET" command to reprogram the HARA memory

NOTE: This process will take approximately 4 minutes and will set-up the HARA with the original alarm limit settings. The HARA enable switch on the card front edge must be "up" (enabled).

D. REMOTE STATION SCANNER IS NOT PICKING UP THE REMOTE STATION INPUTS

Symptoms: - Main Controller appears to be scanning the remote station, but all remote station inputs are being displayed as "out-of-range" on the color CRT pages.
- Main Controller is in the Standby or On-Line modes.

Diagnosis: The connection to the remote station is logically or physically open. The HMHA cards will flash even if the communications channel is down.

Remedy: If the color CRT message in the lower right hand corner of the screen indicates that the respective remote station datalink is "down" (alternatively the Main

7.0 **MAINTENANCE** (Continued)

7.3.2.1 MAIN CONTROLLER (continued)

D. REMOTE STATION SCANNER . . . (continued)

Controller diagnostic command "STATUS" can be used to determine the link status) then:

Remedy:
- Verify the Main Controller datalink wiring. In particular that the correct datalink cable is plugged into the designated HMHA card front edge connector.
- Checkout the datalink isolation module located at the rear of the Master Station cabinet. Specifically, ensure that the module has power to it and the wiring is intact.
- Verify the operation of the remote station.
- With power off, remove the HMHA card, verify jumper settings, correlate card with enclosure slot location and check the HMHA PROM part number to the drawing. Perform this operation for both HMHAs, Main Controller and remote station.

E. MAIN CONTROLLER PERFORMS ONE REMOTE STATION SCAN AND STOPS

Symptoms: A scan is performed, indicated by a flash of all remote station HMHA red LEDs each time the maintenance mode switch is moved from Off-Line to Standby/On-Line.
Note: Unused HMHA outputs will not flash.

Diagnosis: The remote station scanner is stalled. This can be caused by: (1) the Display Controller locking up the Main-Display window,(dual-ported memory), (2) an open Main-Display communication, or (3) the Main Controller improperly accessing window control variables.

Remedy:
- Check the E-bus cable and connector.
- Reset the Main Controller to rebuild Main-Display Window variables.

F. SYSTEM IS NOT COMMUNICATING TO THE REMOTES

Symptoms:
- Main Controller's HMHA "active" LEDs are not flashing.
- Color CRT display shows reasonable input values.
- Main Controller is in STANDBY or ON-LINE.
- System time is incrementing.
- Diagnostics are functional.

7.0 MAINTENANCE (Continued)

7.3.2.1 MAIN CONTROLLER (continued)

F. SYSTEM IS NOT COMMUNICATING TO THE REMOTES (continued)

Diagnosis: The system is in the remote simulator, RSIM, diagnostic mode. The Color CRT will display a message indicating that RSIM is on.

Remedy: Exit the RSIM diagnostics mode.

G. SYSTEM TIME IS NOT BEING INCREMENTED

Symptoms: The system time as seen at the Color CRT and via the Main Controller "STATUS" command is not changing.

Diagnosis: The Main Controller cannot access the system time array, located in the Main-Display window, due to an open or locked up communication channel to the window area.

Remedy: This condition is remedied in like manner to the "Remote Station scanner is not picking up remote station inputs" problem. Note: This problem should not be confused with the normal operating mode of the Display Controller of not updating color CRT time when either of the controllers is not in the On-Line mode.

H. THE MAIN CONTROLLER FAILS TO E'RINT OUT AFTER A RESET

If the teleprinter does not begin printing after the "RESET" button is pushed, then check the following:

1. Teleprinter has power
2. MEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. The RS-232 cable adaptor is present and hooked across the interface.

The first print out is a table listing which proms passed their checksum test. If any of the proms failed, then power down the MEM by opening the breaker marked "MASTER", and check the jumpering on 'the cards starting with the card that has the failed prom. (See prom address table below). If the above seems to have been the problem, power up the MEM by closing the "MASTER" breaker.

If the jumpers are okay, then replace the failed prom, and power up the MEM again.

7.0 MAINTENANCE (Continued)

7.3.2.1 MAIN CONTROLLER (continued)

H. THE MAIN CONTROLLER FAILS TO PRINT OUT AFTER A RESET (continued)

The next print out is a RAM test. If any byte of RAM fails, power down the MEM, and replace the following boards in order until the problem is found and then power up the MEM again.

1. HMPF
2. HXPC
3. HRMA

The third print out is a check of the database proms. They must pass for the system to run properly. If they do not pass, replace the proms that failed (See Main Prom Address table below), after powering down the MEM.

MAIN PROM ADDRESSES

Address	card	Slot	Socket
08000-09FFF LO	HPMA	1B	u14
08000-09FFF HI	HPMA	1B	u22
0A000-0BFFF LO	HPMA	1B	u15
0A000-0BFFF HI	HPMA	1B	U23
0C000-0DFFF LO	HPMA	1B	U16
0C000-0DFFF HI	HPMA	1B	U24
F0000-F1FFF LO	HPMA	1C	u14
F0000-F1FFF HI	HPMA	1C	u22
F2000-F3FFF LO	HPMA	1C	u15
F2000-F3FFF HI	HPMA	1C	U23
F4000-F5FFF LO	HPMA	1C	U16
F4000-F5FFF HI	HPMA	1C	U24
F6000-F7FFF LO	HPMA	1C	u17
F6000-F7FFF HI	HPMA	1C	U25
F8000-F9FFF LO	HPMA	1C	U18
F8000-F9FFF HI	HPMA	1C	U26
FA000-FBFFF LO	HPMA	1C	U19
FA000-FBFFF HI	HPMA	1C	U27
FC000-FDFFF LO	HMPF	1E	U24
FC000-FDFFF HI	HMPF	1E	U25
FE000-FFFFF LO	HMPF	1E	u22
FE000-FFFFF HI	HMPF	1E	U23

7.0 MAINTENANCE (Continued)

7.3.2.2 DISPLAY CONTROLLER

A. THE DISPLAY CONTROLLER FAILS TO PRINT OUT AFTER A RESET

If the teleprinter does not begin printing after the "RESET" button is pushed, then check the following.

1. Teleprinter has power
2. MEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. The RS-232 cable adaptor is present and hooked correctly across the interface.

The first print out is a table listing which proms passed their checksum test. If any of the proms failed, then power down the MEM by opening the breaker marked 'MASTER", and check the jumpering on the cards starting with the card that has the failed prom. (See Display Prom Address table). If the above seems to have been the problem, power up the MEM by closing the "MASTER" breaker.

If the jumpers are okay, then replace the failed prom, and power up the MEM again.

The next print out is a RAM test. If any byte of RAM fails, power down the MEM, and replace the following boards in order until the problem is found and then power up the MEM again.

1. HMPF
2. HXPC
3. HRMA

The third print out is a check of the database proms. They must pass for the system to run properly. If they do not pass, replace the proms that failed, (see Prom Address Table below), after powering down the MEM.

The fourth print out is a check of the CRT database proms. They must also pass for the system to run properly. If they do not pass, check the jumpering on the prom board or replace the prom that failed, (see Display Prom Address Table on the following page), after powering down the MEM.

7.0 MAINTENANCE (Continued)

7.3.2.2 DISPLAY CONTROLLER (continued)

A. THE DISPLAY CONTROLLER FAILS TO PRINT OUT AFTER A RESET (continued1

DISPLAY PROM ADDRESSES

Address	card	Slot	Socket
08000-09FFF LO	HPMA	1H	u14
08000-09FFF HI	HPMA	1H	u22
0A000-0BFFF LO	HPMA	1H	u15
0A000-0BFFF HI	HPMA	1H	U23
0C000-0DFFF LO	HPMA	1H	U16
0C000-0DFFF HI	HPMA	1H	U24
0E000-0FFFF LO	HPMA	1H	u17
0E000-0FFFF HI	HPMA	1H	U25
10000-11FFF LO	HPMA	2L	u14
10000-11FFF HI	HPMA	2L	u22
12000-13FFF LO	HPMA	2L	u15
12000-13FFF HI	HPMA	2L	U23
14000-15FFF LO	HPMA	2L	U16
14000-15FFF HI	HPMA	2L	U24
16000-17FFF LO	HPMA	2L	u17
16000-17FFF HI	HPMA	2L	U25
F0000-F1FFF LO	HPMA	1J	U14
F0000-F1FFF HI	HPMA	1J	U22
F2000-F3FFF LO	HPMA	1J	u15
F2000-F3FFF HI	HPMA	1J	U23
F4000-F5FFF LO	HPMA	1J	U16
F4000-F5FFF HI	HPMA	1J	U24
F6000-F7FFF LO	HPMA	1J	u17
F6000-F7FFF HI	HPMA	1J	U25
F8000-F9FFF LO	HPMA	1J	U18
F8000-F9FFF HI	HPMA	1J	U26
FA000-FBFFF LO	HPMA	1J	U19
FA000-FBFFF HI	HPMA	1J	U27
FC000-FDFFF LO	HMPF	1L	U24
FC000-FDFFF HI	HMPF	1L	U25
FE00U-FFFFF LO	HMPF	1L	u22
FE000-FFFFF HI	HMPF	1L	U23

B. DISPLAY **ON** CRT CANNOT BE CHANGED

Symptoms: Nothing on the CRT can be changed in any way.

Diagnosis: The CRT generator has locked up. This can be caused by disconnecting the cable between the HCMA (PCPA) card and the data generator or by powering down the HCMA (PCPA) and not the data generator.

7.0 MAINTENANCE (Continued)

7.3.2.2 DISPLAY CONTROLLER (continued)

B. DISPLAY ON CRT CANNOT BE CHANGED (continued)

Remedy: Push the "RESET" button on the lower left front of the CRT Electronics Module, see Figure 14 for location of this button. Then put a new page up on the CRT.

7.3.2.3 HCMA CHECK (PTCA)

If the LED on the PTCA (HCMA) card in slot 2G is not lit, then replace the card.

7.3.2.4 HMHA CHECK (PDLA)

If the LEDs on the PDLA (HMHA) cards in slots 2B and 2c are not lit, then replace the card(s).

7.3.2.5 HISTORICAL CONTROLLER

A. THE HISTORICAL CONTROLLER FAILS TO PRINT OUT AFTER A RESET

If the teleprinter does not begin printing after the "RESET" button is pushed, then check the following:

1. Teleprinter has power
2. MEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. The RS-232 cable adaptor is present and hooked correctly across the interface.

The first print out is a table listing which proms passed their checksum test. If any of the proms failed, then power down the MEM by opening the breaker marked "MASTER", and check the jumpering on the cards starting with the card that has the failed prom. (See Prom Address table on the following page). If the above seems to have been the problem, power up the MEM by closing the "MASTER" breaker.

If the jumpers are okay, then replace the failed prom, and power up the MEM again.

The next print out is a RAM test. If any byte of RAM fails, power down the MEM, replace the following boards in order until the problem is found and power up the MEM again.

1. HMPF
2. HXPC
3. HRMA

7.0 MAINTENANCE (Continued)

7.3.2.5 HISTORICAL CONTROLLER (continued)

A. THE HISTORICAL CONTROLLER FAILS TO PRINT OUT AFTER A RESET (continued)

The third print out is a check of the database proms. They must pass for the system to run properly. If they do not pass, replace the proms that failed, see Prom Address table below, after powering down the MEM.

HISTORICAL PROM ADDRESSES

Address	card	Slot	socket
E0000-E1FFF LO	HPMA	2B	u14
E0000-E1FFF HI	HPMA	2B	u22
E2000-E3FFF LO	HPMA	2B	u15
E2000-E3FFF HI	HPMA	2B	U23
E4000-E5FFF LO	HPMA	2B	U16
E4000-E5FFF HI	HPMA	2B	U24
E6000-E7FFF LO	HPMA	2B	u17
E6000-E7FFF HI	HPMA	2B	U25
E8000-E9FFF LO	HPMA	2B	U18
E8000-E9FFF HI	HPMA	2B	U26
EAUUU-EBFFF LO	HPMA	2B	U19
EAUUU-EBFFF HI	HPMA	2B	u27
ECUUU-EDFFF LO	HPMA	2B	U20
ECUUU-EDFFF HI	HPMA	2B	U28
EE000-EFFFF LO	HPMA	2B	u21
EE000-EFFFF HI	HPMA	2B	u29
FOOOO-F1FFF LO	HPMA	2A	u14
F0000-F1FFF HI	HPMA	2A	u22
F2000-F3FFF LO	HPMA	2A	u15
F2000-F3FFF HI	HPMA	2A	U23
F4000-F5FFF LO	HPMA	2A	U16
F4000-F5FFF HI	HPMA	2A	U24
F6000-F7FFF LO	HPMA	2A	u17
F6000-F7FFF HI	HPMA	2A	U25
F8000-F9FFF LO	HPMA	2A	U18
F8000-F9FFF HI	HPMA	2A	U26
FAUUU-FBFFF LO	HPMA	2A	U19
FAUUU-FBFFF HI	HPMA	2A	U27
FCUUU-FDFFF LO	HMPF	1B	U24
FCUUU-FDFFF HI	HMPF	1B	U25
FEUUU-FFFFF LO	HMPF	1B	u22
FEUUU-FFFFF HI	HMPF	1B	U23

7.3.2.6 EXPLANATION OF DISK CRT MESSAGES

A. DISK FAULT - A hardware error has occurred upon recording to the disk. Refer to "STATUS" diagnostics command

7.0 MAINTENANCE (Continued)

7.3.2.6 EXPLANATION OF DISK CRT MESSAGES (continued)

B. DISK WRITE PROTECT ON - Write Protect Switch on, preventing recording to the disk

C. OVERLAP IN TIME - Recording dates on logs are not in chronological order

D. DISK DISMOUNTED - Maintenance Module "DISK" Switch is switched to Dismounted, Recording to Disk Ceased

E. DISK READY - Disk ready to receive logs from "MAIN." Normal state of disk.

F. DISK UNIT NOT READY - A command was sent to the disk before the disk has powered up

G. DISK POWER UP INCOMPLETE - There was an error during power up initialization. Refer to "DISKPOWER" diagnostics command

H. INVALID HIST. LOG DATE - The system date/time which is shown on the CRT has not been set.

I. HISTORICAL OFFLINE - Maintenance module switch mode is turned to "OFFLINE". All recording to tape and disk is stopped

J. OVERLAP RECOVERY - Disk is deleting out of order logs after a time overlap

7.3.2.7 EXPLANATION OF TAPE CRT MESSAGES

A. END OF TAPE NEAR - Displayed when current tape has recorded 13,500 logs. Operator should prepare to dismount current Tape and remount New Tape.

B. TAPE FULL - Displayed when end of tape marker has been reached and current logs are no longer being recorded. Operator must now replace the tape.

C. TAPE REWINDING - The tape is rewinding (at a slow pace).

D. TAPE HARDWARE ERROR - May occur when tape is bad, controller board fails or other hardware related problem. Refer to "STATUS" diagnostics command.

7.0 MAINTENANCE (Continued)

7.3.2.7 EXPLANATION OF TAPE CRT MESSAGES (continued)

E. MOUNTED TAPE FAULTY - A series of unsuccessful1 accesses were attempted to the tape due to bad spaces on the tape. Refer to "STATUS" diagnostics command.

F. TAPE CONTOLLER ERROR - Tape controller board not functioning properly. It is either missing interrupts or failing to initialize. Refer to "STATUS" diagnostics command.

G. WRITE PROTECTED TAPE - An attempt was made to write to a tape that does not have a write-protect ring intact.

H. TAPE OFFLINE - Tape drive is offline.

I. TAPE DISMOUNTED - Maintenance module "TAPE MOUNT" switch is set to dismount.

J. TAPE READY - Tape ready to receive logs from "MAIN". Normal recorder state.

K. TAPE RECORDING FROM DISK - Tape recording back logs from the disk.

L. DISK RECORDING FROM TAPE - A tape read is in progress to the disk.

M. OVERLAP RECOVERY - Tape will rewind and search for appropriate log date from which to continue recording after a time overlap.

N. EOT NOT MARKED - "TAPE" switch is set to "USED" but file markers could not be found to mark last record recorded on tape. Therefore this is either a new or a full tape, or perhaps it was not properly dismounted as explained in Section 3.3.

0. TAPE DMA OVERRUN - Bytes were added to log

P. TAPE DMA INCOMPLETE - Bytes were missing from log

7.3.2.7.1 TAPE DUMP CRT MESSAGES

A. MOUNT A LATER TAPE - The tape on which the tape read was attempted contained dates earlier than the requested date.

7.0 MAINTENANCE (continued)

7.3.2.7.1 TAPE DUMP CRT MESSAGES (continued)

B. MOUNT AN EARLIER TAPE - The tape on which the tape read was attempted contained dates later than the requested date.

7.3.2.8 DIAGNOSTIC COMMANDS

To use diagnostics commands, attach a serial link from a terminal to the HMPF. Through the keyboard, commands can be entered to help the operator or maintenance personnel solve problems or determine status of the system.

THE COMMANDS ARE:

1. CONF(IDENCE)
2. DRA(NGE)
3. LAB(EL)
4. LAM(P)
5. H(ELP)
6. MON(ITOR)
7. STAT(US)
8. QU(EUE)

7.3.2.9 HISTORICAL CONTROLLER DIAGNOSTICS COMMANDS

A. STATUS

1. DESCRIPTION: This command will display to the terminal the status of the system.

2. PRIVILEGE REQUIRED: No

3. FORMAT: STAT(US)

4. EXIT: Automatic exit to $ prompt

B. CONFIDENCE TEST

1. DESCRIPTION: This runs a RAM and ROM check test on the system and reports any errors to the terminal.

2. PRIVILEGE REQUIRED: No

3. FORMAT: CONF(IDENCE)

4. EXIT: Automatic exit to $ prompt

7.0 MAINTENANCE (Continued)

7.3.2.9 HISTORICAL CONTROLLER DIAGNOSTICS COMMANDS (continued)

C. LAMP TEST

1. DESCRIPTION: This test will turn the lights on for a short period of time to check for inoperative bulbs.

2. PRIVILEGE REQUIRED: No

3. FORMAT: LAM(P)

4. EXIT: Automatic exit to $ prompt

D. MONITOR MODE

1. DESCRIPTION: This command displays the Hex value dates of logs and error/status codes to the diagnostics terminal as logs are recorded to the tape and disk.

2. PRIVILEGE REQUIRED: No

3. FORMAT: MON(ITOR)

a. PROMPT: "WHICH DEVICE(S) DO YOU WANT TO MONITOR?" "TAPE?"
RESPONSE: Input a "Y" to indicate yes, else return.
b. PROMPT: "DISK?"
RESPONSE: Input a "Y" to indicate yes, else return.
c. PROMPT: "FILE MANAGER ERROR/STATUS CODES?"
RESPONSE: Input a "Y" to indicate yes, else return.

4. EXIT: (CONTROL) Y

7.3.2.10 DISK FUNCTIONS

A. DISK RANGE

1. DESCRIPTION: This command will display the date range of each directory on the disk. This is helpful in determining if a tape dump is needed to obtain the historical log desired.

2. PRIVILEGE REQUIRED: No

3. FORMAT: DRA(NGE)

4. EXIT: Automatic exit to $ prompt

7.0 MAINTENANCE (Continued)

7.3.2.11 TAPE FUNCTIONS

A. TAPE QUEUE COMMAND

1. DESCRIPTION:

   This diagnostics command is particularly useful in reassuring the user that the tape update software is running properly, since it queues twelve logs before recording them to tape. At a five minute recording rate, it will be an hour before each recording. By knowing the recording rate and the number of outstanding logs, the time of the next tape movement can be calculated.

2. PRIVILEGE REQUIRED: No

3. FORMAT: **TQU(EUE)**

4. EXIT: view data, and it will automatically exit to $ prompt.

B. TAPE LABELS

1. DESCRIPTION: This command displays the first and last dates of the last recorded tape. This is for the operator's convenience in recording the date range of a tape for archival storage. The operator must record this information before mounting the next tape.

2. PRIVILEGE REQUIRED: No

3. FORMAT: **LAB(EL)**

4. EXIT: View data, and it will automatically exit to $

7.3.3 TSI REMOTE CABINET

7.3.3.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT

If the teleprinter does not begin printing after power is applied through the breaker marked "CONTROL", check the following:

1. Teleprinter has power
2. SEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. All RS-232 connections are firm
6. The RS-232 cable adaptor is present and hooked correctly across the interface.

7.0 MAINTENANCE (Continued)

7.3.3.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT (continued)

The first printout is a table listing which proms passed their checksum test. If any of the proms failed, then power down the SEM by opening the breaker marked "CONTROL" and check the jumpering on the HMPF, HPMA and HXPC cards. If the above seems to have been the problem, power up the SEM by closing the "CONTROL" breaker.

If the jumpers are okay, then replace the failed prom and power up the SEM again.

The next printout is a RAM test. If any byte of RAM fails, power down the SEM, replace the following boards in order until the problem is found and power up the SEM again.

1. HMPF
2. HXPC

The third print out is a check of the database proms. They must pass fbr the system to run properly. If they do not pass, replace the proms that failed, see Prom address table below, after powering down the SEM.

After the "$" prompt comes on the teleprinter, type "STAT" followed by a carriage return. The teleprinter should print the proper remote name and that there are no errors. The input status is not important at this time. If the wrong remote name is printed, check the jumpers on the HXPC card. If there are any other errors, try to correct them according to the message.

TSI PROM ADDRESSES

Address	Card	Slot	Socket
F0000-F1FFF LO	HPMA	1B	u14
F0000-F1FFF HI	HPMA	1B	u22
F2000-F3FFF LO	HPMA	1B	u15
F2000-F3FFF HI	HPMA	1B	U23
F4000-F5FFF LO	HPMA	1B	U16
F4000-F5FFF HI	HPMA	1B	U24
F6000-F7FFF LO	HPMA	1B	u17
F6000-F7FFF HI	HPMA	1B	U25
F8000-F9FFF LO	HPMA	1B	U18
F8000-F9FFF HI	HPMA	1B	U26
FA000-FBFFF LO	HPMA	1B	U19
FA000-FBFFF HI	HPMA	1B	U27
FC000-FDFFF LO	HMPF	1A	U24
FC000-FDFFF HI	HMPF	1A	U25
FE000-FFFFF LO	HMPF	1A	u22
FE000-FFFFF HI	HMPF	1A	U23

7.0 MAINTENANCE (Continued)

7.3.3.2 ADJUSTMENT OF OVERVOLTAGE PROTECTORS

If the vibration readings appear to go "dead", it could be due to a power supply trip. The +/- 15 VDC power supplies in the analog section of TSI have an overvoltage circuit that will cause the output to suddenly drop if the overvoltage setpoint is detected.

Before adjusting the overvoltage protectors, disconnect all load from the power supply by either disconnecting the cabling at the power supply, or pulling all the cards.

Initially set the adjusting screws on the Overvoltage Protectors, OVP1 and 2, fully clockwise.

The system has two DC power supplies, DCPS1 and 2. Turn system power OFF and apply a DC current limited, 30ma, adjustable voltage source to the output terminals of one supply, observing polarity. While monitoring the buffer supplies output with a DVM, carefully increase the output from zero to 18 VDC.

Turn the Overvoltage Protectors adjustment counterclockwise until a sudden drop in voltage is detected.

Repeat the above procedure for the second DC supply.

Once adjusted, do not change these settings.

7.3.4 TURBINE REMOTE CABINET

7.3.4.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT

If the teleprinter does not begin printing after power is applied through the breaker marked "CONTROL", check the following:

1. Teleprinter has power
2. SEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. All RS-232 connections are firm
6. The RS-232 cable adaptor is present and hooked correctly across the interface.

If any of these seems to have been the problem, power down the SEM, check the set up and power up the SEM again.

7.0 MAINTENANCE (Continued)

7.3.4.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT (continued)

The first print out is a table listing which proms passed their checksum test. If any of the proms failed, then power down the SEM by opening the breaker marked "CONTROL" and check the the jumpering on the HMPF, HPMA and HXPC cards. If the above seems to have been the problem, power up the SEM by closing the "CONTROL" breaker.

If the jumpers are okay, replace the failed prom, and power up the SEM again.

The next print out is a RAM test. If any byte of RAM fails, power down the SEM, replace the following boards in order until the problem is found and power up the SEM again.

1. HMPF
2. HXPC

The third print out is a check of the database proms. They must pass for the system to run properly. If they do not pass, replace the proms that failed, see Prom Address table below, after powering down the SEM.

After the "$" prompt comes on the teleprinter, type "STAT" followed by a carriage return. The teleprinter should print the proper remote name and that there are no errors. The input status is not important at this time. If the wrong remote name is printed, check the jumpers on the HXPC card. If there are any other errors, try to correct them according to the message.

TURBINE PROM ADDRESSES

Address	Card	Slot	Socket
F0000-F1FFF LO	HPMA	1B	U14
F0000-F1FFF HI	HPMA	1B	u22
F2000-F3FFF LO	HPMA	1B	U15
F2000-F3FFF HI	HPMA	1B	U23
F4000-F5FFF LO	HPMA	1B	U16
F4000-F5FFF HI	HPMA	1B	U24
F6000-F7FFF LO	HPMA	1B	U17
F6000-F7FFF HI	HPMA	1B	U25
F8000-F9FFF LO	HPMA	1B	Ul.8
F8000-F9FFF HI	HPMA	1B	U26
FAUOO-FBFFF LO	HPMA	1B	U19
FAUOO-FBFFF HI	HPMA	1B	U27
FC000-FDFFF LO	HMPF	1A	U24
FC000-FDFFF HI	HMPF	1A	U25
FE000-FFFFF LO	HMPF	1A	u22
FE000-FFFFF HI	HMPF	1A	U23

7.0 MAINTENANCE (Continued)

7.3.5 GENERATOR REMOTE CABINET

7.3.5.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT

If the teleprinter does not begin printing after power is applied through the breaker marked "CONTROL", check the following:

1. Teleprinter has power
2. SEM has power
3. HMPF inserted firmly and jumpered properly
4. Teleprinter switches are configured properly
5. All RS-232 connections are firm
6. The RS-232 cable adaptor is present and hooked correctly across the interface.

The first print out is a table listing which proms passed their checksum test. If any of the proms failed, then power down the SEM by opening the breaker marked "CONTROL" and check the the jumpering on the HMPF, HPMA and HXPC cards. If the above seems to have been the problem, power up the SEM by closing the "CONTROL" breaker.

If the jumpers are okay, replace the failed prom and power up the SEM again.

The next print out is a RAM test. If any byte of RAM fails, power down the SEM, replace the following boards in order until the problem is found and power up the SEM again.

1. HMPF
2. HXPC

The third print out is a check of the database proms. They must pass for the system to run properly. If they do not pass, power down the SEM and replace the failed proms according to the Prom Address table on the following page.

After the "$" prompt comes on the teleprinter, type "STAT" followed by a carriage return, The teleprinter should print the proper remote name and that there are no errors. The input status is not important at this time. If the wrong remote name is printed , check the jumpers on the HXPC card. If there are any other errors, try to correct them according to the message.

7.0 MAINTENANCE (Continued)

7.3.5.1 CONTROLLER VERIFICATION - FAILURE TO PRINT OUT (continued)

GENERATOR PROM ADDRESSES

Address	Card	Slot	Socket
F0000-F1FFF LO	HPMA	1B	u14
F0000-F1FFF HI	HPMA	1B	u22
F2000-F3FFF LO	HPMA	1B	u15
F2000-F3FFF HI	HPMA	1B	U23
F4000-F5FFF LO	HPMA	1B	U16
F4000-F5FFF HI	HPMA	1B	U24
F6000-F7FFF LO	HPMA	1B	u17
F6000-F7FFF HI	HPMA	1B	U25
F8000-F9FFF LO	HPMA	1B	U18
F8000-F9FFF HI	HPMA	1B	U26
FA000-FBFFF LO	HPMA	1B	U19
FA000-FBFFF HI	HPMA	1B	U27
FC000-FDFFF LO	HMPF	1A	U24
FC000-FDFFF HI	HMPF	1A	U25
FE000-FFFFF LO	HMPF	1A	u22
FE000-FFFFF HI	HMPF	1A	U23

7.3.6 CONTROL CONSOLE (Pushbutton Style) - Refer to separately supplied manual.

7.3.7 CRT MONITOR - Refer to separately supplied manual.

7.3.8 TELEPRINTER (Optional)

Type a carriage return on the teleprinter. It should return with an "*" prompt. Type H(ELP) followed by a carriage return. The teleprinter should respond with a help massage followed by an "*" prompt again. If the port does not work properly, then do the following:

1. Check the Berg jumpers, little blue plastic pieces, on the PCPA (HCMA) card, slot 2H, thaat has two connectors.

2. Replace the PCPA (HCMA) card.

3. Bypass modems by connecting a teleprinter (RS232C Interface) directly to PCPA Port and try again. Check modems and cabling if port works.

7.3.9 MODEM (DS512)

Below is a procedure for troubleshooting the full duplex modem.

1. Varify that the strap options are correct:

   A. The Carrier Switch should be in the "CONTROLLED" position (Leased Line Configuration).

7.0 MAINTENANCE (Continued)

7.3.9 MODEM (continued)

B. The Half Duplex strap should not be used. It should have no jumper across its terminals.

C. The Ground Strap should not be used. It should not have a jumper.

D. The CTS, Clear To Send, Delay Strap should not be used. It has three terminals and none should be jumpered,

2. At location A:

A. Verify that four wires, two pair, are available. Connect one pair to S1 - S2 and the other pair to R1 - R2.

B. Put the Loopback Switch in the LOOPBACK position.

3. At location B:

A. Place the Loopback Switch in the LOOPBACK position.

B. Connect each pair in turn to R1 - R2. One pair will cause the CARRIER lamp to light, the other will not, If the CARRIER light will not light by connecting either pair, the range of the modem may have been exceeded or a line fault may be present.

C. Connect the pair that causes the CARRIER lamp to light, but not the DATA lamps. The reverse connection should cause the DATA lamps to light.

D. Turn the POWER off by disconnecting the data set from the AC outlet. Connect the other pair to S1 - S2. Plug the data set back in. If the DATA lamps are off, the polarity is correct. If the DATA lamps are on, turn the POWER back off, reverse S1 - S2, and turn the POWER back on. DATA lamps should then be off.

NOTE: It is necessary to turn off the POWER when performing this test in order to reset the system.

E. Place the Loopback Switch in the NORMAL position.

4. At location A:

A. Place the Loopback Switch in the NORMAL position.

B. The data set is now ready for use.

7.0 MAINTENANCE (Continued)

7.3.10 DS512 D.C. - SELF TEST

DC operation of the DS512 can be verified by performing the procedure below:

1. Turn DS512 power off. Connect Rl to S1 and R2 to S2. When the power is turned back on, CARRIER, TX DATA and RX DATA lamps should be off.

2. Turn DS512 power off. Place Loopback Switch in LOOPBACK position. When the power is turned back on, CARRIER lamp should be on and TX DATA and Rx DATA lamps should be off.

3. Turn DS512 power off. Connect R1 to S2 and R2 to S1. When the power is turned back on, CARRIER, TX DATA and RX DATA lamps should be on.

7.3.11 KOOLTRONIC ELECTRONIC CABINET AIR CONDITIONER (MODEL KAC8)

7.3.11.1 ROUTINE MAINTENANCE

The KAC series air conditioners are designed to require no routine maintenance other than the cleaning of ambient air filters as necessary to permit unimpeded air flow through the condenser heat exchanger. It is not possible to recommend filter cleaning intervals since the level of contaminant concentration and the nature of airborne particulate matter is variable and may differ widely with each installation. It is generally sufficient to remove and wash the reuseable aluminum air filters when the entering surfaces of these filters appear covered with a thin layer of dust and lint. If filter servicing is delayed excessively, the maximum ambient temperature at which the air conditioner will provide design performance decreases. The first indication of excessively clogged air filters is usually loss of ability to control temperature within the electronic equipment cabinet. Air outlet temperature gradually increases to, perhaps, 10° above set point. If operation is continued under these conditions, the compressor, protected by an internal thermal overload device, may then shut itself off. The compressor will restart when internal temperatures drop below the protector threshold, and compressor operation will continue on a sporadic basis. Operation under these conditions may be damaging to the compressor and should be guarded against by the user.

Air filters are removed from the exterior surfaces of the air conditioner by removing the machine screws visible on filter mounting angles. The filters may then be back flushed under running water with the clean side up, driving contaminants out of the dirty side of the filter. The filters may be recoated with Kooltronic filter coating material to operate as viscous impingement filters where fine airbourne particulates are present, or may simply be dried and reinstalled as strainer type filters. While the air conditioners may be operated briefly with the filters removed, continued operation without the air filters is not

7.0 MAINTENANCE (Continued)

7.3.11 KOOLTRONIC Electronic cabinet Air conditioner (continued)

7.3.11.1 ROUTINE MAINTENANCE (continued)

recommended since the entering face of the condenser heat exchanger, which is not visible from the exterior of the air conditioner, may become clogged with dust or lint. A clogged heat exchanger is not readily detected and results in reduction of maximum ambient temperature capability.

Condenser and evaporator blowers are equipped with motors incorporating permanently lubricated precision ball bearings and, like the compressor, U/L recognized automatic reset thermal overload protectors. The design life of the blowers employed in all KAC air conditioners is substantially in excess of 20,000 hours. If field replacement of a blower motor is necessary, the entire blower assembly and its mounting plate is readily removed as a complete slide-out assembly. Condenser blower failure is detected by loss of ability to control the cabinet heat load, and is verified by the absence of air exhausting from the condenser heat exchanger outlet. Failure of the evaporator blower is indicated by inability to cool the cabinet enclosure combined with continuous operation of the compressor. The hot gas bypass would be open, therefore, the line between the hot gas bypass valve and the evaporator inlet would be at a relatively high temperature.

Access to both blowers is obtained by removing the back panel of the air conditioner package. It is not necessary to remove the air conditioner from the electronic equipment enclosure. Each of the blower mounting plates is held to the air conditioner cabinet structure by screws and nuts. Electrical connections may be broken at the terminal block, located near the compressor, or power leads may simply be cut as convenient and spliced together on installation of the replacement blower.

If failure involving the compressor or the hot gas bypass valve is encountered, it is strongly recommended that the air conditioner be returned prepaid to the Kooltronic plant.

7.3.11.1.1 UNUSUAL CONDITIONS

Momentary power interruption can result in a period of intermittent compressor operation. This condition is most likely to be encountered if the air conditioner is operating in full load and near maximum temperature; the power interruption is on the order of 1/2 to 5 seconds and the air conditioner is operating near the limit of its pumping capacity, the compressor may not be able to restart at the instant of power reapplication. The stalled compressor will cycle out under thermal overload protector, and remain off for, perhaps, 20 to 40 seconds. During this period, pressure differential across the compressor will tend to equalize and as the thermal overload protector closes, the compressor

7.0 MAINTENANCE (Continued)

7.3.11 KOOLTRONIC Electronic cabinet Air Conditioner (continued)

7.3.11.1.1 UNUSUAL CONDITIONS (continued)

should start. Under extreme conditions, overload cycling may take place two or three times before a clean start.

7.3.11.2 REFRIGERANT LOSS

Kooltronic air conditioners are subjected to a series of tests during and after manufacture, to prevent refrigerant leaks. It is possible, however, that concealed shipping damage or, over a sufficiently long period, the effects of possible microscopic leaks, below the threshold of the sensitive electronic leak detectors employed during manufacture, may result in the need for replenishment of refrigerant charge after an air conditioner has been placed in service is revealed by inability to cool the load to design temperature. When it has been verified by a qualified serviceman that a refrigerant shortage does exist, the unit may be recharged in the field using local purchase items or a recharge kit supplied by Kooltronic.

The minimum materials required for replenishment of refrigerant charge are: vacuum pump, gauges, charging hose and flask of appropriate refrigerant.

7.3.11.3 CHARGING PROCEDURE

1. Remove rear cover. After repairing the leak, the system should be evacuated to 26 in Hg vacuum.

2. Start unit. Enclosure door should be open to ensure that temperature at the air conditioner inlet port is no lower than room ambient.

3. Remove charge port valve cap.

4. Screw fitting and charging hose into light engagement at compressor fill valve; no refrigerant should be allowed to escape from compressor fill valve.

5. Remove protective cap from refrigerant flask.

6. Screw fitting on charging hose assembly, closest to needle valve, firmly in place on refrigerant flask.

7. Verify that charging flask is standing upright on base to ensure that only refrigerant in gas phase will be permitted to leave flask.

7.0 MAINTENANCE (Continued)

7.3.11 KOOLTRONIC ELECTRONIC CABINET AIR CONDITIONER (continued)

7.3.11.3 CHARGING PROCEDURE (continued)

8. Crack open needle valve, allowing refrigerant gas to pass through charging hose assembly and escape into atmosphere through loose charging hose fitting at compressor fill port.

9. Continue permitting refrigerant gas to escape through fitting for 10 - 15 seconds, sufficient to ensure that all air has been purged from charging hose assembly.

10. While gas is escaping at compressor fill port, tighten fitting firmly in place.

11. close needle valve firmly; pressure gauge will now show saturation pressure of refrigerant in air conditioner.

12. Open needle valve of charging hose assembly and permit refrigerant to enter air conditioner until proper weight of refrigerant is added. Refer to specifications for correct charge.

13. When proper charging has been achieved, the charging hose assembly fitting should be rapidly removed from the compressor fill port. There may be some loss of refrigerant during the removal of this fitting, consequently, removal should be done in a smooth and rapid manner. The Schrader valve is self-sealing, Check gasket inside brass compressor charging port cap to ensure that it has not been torn or lost. If gasket is damaged, discard cap and replace with cap from spares. Install cap firmly at compressor fill port.

14. Replace air filters.

7.3.11.4 SPECIFICATIONS

UNIT	KAC8	UNIT	KAC8
BTU RATING	8,000	HEIGHT	52 7/16
DEPTH	8 13/14	RUNNING CURRENT	12A
MAXIMUM AMBIENT	125°F	REFRIGERANT	R500
REFRIGERANT WT. ozs.	33	INSTALLED WEIGHT	177 lb.
VOLTAGE	208/230	FREQUENCY	60 Hz
PHASE	1	CONDENSATION DRAIN	1/8 - 27 NPT

7.0 MAINTENANCE (Continued)

7.3.11 KOOLTRONIC ELECTRONIC CABINET AIR CONDITIONER (continued)

7.3.11.5 BASIC RENEWAL PARTS

Basic Renewal Parts may be ordered direct from the manufacturer at the following address:

KOOLTRONIC, INC.
Hamilton Avenue
Hopewell, NJ 08525
Telephone: (609) 466-3400

7.3.11.5.1 BASIC RENEWAL PARTS LISTING

UNIT	RAC8
AIR FILTER, CONDENSER SET (STANDARD)	KCF 8/12
FILTER RECOAT AEROSOL CAN, PT.	A - 16
BLOWER, CONDENSER	KCC 8/12
BLOWER, EVAPORATOR	KCE 8/12
MOUNTING HINGE	KCH 8/12
REFRIGERANT CHARGING KIT	KCR 500

7.4 CARD AND MODULE FUNCTION LISTS

7.4.1 GENERAL - DIRECTO-MATIC* PLUS

DIRECTO-MATIC Plus *is* the name given to the family of electronics (cards) built by GE. The basic card size is 7.35 inches tall by 9 inches deep, with 80 pins at the back of the card.

A. characteristics of DIRECTO-MATIC Plus include:

1. Functional modules
2. Integrated circuits
3. Plastic coated circuit runs
4. Interchangeable without requiring tuneup
5. Optimum use of analog, digital, microprocessor technology
6. Automatic assembly - consistent high quality

* Registered trademark of General Electric company, U. S. A.

7.8 MAINTENANCE (Continued)

7.4 CARD AND MODULE FUNCTION LISTS (continued)

7.4.1 GENERAL – DIRECTO-MATIC PLUS (continued)

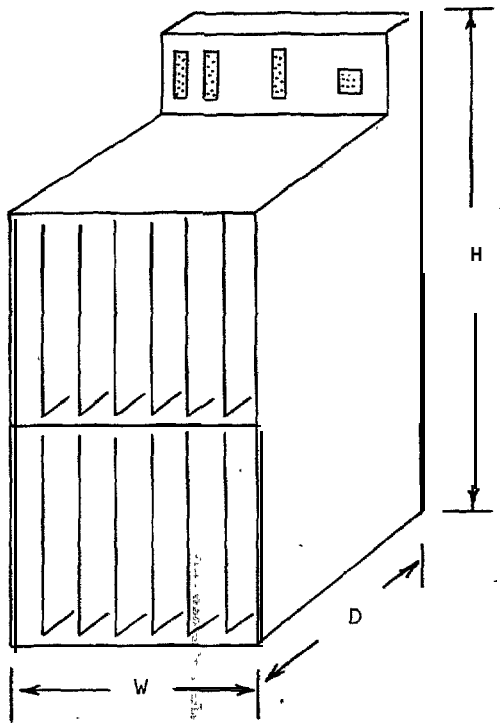
B. Features of DIRECTO-MATIC Plus cards include:

1. Self checking "I'MOK" (yellow LED)
2. Status indication (red LED)
3. Daughter boards – specific drive adjustments
4. Test mode switches
5. Test Rings
6. Convenient extraction levers

Modules are named according to size. A TEM is a Tiny Electronics Module, SEM is a Small Electronics Module, and MEM is a Medium Electronics Module. A diagram of the number of cards in, and the size of, these modules is given in Figure 23.

DIRECTO-MATIC (R) PLUS MODULES

STANDARD MODULES



		CARDS	H	W	D
1 X 6 "TEM"		6	17.45"	8"	11.5"
2 X 6 "SEM"		12	25.5	8	11.5
2 X 12 "MEM"		24	25.5	16	11.5

Figure 23: DIRECTO-MATIC* PLUS - STANDARD MODULES

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST

Below is an alphabetical listing of the cards used in the T-GSI system. The CMBA and CMBB cards are of the DS3810 series, and the rest are of the DS3800 series.

A. CMBA - Tape Controller - Historical

The tape controller card is used to interface the Historical, host, controller to the magnetic tape drive. The card has on-board memory and registers which the host must move data to or from for data transfers. control registers are used for the host to specify the action to be completed by the controller. Included is such information as the function (read, write, direction, write file mark, erase), as well as density, parity, and drive selection. Other registers are used to provide the host with status information and error indications of the last operation.

B. CMBB - Disk Controller - Historical

The disk controller is used to interface the Historical, host, controller to the Winchester disk drive unit. Any and all communication with the disk unit must go through this card. The host memory contains controller communication blocks and data buffers. The controller blocks contain information to describe the action the card should perform. Included are the type of function (initialize card, transfer drive status, format track, read, write, and others), drive unit, cylinder, sector, head, and address of data buffer. The host must signal the controller on the card to start activity. It can then read and write to the blocks and data buffer from host memory to make data transfers between the disk drive and memory in the Historical controller.

C. DMPC - E-BUS Generator

This is a daughter board that interfaces with the HMPF processor card. This card generates a fully arbitrated processor bus to be used in multiprocessor systems.

D. HAIA - Analog to Digital (A/D) Input

This card is intended to be used in an intelligent interface module for analog signals. Analog values in the range of +10/-10 volts are selected and converted to digital values.

This single-sized card accepts 16 analog input channels, These inputs are multiplexed into 1 line and converted to a two byte digital word (1 sign, 11 data bits), after waiting two programmable delays for settling.

7.0 MAINTENANCE (continued)

7.4.2 CARD FUNCTION LIST (continued)

It also routes general-purpose, one byte, digital inputs and outputs with respect to the microprocessor data bus. Both functions occur under the direct control of a microprocessor on the bottom byte of the data bus. This card uses both a busy flag and one interrupt to interact with the microprocessor.

E. HARA - Alterable Read-Only Memory (ROM)

A single sized, non-volatile memory card containing four 1024 X 4 EAROM chips that can be configured either as 1024 X 16 or 2048 X 8. The EAROM chips will have a ten year data retention with or without power supplied assuming the following limitations have not been exceeded:

1. 1000 Erase/Write Cycles per location

2. 70 Degree C Ambient Temperature

One other limitation is that each location may be read 1 X El0 times between refreshes. A write cycle is considered a refresh. The information in the HARA card is transferred to RAM at initialization or power up so any references by the microprocessor will not jeopardize the HARA's retention. This card uses the full 16 bit data bus and one interrupt with the microprocessor.

F. HCM_ (PCPA, PTCA, COME - PROGRAMMED) - Dual Communication

An 8085 microprocessor-based, dual-channel, serial communication subsystem. It has two RS-232C full or half duplex channels that can be programmed for baud rates up to 19.2 k bits/sec., synchronous or asynchronous. Interfaces to the L-Bus take place via a 2 k byte dual port RAM. It accepts either a 2 k or 4 k byte EPROM to program communication protocol. This card uses the full 16 bit data bus and two interrupts, one per channel, to interact with the microprocessor.

G. HIOD - Digital Input/Output (I/O)

A general purpose digital (32 bit input, 32 bit output) card. It can be used to sense the output of open collector devices because it provides an open collector pull-up line. The inputs can also sense standard TTL drivers. Both input and output functions occur under the direct control of a microprocessor in the bottom byte of the data bus.

H. HIMA - SDLC Isolation Card

An isolation board to be used in the Serial Data Link Isolation Module (LIM). It includes a 28 Vdc to 5 Vdc isolated power supply a yellow

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUCTION LIST (continued)

H. HIMA SDLC Isolation Card (continued)

"IMOK" LED. It contains a single-channel RS-422 channel which operates in a full duplex mode. Working common mode signal isolation is up to 200 volts. Signals are optically coupled for isolation. This card is used in the DS3820 LIMA module to interface and terminate on channel of an HMHA/SDLC data link card.

I. HMBA - Multibus* Mother Board

This is a double-sized board with a connector that interfaces a standard multibus board with a wire-wrapped backplane.

J. HMCA - Multibus Creator

This is a double-sized board that creates a multibus from an E-bus. The HMCA is a bi-directional, transparent interface used to drive multibus tape and disk controller boards in T-GSI historical electronics modules.

There are two Berg jumpers on the HMCA: J1 is used to either place the on board 14.7456 MHz oscillator in or out of the circuit . It is only used in test. J2 is used to enable or disable the 2.3 MHz arbitrator clock. Also used only in test. Both J1 and J2 should normally be in the "A" or in position.

K. HMHA (PDLA, PDLC - PROGRAMMED) - Serial Highway Master

An 8085 microprocessor-based, dual-channel, serial communication subsystem. Has high speed DMA, as well as a two SDLC/RS-422 parity line serialchannels that can be programmed for baud rates up to 48 K bits/sec., synchronous; interfaces to the L-Bus via a 2 K byte dual port RAM; accepts either a 2 K or 4 K byte EPROM to program commuication protocol. The card uses the full 16 bit data bus and two interrupts, one per channel, to interface with the microprocessor.

L. HMPF, MPFA (Programmed) - Microprocessor Controller

A single-size complete microcomputer, needing only an external power supply and a user oriented I/O device. Contains a 16 bit 8086 microprocessor with 8 K of random access memory (RAM) and provision for 16 K or 32 K programmable read-only memory (PROM). User programmable options include time base and baud rate generator. Other features include Central Processing Unit (CPU), stall detection and bus-hog detection. A total of four user defined interrupts are wired to a backplane connector with provision for external expansion up to 32.

* Multibus is a registered trademark of the Intel Coropration.

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST (continued)

L. HMPF - Microprocessor Controller (continued)

The card also has one RS-232C channel and an interface to a daughter board, which will generate a multi-processor Data/IO Bus.

M. HPMA - 64K Byte PROM

A single-size board with up to 64K bytes of programmable read only memory (PROM). Memory is addressable by either byte or word. Sixteen 24-pin sockets are provided. Jumpers are provided such that 2K or 4K byte PROMs may be used. Four different access times may be selected via jumpers on the board. This card uses the full 16 bit data bus to interface with the microprocessor.

N. HPRA - Pulse Rate Input

An HMPF (L-Bus) peripheral board which provides pulse rate measurement capability under software control. Signal conditioning and counters are provided for four tachometer inputs (3040A type) and generator frequency and/or phase measurement when used with a Sync. Module.

The card uses only the bottom byte of the data bus and one interrupt line, for a slave interrupt chip, to interface with the microprocessor.

0. HRDB - Relay Driver

A single-size, logic to load driver card, primarily for relay loads. It contains 16 voting, 16 non-voting and 3 discrete circuits. Each driver output turns on low and can interface TTL voltage levels to 28 V at 90 mA load current. Two external power supplies are diode "OR"-ed to form one power bus. This card is used in a DS3820 RDM module.

P. HRMA - Dual-Port 16K Random Access Memory (RAM)

A single-size board with 16K bytes of static random access memory (RAM). It is addressable by either byte or word.

Q. HSCG - High Level Isolation

A 16-channel isolated voltage sensing card which has several selectable ranges of input voltage. Major circuit functions include high impedance sense, threshold detection and optical isolation. The board can be used to isolate at 125V, 75V, and 28V with the proper jumpers in place. This single-size board receives inputs from the backplane wiring and maintains digital outputs at a front edge connector. This card is used in a DS3820 DMCB module.

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST (continued)

R. HXPC - CPU Expander

Expander board for the HMPF which provides the following L-Bus functions:

1. RAM (4K bytes)
2. PROM (4 sockets)
3. Digital I/O (Front edge connector)
4. Interrupts (8)
5. counters
6, Programmable Stall Timer

S. NAIA - 4-20mA Analog Point Input

A single-size, 16 points to 1 channel, multiplexed current loop to voltage board. Input range for all points is 4-20 mA and the output range is 2-10V. Accuracy is 0.1%. Has 1 enable and 4 address lines for channel selection and is designed for use with HAIA. Has both differential and common mode input filtering.

T. NAIB - +10/-10 Volt Analog Input

Same as NAIA except input and output ranges are +10/-10 volts.

U. NCLA - Current Loop to RS-232C I/F

This is an RS-232 to 20 mA converter that will accept I/O from an RS-232C port and convert it to a 20 mA loop interface. This card will work up to 9600 baud and can be jumpered to source or sink current on the transmit data and receive data loops. when working in the sink current mode, the 20 mA side is isolated from the RS-232 side. This card is used in module DS3820CLMA.

V. NDAC - 4-Channel Digital to Analog (D/A) Output

A digital to analog converter board with 4 addressable outputs. Each output is loaded by sending two consecutive bytes of data to the board, the first byte being the four least significant bits required by the D/A, and the second byte being the eight most significant bits. The outputs are bipolar, +/- 10 V maximum.

W. NPSL - Isolated Power supply

This card develops 8 isolated power buses to interface with the RTD input cards NRTA, NRTB, NRTC or NRTD.

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST (continued)

X. NPTA - Complex Power Analog Input

This card inputs 3-phase line currents and 3-phase line-to-line voltage, where an analog process converts them into DC signals proportional to line currents, line-to-line voltages and power, both Watts and VAR's. This card then multiplexes these 8 points into one channel to interface with the HAIA. The inputs for the NPTA come from the power conditioning module DS3820 PCNA.

Y. NRT__ RTD Analog Point Input

This card inputs up to eight RTD's into one channel to interface with the HAIA. This card uses a constant current source method to sense the RTD and will compensate each RTD for the loop resistance in the cables interfacing the card to the sensor. Possible card types and ranges are as follows:

CARD	RTD	RANGE
NRTA	10 ohm copper @ 25°C	-50 to 260°C
NRTB	100 ohm platinum "	-50 to 260°C
NRTC	100 ohm platinum "	-50 to 660°C
NRTD	120 ohm nickel "	-50 to 260°C

Z. NSCE - Passive Signal Conditioning

This card is used in the DS3820 PCNA module to interface and filter 3 PT isolation transformers and 3 CT isolation transformers to the NPTA card.

AA. NTBA, NTBG - Analog Terminal Board

This card is used to go from terminal boards to a ribbon cable for the NAI__ card. All signals on this card are decoupled to chassis ground. The card is used in the DS3820 ATA__ module.

AB. NTBC, NTBE - RTD Termination Board

This card performs the same function as the analog terminal board listed above, but with RTD inputs. The card is used in the DS3820 RTD module.

AC. NTCA - Thermocouple Input

A thermocouple signal processing card, using K type thermocouples. It accepts 16 differential input signal line pairs and performs the following functions:

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST (continued)

AC. NTCA - Thermocouple Input (continued)

1. Filtering: - differential noise filtering, line to line, and common mode filtering, each line per pair to ground.

2. Multiplexing: - multiplex the filtered 16 differential line pairs down to one differential line pair under external addressing control, originating off card.

3. Amplification: - to amplify the multiplexed millivolt thermocouple signal to +/- 10 V full scale, while providing very high input impedance, line-to-line and line-to-ground.

Physically, NTCA is a standard DIRECTO-MATIC Plus single slot, double layer printed wire board, PWB.

The thermocouple signal line pairs come on board through a card front connector. All other signals arrive via the backplane.

AD. NTCB - Thermocouple Input

This card is the same as NTCA except it uses a T type thermocouple.

AE. NTCC - Thermocouple Input

This card is the same as NTCA except it uses a J type thermocouple.

AF. NTCD - Thermocouple Input

This card is the same as NTCA except it uses an E type thermocouple.

AG. XPER - Isolation Module Termination Board

This card is used in the Isolation Module, DS3820 LIM__, accepting both isolated and non-isolated signals. Signals on the non-isolated or low level side are connected to a 10-pin ribbon cable connector and terminated with an RC filter. Signals on the isolated or high level side are connected to a 12-pin terminal board. This card may be used in the LIM with the HIMA or it may be used without the HIMA as a termination board.

AH. XPEV - 125 VDC Contact Input Mother Board; XCIA - (new plug in terminal board, uses DMCB)

This card is used in the contact input module, DS3820 DMCA, as a mother board for terminal boards and HSCG input cards. This mother board has jumpering on it to select the 125 VDC input level on the HSCG.

7.0 MAINTENANCE (Continued)

7.4.2 CARD FUNCTION LIST (continued)

AI. XPEX - Relay Driver Mother Board

This card is used in the relay driver module, DS3820 RDMA, as a mother board.

AJ. XPEY,XTIA - Isothermal Module Termination Board XTIA

This card is the key element in an isothermal module, converting thermocouple wire to copper wire. A temperature sensing element inside the module monitors ambient temperature and generates a proportional voltage to leave the card.

7.4.3 MODULE FUNCTION LIST

Below is an alphabetical listing of the modules used in the T-GSI system. The modules are of the DS3820 series.

A. ATA__ - Analog Termination Module

This module consists of a termination board, NTBA, mounted on brackets. On these brackets are sixteen screws along one side. Analog signals are brought into the module on the terminal boards and leave by one of the ribbon cables.

B. CLMA - Current Loop Interface Module

This module consists only of a current loop board, NCLA, on mounts.

C. DMC__ - Contact Digital Input Module

This input module consists of two isolation boards, HSCGs, mated to a backplane, XPEV or XCIA.

D. LIMA - Data Link Isolation Module

Isolation module consisting of the isolation board, HIMA, backplane termination board, XPER, phone jack and mounting hardware. This module is used to terminate and isolate signals on the control party line, CPL, synchronous communication link.

E. PCNA - Complex Power Module

This module converts the PT and CT, power transformer and current transformer, signals into voltage signals for generator voltage, current, real power, and reactive power.

7.0 MAINTENANCE (Continued)

7.4.3 MODULE FUNCTION LIST (continued)

F. PS5A - Power Supply Module

Provides four regulated DC output voltages derived from a single-phase AC input voltage. The four output voltages are +5V, +28V, +15V, and -15V. These are available at connectors and on a terminal strip located on the end of the module. The module checks for low output, high output, loss of AC input and missing output, and displays a status "ready" light,

G. PSFC - Power Supply Module

Provides a +28V, 6A output. The input can be 103VAC to 127VAC which can be floated or grounded.

H. PSFD - Power Supply Module

Provides one of the following output voltage families:

+5V 20A, +15V 1A, -15V 1A
or
+5V 15A, +15V 1.5A, -15V 1.5A

The input can be 103VAC to 127VAC which can be floated or grounded.

I. RDMA - Relay Driver Module

This is a module consisting of 4 relay driver cards, HRD__.

J. TIM__ - Thermocouple Isothermal Module

This is primarily a junction module that also houses a solid state temperature transducer for cold junction readings from the thermocouples.

7.4.4 OTHER MODULES

A. RELAY MODULE

This is a standard Directo-Matic* relay module consisting of eight terminal boards with 12 screws per board, two cards, and 30 relays. The entire module is wired on the backplane to provide 30 form C contacts.

*DIRECTO-MATIC is a registered trademark of General Electric Company, U. S. A.

7.0 MAINTENANCE (Continued)

7.4.4 OTHER MODULES (continued)

B. SYNCH MODULE

This module takes 115VAC and converts it to a TTL 0 - 5V square wave signal which is used in the master cabinet and the generator cabinet. In the master cabinet this signal runs the timing clock. In the generator cabinet the AC input comes from the PT and the TTL signal is used to measure the line frequency.

8.0 RENEWAL PARTS

Spare parts may be ordered through the nearest GE Apparatus and Engineering Services Organization office. Specify the drawing number or description of the parts. Also specify the system number and turbine number.

INDEX

Accuracy, see SPECIFICATIONS
Alarm Set Point Instructions see OPERATION

Cable Access, see SPECIFICATIONS
  Cables
Cables, see SPECIFICATIONS
Card Function List see MAINTENANCE
Connecting Cables
  maximum permissible length see SPECIFICATIONS
Control Console see INSTALLATION
Control Console Instructions see OPERATION
CRT Control Console see SPECIFICATIONS
CRT Control Console Keys see OPERATION
CRT Display, see SPECIFICATIONS
CRT Monitor see INSTALLATION

Data and Display Descriptions see OPERATION
Data Link see SPECIFICATIONS
Data Storage Instructions see OPERATION

Electrical Noise
  susceptability to see SPECIFICATIONS
Environment see SPECIFICATIONS
EQUIPMENT DESCRIPTION, 5
  THE MASTER STATION, 5
  REMOTE STATIONS, 7
    THE GENERATOR REMOTE STATION, 9
    THE TSI REMOTE STATION, 8
    THE TURBINE REMOTE STATION, 8
  THE CONTROL CONSOLE, 9
  THE CRT MONITOR, 10
  THE TELEPRINTER, 10
Equipment Location see SPECIFICATIONS

Generator Remote Cabinet see INSTALLATION

Humidity see SPECIFICATIONS

INDEX (continued)

I/O Devices see SPECIFICATIONS
Input Power see SPECIFICATIONS
INSTALLATION, 17
  CONTROL CONSOLE, 44
    CHECKOUT AND TEST, 44
    EQUIPMENT INSTALLATION, 44
  CRT MONITOR, 44
    CHECKOUT AND TEST, 44
    EQUIPMENT INSTALLATION, 44
  GENERAL, 17
    DIAGNOSTICS, 18
  GENERATOR REMOTE CABINET, 60
    CHECKOUT AND TEST, 61
    EQUIPMENT INSTALLATION, 60
  MASTER CABINET, 21
    CHECKOUT AND TEST, 21
    EQUIPMENT INSTALLATION, 21
  TELEPRINTER, 44
    CHECKOUT AND TEST, 44
    EQUIPMENT INSTALLATION, 44
  TSI REMOTE CABINET, 45
    CHECKOUT AND TEST, 45
    EQUIPMENT INSTALLATION, 45
  TURBINE REMOTE CABINET, 54
    CHECKOUT AND TEST, 57
    EQUIPMENT INSTALLATION, 54
Interconnecting Cable see SPECIFICATIONS
INTRODUCTION, 1

Magnetic Disk Unit see SPECIFICATIONS
MAINTENANCE, 114
  ROUTINE MAINTENANCE, 123
  SYSTEM OVERVIEW, 114
  TROUBLESHOOTING, 123
    AIR CONDITIONER, 144
    CARD AND MODULE FUNCTION LISTS, 148
    CONTROL CONSOLE, 142
    CRT MONITOR, 142
    GENERATOR REMOTE CABINET, 141
    MASTER CABINET, 124
    MODEM, 142
    TELEPRINTER, 142
    TSI REMOTE CABINET, 137
    TURBINE REMOTE CABINET, 139
Master Cabinet see INSTALLATION
Memory Capacity see, SPECIFICATIONS
Modem, see MAINTENANCE,
  TROUBLESHOOTING
Module Function List see MAINTENANCE

INDEX (continued)

OPERATION, 68
ALARM SET POINT INSTRUCTIONS, 110
CONTROL CONSOLE INSTRUCTIONS, 77
DATA STORAGE INSTRUCTIONS, 107
INTRODUCTION, 68
CONTROL CONSOLE KEY DESCRIPTION, 68
DATA & DISPLAY DESCRIPTIONS, 70
ALARM AND TRIP PROVISIONS, 73
ALARM LOGIC: TURBINE-GENERATOR SHAFT VIBRATION, 74
ALARM SUMMARY PAGE, 73
COMPOSITE GRAPH DATA PAGE, 70
CRT DISPLAY - GENERAL, 70
DATA GROUP PAGES, 71
BEARING DATA PAGE, 71
COOLING GAS TEMP. PAGE, 71
EXCITATION DATA PAGE, 72
LP TURBINE DATA PAGE, 71
STATOR RTD AND TC TEMPERATURES PAGE, 72
STATOR WATER DATA PAGE, 72
TURBINE TEMPERATURE DATA PAGE, 71
DATA TABLE (LOG) DISPLAY, 73
GENERAL DATA PAGE, 71
JOURNAL BEARING TEMPERATURE SPIKE MONITOR, 76
RATE OF MAIN & REHEAT STEAM TEMPERATURE CHANGE ALARM, 76
SELECTIVE DATA PAGE, 73
STATOR TEMPERATURE VARIABLE ALARM LIMIT, 76
SYSTEM MESSAGES, 74
DISPLAY MESSAGES, 74
HISTORICAL DISK MESSAGES, 75
HISTORICAL TAPE MESSAGES, 75
LIGHTS, 75
MAIN MESSAGES, 75
TAPERED LAND THRUST BEARING TEMPERATURE MONITOR, 76
TREND CHANNEL DIRECTORY PAGE, 73
TREND DISPLAY, 72
TELEPRINTER INSTRUCTIONS, 99

RECEIVING, HANDLING AND STORAGE, 3
GENERAL, 3
RECEIVING AND HANDLING, 3
STORAGE, 3
LONG TERM, 3
SHORT TERM, 3
Recording Interval see SPECIFICATIONS
RENEWAL PARTS, 160
Routine Maintenance see MAINTENANCE

INDEX (continued)

Scales, SEE SPECIFICATIONS
Sensor Detection, bad see SPECIFICATIONS
SIGNAL FLOW, 114
SPECIFICATIONS, 12
ACCURACY, 13
BAD SENSOR DETECTION AND INHIBITING, 14
ELECTRICAL NOISE, SUSCEPTABILITY TO, 13
INPUT POWER, 13
CABLES, 12
ATTACHMENTS AND CABLE ACCESS, 12
INTERCONNECTING CABLE BETWEEN CABINETS AND EQUIPMENT, 12
MAXIMUM PERMISSIBLE LENGTH OF CONNECTING CABLES, 12
ENVIRONMENT, 12
HUMIDITY, 13
TEMPERATURE, 12
EQUIPMENT LOCATION, 12
I/O DEVICES, 16
CRT DISPLAY, 16
CRT, 16
CRT CONTROL CONSOLE, 16
DATA LINK TO CUSTOMER COMPUTER, 16
MAGNETIC DISK UNIT, 16
TAPE RECORDER, 16
TELEPRINTER PORT, 16
MEMORY CAPACITY, 15
DISK DRIVE, 15
TAPE, 15
TREND DISPLAY, 15
RECORDING INTERVAL, 15
LOG RECORDING INTERVAL, 15
UPDATING, 15
SCALES, 14
SYSTEM CHECKS, 66
DATA RETRIEVAL, 66
EXERCISING REMOTES AND MASTER, 66

Tape Recorder see SPECIFICATIONS
Teleprinter Connections see SPECIFICATIONS
Teleprinter Instructions see OPERATION
Teleprinter, see INSTALLATION
Temperature see SPECIFICATIONS
Troubleshooting see MAINTENANCE
TSI Remote Cabinet see INSTALLATION
Turbine Remote Cabinet see INSTALLATION

Updating, see SPECIFICATIONS

ADDENDUM

This addendum covers the disk initialization routine and the substitution of a touch screen control console in place of the older pushbutton style.

Disk Initialization Command

Refer to the historical controller's "diagnostic commands" under the maintenance section for procedure on using a diagnostic terminal.

1. Description:	Disk initialization need only be run on the initial power-up of a new disk drive or if some system error has* allowed an erroneous log to be recorded. This command will destroy all records recorded to the disk
2. Maintenance Panel Setup:	Disk Dismounted
3. Privilege Required:	**Yes**
4. Format:	DIN(it)
5. Prompt: Response:	Password Enter the Password "1D7C"
6. Exit:	Wait for message stating "Initialization Successful" on the diagnostic teleprinter, This will take approximately 5 minutes. Then enter (Control) Y to get back to $ prompt.

Touch Screen Control

The touch screen control circuitry differs from the pushbutton version supplied on earlier TGSI systems only in hardware. The layout of the touch screen is similar to the panel layout shown in Figure 2. After initialization the command sequences are the same. The hardware differs in that the pushbuttons have been replaced with a touch sensitive display screen and the interface circuitry located in the TGSI Master Panel is different (refer to Fig. Al).

The touch screen control consist of two panels. One contains the touch screen and associated keyswitch and pushbuttons while the other has the system fault lights. The keyswitch and execute button are not used in this application. Only the reset button is used. These two panels are designed for panel or rack mounting. Refer to the outline drawing supplied. Connections consist of two cables that originate from the master station. Refer to customer connection diagram supplied for details.

The console's control circuitry is tied to the Master Panel via modems (Line Drivers) using RS232 protocol at 9600 baud, ASCII-8, no parity with 1 stop bit. The interface circuitry to the TGSI display processor uses RS232 protocol at 1200 baud, ASCII-7, even parity with 1 stop bit.

The interface circuitry, located in the rear of the Master Station's right hand bay, consist of a Tiny Electronics Module (TEM) and its associated power supply (PSFD-1).

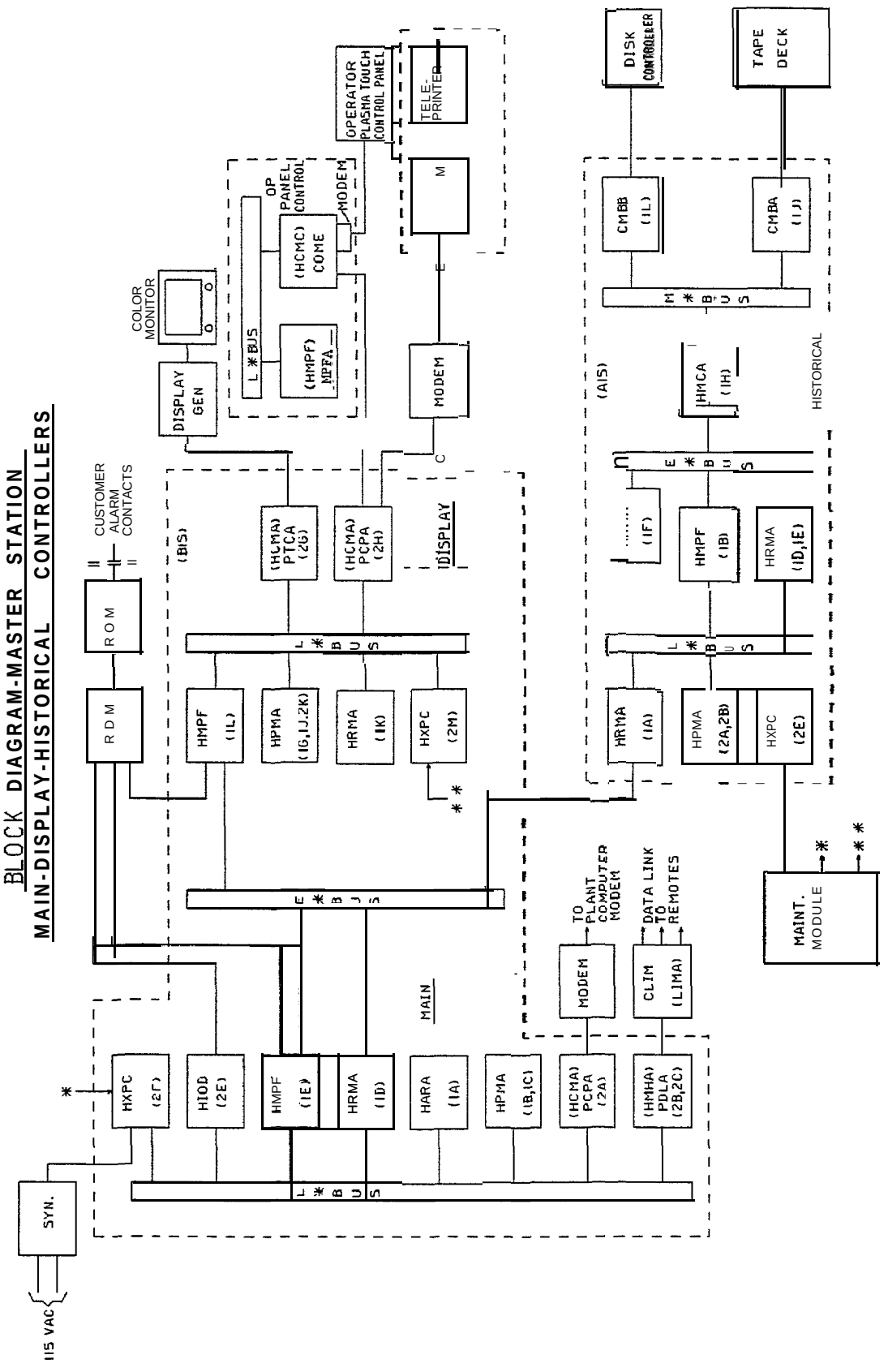


Figure A1. Block Diagram - Master Station with Touch Screen Style Control Console

Power Supply Check

Power is applied to PSFD-1 when power is applied to the supplies associated with the MEMs located in the Master Station. This is done by closing the two main breakers marked "Main Disconnect" and "Master". To apply power to the TEM close the breaker on the front of the power supply.

Using the test rings at the top of the TEM, measure the following voltages:

Test Rings	Voltage	PSFD-1 Adiust
P5 to DCOM	+5 +/- 0.1 Vdc	R1 (NPPH/C)
P15 to ACOM	+15 +/- 0.3 vdc	R6 (NPSK)
N15 to ACOM	-15 +/- 0.3 Vdc	R7 (NPSK)
ACOM to DCOM	0 +/- 0.1 Vdc	

(Readjust as necessary)

Control Console Power-Up

Verify connections per customer connection diagram supplied with the system.

Power-up the control console touch screen by applying power to it's power supply mounted on the rear of the unit. A bar will appear down each side of the display. After a few seconds a "-" prompt character will appear in the upper left comer of the screen. Now press the reset button on bottom of control panel. A set-up screen will appear. Entries can be changed by touching the desired setup item. The correct modes are:

End With	-	New Line
Operating Modes		ANSI
		Normal/Rel
Wrap		Off
Scroll		Off
Page		Blank
Main Port		9600
		8N1
		232N
Auxiliary Port		Don't care
Touch Overlay		Softkey
Parallel Keyboard		Off
Display Brightness		(1)
Display Size		512/XXX
Xon/Xoff	-	Off

When the set-up is correct, press "Save" on bottom left of screen. Then, press "Reset". The display is now ready to be initialized by the Interface Controller.

Interface Power-Up

To power-up the interface circuitry, turn switch on front of the PSFD-1 power supply "On". The processor will immediately start to write initialization information to the touch display processor, indicated by the fact the "AEN" led on top front of the communication card in slot 1D is lit during the transfer. After a few seconds the I'MOK led on the processor card in slot 1F will come "On" and the "BEN" led on the communication card will start to blink

at a two second rate, indicating a healthy link to the display processor in mem "B1S".

The touch screen should now display a matrix of operating keys similar to that shown in Fig. 2 And an area at top center of screen for interactive messages. The keys are activated by touching the screen over the key with a finger or other opaque object. Touching two keys, or a key and a non-key area, or even two non-key areas at the same time will result in an error. Touching a single non-key area will have no affect.

The system is now ready to accept commands as described for the pushbutton style in other sections of this manual.

Maintenance

If the touch screen controller fails to respond. Check to see if "Control Console Link Bad" message is displayed on the CRT. If so, reset interface control circuitry (Master Station) by turning power supply PSFD-1 off then back on. Otherwise, reset touch screen by pressing the Reset Button, on front of the console, to obtain the set-up page. Now press "Recall". Then, press the on screen "Reset" before resetting the interface circuitry. Verify that after pressing reset the "-" prompt appears in the upper left corner of screen before resetting the interface circuitry.

If the prompt does not appear after resetting, power down the touch screen then, depress and hold the Reset button until the set-up screen appears. Select modes as required (refer to list under Control Console Power-Up), press "Save", then press "Reset". If prompt still does not appear, replace touch screen Power supply. Then replace the touch panel itself.

If prompt appears but communications still cannot be obtained, check wiring and modems (Line Drivers) between the touch screen and interface circuitry.